
Nonlinear Regression Models

10.1 Introduction

The regression models of Chapters 8 and 9 involve a continuous response that depends linearly on the parameters. Linear models remain the backbone of most statistical data analysis, but they have their deficiencies. In many applications, response variables are discrete, or statistical or substantive considerations suggest that covariates will appear nonlinearly. Models of this sort appeared on a somewhat *ad hoc* basis in the literature up to about 1970, since when there has been an explosion of generalizations to the linear model. Two important developments were the use of iterative weighted least squares for fitting, and the systematic use of exponential family response distributions. The iterative weighted least squares algorithm has wide applicability in nonlinear models and we outline its properties in Section 10.2, giving also a discussion of likelihood inference in this context. Exponential family response densities play a central role in generalized linear models, which we describe in Section 10.3, turning to the important special cases of binomial and Poisson responses in Sections 10.4 and 10.5. These models are widely used, but real data often display too much variation for them to be taken at face value. In Section 10.6 we outline remedies for this, based on the discussion of estimating functions in Section 7.2.

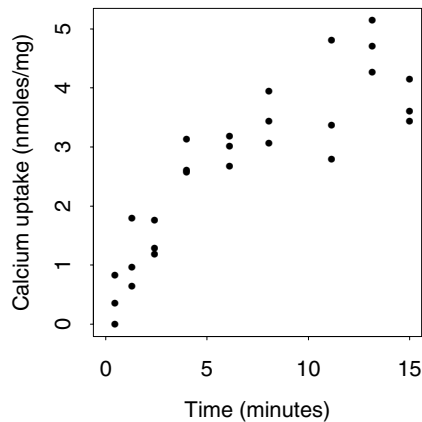
In each of these generalizations of the linear model our key notion that a few parameters summarize the entire model is retained. Section 10.7 branches out in a different direction, taking the viewpoint that the regression curve itself is more central than the parameters that summarize it. This leads to the idea of semiparametric modelling, particularly useful in exploratory analysis and assessment of model fit. Finally Section 10.8 outlines how the special features of survival data described in Section 5.4 may be dealt with in regression settings. The remainder of this section briefly motivates later developments.

Below we mostly assume that the responses $y_1, \dots, y_n$ are independent, and that the density of y_j depends on a parameter η_j , through which systematic variation enters. The η_j depend on parameters β and explanatory variables, so we can write $\eta_j = \eta_j(\beta)$. In some models there is an incidental parameter ϕ which controls the

Table 10.1 Calcium uptake (nmoles/mg) of cells suspended in a solution of radioactive calcium, as a function of time suspended (minutes) (Rawlings, 1988, p. 403).

Time (minutes)	Calcium uptake (nmoles/mg)		
0.45	0.34170	−0.00438	0.82531
1.30	1.77967	0.95384	0.64080
2.40	1.75136	1.27497	1.17332
4.00	3.12273	2.60958	2.57429
6.10	3.17881	3.00782	2.67061
8.05	3.05959	3.94321	3.43726
11.15	4.80735	3.35583	2.78309
13.15	5.13825	4.70274	4.25702
15.00	3.60407	4.15029	3.42484

Figure 10.1 Calcium uptake (nmoles/mg) of cells suspended in a solution of radioactive calcium, as a function of time suspended (minutes).



shape or spread of the distribution. Usually the aim is inference about the parameters β or prediction of future responses. For the normal linear model y_j is normally distributed with mean η_j and variance $\phi = \sigma^2$ and $\eta_j = x_j^T \beta$, but much wider classes of models can be put into this framework.

Example 10.1 (Calcium data) Table 10.1 contains data on the uptake of calcium by cells that had been in “hot” calcium suspension. There are three observations at each of nine times after the start of the experiment. The data, plotted in Figure 10.1, show nonlinear dependence of calcium uptake on time.

Let y denote calcium uptake at time x after the start of the experiment. Then a differential equation that might describe how y depends on x is

$$\frac{dy}{dx} = (\beta_0 - y)/\beta_1,$$

with initial condition $y = 0$ when $x = 0$ and solution $y = \beta_0\{1 - \exp(-x/\beta_1)\}$. Allowing for measurement error, which seems to be similar at all levels of y , we might write

$$y = \beta_0\{1 - \exp(-x/\beta_1)\} + \varepsilon,$$

	<i>Aphrissa boisduvalli</i> N/S/E	<i>Phoebis argante</i> N/S/E	<i>Dryas iulia</i> N/S/E	<i>Pierella luna</i> N/S/E	<i>Consul fabius</i> N/S/E	<i>Siproeta stelenes</i> † N/S/E
Unpainted	0/0/14	6/1/0	1/0/2	4/1/5	0/0/0	0/0/1
Brown	7/1/2	2/1/0	1/0/1	2/2/4	0/0/3	0/0/1
Yellow	7/2/1	4/0/2	5/0/1	2/0/5	0/0/1	0/0/3
Blue	6/0/0	0/0/0	0/0/1	4/0/3	0/0/1	0/1/1
Green	3/0/1	1/1/0	5/0/0	6/0/2	0/0/1	0/0/3
Red	4/0/0	0/0/0	6/0/0	4/0/2	0/0/1	3/0/1
Orange	4/2/0	6/0/0	4/1/1	7/0/1	0/0/2	1/1/1
Black	4/0/0	0/0/0	1/0/1	4/2/2	7/1/0	0/1/0

† includes *Philaethria dido* also.

Table 10.2 Response of a rufous-tailed jacamar to individuals of seven species of palatable butterflies with artificially coloured wing undersides. (N=not sampled, S = sampled and rejected, E = eaten)

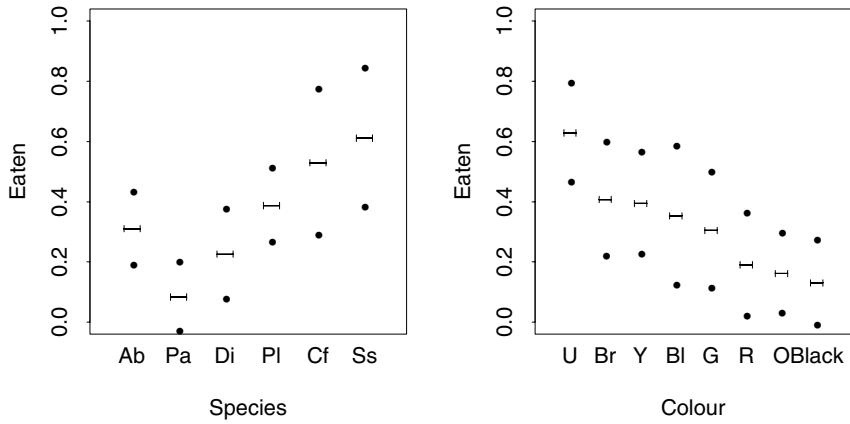


Figure 10.2 Proportion of butterflies eaten ($\pm 2SE$) for different species and wing colour.

where ε is normal with mean zero and variance σ^2 . This fits into the general framework with $\eta(\beta) = \beta_0\{1 - \exp(-x/\beta_1)\}$, if y is normal with mean η and variance σ^2 . ■

Example 10.2 (Jacamar data) As part of a study of the learning ability of tropical birds, Peng Chai of the University of Texas at Austin collected data on the response of a rufous-tailed jacamar, *Galbula ruficauda*, to butterflies. He used marker pens to paint the underside of the wings of eight species of butterflies, and then released each butterfly in the cage where the bird was confined. The bird responded in three ways: by not attacking the butterfly (N); by attacking the butterfly, then sampling but rejecting it (S); or by attacking and eating the butterfly, usually after removing some or all of the wings (E).

Table 10.2 gives the data in the form of an 8×6 layout, but the response is a triplet of counts. Figure 10.2 shows strong variation in the proportion of the different species and colours eaten.

If we take the number of butterflies eaten as the response, we might consider a model where the probability of the jacamar eating a butterfly would depend on its species and on its wing colour. However the low numbers in most cells make it likely that a linear model would give negative fitted probabilities, and this is clearly

undesirable. In the better models described in Section 10.3 the probabilities have logistic form $\exp(\eta)/\{1 + \exp(\eta)\}$, where $\eta(\beta) = x^T\beta$, so once again we have a model that is nonlinear in the covariates. ■

One time-honoured way to treat examples like these is to transform the responses so that a linear model can be applied. The transformation may be chosen to stabilise the variance of the response, or better, for simplicity of interpretation. Although transformations remain useful for exploratory analysis and for plotting, they have largely been superseded by the use of likelihood estimation and more realistic modelling. In the next section we outline how this may be applied in nonlinear regression.

10.2 Inference and Estimation

10.2.1 Likelihood inference

Inference for nonlinear models is usually based on the large-sample likelihood theory described in Chapter 4. Under mild regularity conditions the asymptotic chi-squared distributions of likelihood ratio statistics and the joint normal distribution of the maximum likelihood estimates of β and ϕ form the basis for tests and confidence intervals, though the adequacy of these approximations needs to be considered in applications.

Comparisons between nested models are often based on a form of likelihood ratio statistic known as the *deviance*. Suppose that the log likelihood for a model for independent observations $y_1, \dots, y_n$ is

$$\ell(\beta, \phi) = \sum_{j=1}^n \log f\{y_j; \eta_j(\beta), \phi\}, \quad (10.1)$$

and that ϕ is known; for now we suppress ϕ . If the maximum likelihood estimate of the $p \times 1$ parameter vector β is $\hat{\beta}$, then the maximum likelihood estimate of η_j is $\hat{\eta}_j = \eta_j(\hat{\beta})$ and the maximized log likelihood is $\ell(\hat{\beta})$. This is obtained by maximizing over p parameters, which connect the η_j for the different observations y_j , and thereby constrain them: larger p would lead to a larger log likelihood. The *saturated model* with no constraints on the η_j gives the largest possible log likelihood. Then β has dimension n and if the map between β and η is 1–1, the maximized log likelihood is simply the sum of the maxima of the individual terms on the right of (10.1). Let $\tilde{\eta}_j$ be the value of η_j that maximizes $\log f\{y_j; \eta_j\}$. Then the *scaled deviance* is defined as

$$D = 2 \sum_{j=1}^n \{\log f(y_j; \tilde{\eta}_j) - \log f(y_j; \hat{\eta}_j)\}, \quad (10.2)$$

which is always non-negative. This will be small when the $\hat{\eta}_j$ and $\tilde{\eta}_j$ are close, suggesting that the model fits well. Large D suggests poor fit, analogous to the sum of squares in a linear model.

Suppose we want to test whether $\beta_{q+1}, \dots, \beta_p$ take specified values. Let A denote the model in which all p components of β vary freely and let B denote the model

with $q < p$ parameters nested within A , with maximum likelihood estimates $\hat{\eta}^A$ and $\hat{\eta}^B$ and deviances D_A and D_B . Then the likelihood ratio statistic for comparing the models is

$$2 \sum_{j=1}^n \{ \log f(y_j; \hat{\eta}_j^A) - \log f(y_j; \hat{\eta}_j^B) \} = D_B - D_A, \quad (10.3)$$

and provided the models are regular, this has an approximate χ^2_{p-q} distribution when model B is correct. Hence differences between scaled deviances are often used to compare nested models.

Example 10.3 (Normal deviance) Suppose that the y_j are normal with means η_j and known variance ϕ . Then

$$\log f(y_j; \eta_j, \phi) = -\frac{1}{2} \log(2\pi\phi) - (y_j - \eta_j)^2 / \phi$$

is maximized with respect to η_j when $\tilde{\eta}_j = y_j$, giving $\log f(y_j; \tilde{\eta}_j, \phi) = -\frac{1}{2} \log(2\pi\phi)$. Therefore the scaled deviance for a model with fitted means $\hat{\eta}_j$ is

$$D = \phi^{-1} \sum_{j=1}^n (y_j - \hat{\eta}_j)^2,$$

which is just the residual sum of squares for the model, divided by ϕ . If $\eta_j = x_j^T \beta$ is the correct normal linear model, we saw in Section 8.3 that the distribution of the residual sum of squares is $\phi \chi^2_{n-p}$, so values of D extreme relative to the χ^2_{n-p} distribution call the model into question.

The difference between deviances for nested models A and B in which β has dimensions p and $q < p$,

$$D_B - D_A = \phi^{-1} \sum_{j=1}^n \{ (y_j - \hat{\eta}_j^B)^2 - (y_j - \hat{\eta}_j^A)^2 \} \sim \chi^2_{p-q}$$

when model B is correct. Results from Section 8.5.1 show that this distribution is exact for linear models. ■

If ϕ is unknown, it is replaced by an estimate. The large-sample properties of deviance differences outlined above still apply, though in small samples it may be better to replace the approximating χ^2 distribution by an F distribution with numerator degrees of freedom equal to the degrees of freedom for estimation of ϕ .

10.2.2 Iterative weighted least squares

In order to apply large-sample likelihood approximations we must obtain the maximum likelihood estimates $\hat{\beta}$ and their standard errors. A general procedure for this may be obtained by a variant Newton–Raphson method, an *iterative weighted least squares* algorithm widely used for nonlinear estimation. We now discuss this and some of its ramifications.

Assuming for now that ϕ is fixed, we write the log likelihood for β as

$$\ell(\beta) = \sum_{j=1}^n \ell_j\{\eta_j(\beta), \phi\},$$

where $\ell_j\{\eta_j(\beta), \phi\}$ is the contribution made by the j th observation. The maximum likelihood estimates $\hat{\beta}$ usually satisfy the equations

$$\frac{\partial \ell(\hat{\beta})}{\partial \beta_r} = 0, \quad r = 1, \dots, p,$$

which we put in matrix form to give the likelihood equation

$$\frac{\partial \ell(\hat{\beta})}{\partial \beta} = \frac{\partial \eta^\top}{\partial \beta} u(\hat{\beta}) = 0, \quad (10.4)$$

where $u(\beta)$ is the $n \times 1$ vector whose j th element is $\partial \ell(\beta)/\partial \eta_j$ because η_j only enters the log likelihood through the contribution made by the j th observation.

To find the maximum likelihood estimate $\hat{\beta}$ starting from a trial value β , we make a Taylor series expansion in (10.4), to obtain

$$\frac{\partial \eta^\top(\beta)}{\partial \beta} u(\beta) + \left\{ \sum_{j=1}^n \frac{\partial \eta_j(\beta)}{\partial \beta} \frac{\partial^2 \ell_j(\beta)}{\partial \eta_j^2} \frac{\partial \eta_j(\beta)}{\partial \beta^\top} + \sum_{j=1}^n \frac{\partial^2 \eta_j(\beta)}{\partial \beta \partial \beta^\top} u_j(\beta) \right\} (\hat{\beta} - \beta) \doteq 0. \quad (10.5)$$

If we denote the $p \times p$ matrix in braces on the left by the $p \times p$ matrix $-J(\beta)$, assumed invertible, we can rearrange (10.5) to obtain

$$\hat{\beta} \doteq \beta + J(\beta)^{-1} \frac{\partial \eta^\top(\beta)}{\partial \beta} u(\beta). \quad (10.6)$$

This suggests that maximum likelihood estimates may be obtained by starting from a particular β , using (10.6) to obtain $\hat{\beta}$, then setting β equal to $\hat{\beta}$, and iterating (10.6) until convergence. This is the Newton–Raphson algorithm applied to our particular setting. In practice it can be more convenient to replace $J(\beta)$ by its expected value

$$I(\beta) = \sum_{j=1}^n \frac{\partial \eta_j(\beta)}{\partial \beta} E \left(-\frac{\partial^2 \ell_j}{\partial \eta_j^2} \right) \frac{\partial \eta_j(\beta)}{\partial \beta^\top};$$

the other term vanishes because $E\{u_j(\beta)\} = 0$. We write

$$I(\beta) = X(\beta)^\top W(\beta) X(\beta), \quad (10.7)$$

where $X(\beta)$ is the $n \times p$ matrix $\partial \eta(\beta)/\partial \beta^\top$ and $W(\beta)$ is the $n \times n$ diagonal matrix whose j th diagonal element is $E(-\partial^2 \ell_j/\partial \eta_j^2)$.

If we replace $J(\beta)$ by $X(\beta)^\top W(\beta) X(\beta)$ and reorganize (10.6), we obtain

$$\hat{\beta} = (X^\top W X)^{-1} X^\top W (X\beta + W^{-1}u) = (X^\top W X)^{-1} X^\top W z, \quad (10.8)$$

say, where the dependence of the terms on the right on β has been suppressed. That is, starting from β , the updated estimate $\hat{\beta}$ is obtained by weighted linear regression of the vector $z = X(\beta)\beta + W(\beta)^{-1}u(\beta)$ on the columns of $X(\beta)$, using weight matrix $W(\beta)$. The maximum likelihood estimates are obtained by repeating this step until

$\partial g/\partial \beta$ denotes the $p \times 1$ vector whose r th element is $\partial g/\partial \beta_r$, $\partial^2 g/\partial \beta \partial \beta^\top$ denotes the $p \times p$ matrix whose (r, s) element is $\partial^2 g/\partial \beta_r \partial \beta_s$, $\partial \eta/\partial \beta^\top$ denotes the $n \times p$ matrix with (j, r) element is $\partial \eta_j/\partial \beta_r$, and so forth. Note that $\partial \eta^\top/\partial \beta = (\partial \eta/\partial \beta^\top)^\top$.

the log likelihood, the estimates, or more often both are essentially unchanged. The variable z plays the role of the response or dependent variable in the weighted least squares step and is sometimes called the *adjusted dependent variable*.

Often the structure of a model simplifies the estimation of an unknown value of ϕ . It may be estimated by a separate step between iterations of $\hat{\beta}$, by including it in the step (10.6), or from the profile log likelihood $\ell_p(\phi)$.

Example 10.4 (Normal linear model) In the normal linear model, we write $\eta_j = x_j^T \beta$. If the y_j are independently normally distributed with means η_j and variances $\phi = \sigma^2$, we have

$$\ell_j(\eta_j, \sigma^2) \equiv -\frac{1}{2} \left\{ \log \sigma^2 + \frac{1}{\sigma^2} (y_j - \eta_j)^2 \right\},$$

so

$$u_j(\eta_j) = \frac{\partial \ell_j}{\partial \eta_j} = \frac{1}{\sigma^2} (y_j - \eta_j), \quad \frac{\partial^2 \ell_j}{\partial \eta_j^2} = -\frac{1}{\sigma^2};$$

the j th element on the diagonal of W is the constant σ^{-2} . The (j, r) element of the matrix $\partial \eta / \partial \beta^T$ is $\partial \eta_j / \partial \beta_r = x_{jr}$, so $X(\beta)$ is simply the $n \times p$ design matrix X . We see that $z = X(\beta)\beta + W^{-1}(\beta)u(\beta) = y$, because in this situation $X(\beta)\beta = X\beta$ and $W^{-1}(\beta)u(\beta) = \sigma^2(y - X\beta)/\sigma^2$.

Here iterative weighted least squares converges in a single step.

The maximum likelihood estimate of σ^2 is obtained as in Section 8.2.1, and is $\hat{\sigma}^2 = SS(\hat{\beta})/n$, where $SS(\beta)$ is the sum of squares $(y - X\beta)^T(y - X\beta)$. ■

Example 10.5 (Normal nonlinear model) Here the mean of the j th observation is $\eta_j = \eta_j(\beta)$. The log likelihood contribution $\ell_j(\eta_j)$ is the same as in the previous example, so u and W are the same also. However, the j th row of the matrix $X = \partial \eta / \partial \beta^T$ is $(\partial \eta_j / \partial \beta_0, \dots, \partial \eta_j / \partial \beta_{p-1})$, and as η_j is nonlinear as a function of β , X depends on β . After some simplification, we see that the new value for $\hat{\beta}$ given by (10.8) is

$$\hat{\beta} \doteq (X^T X)^{-1} X^T (X\beta + y - \eta), \quad (10.9)$$

where X and η are evaluated at the current β . Here $\eta \neq X\beta$ and (10.9) must be iterated.

The log likelihood is a function of β only through the sum of squares, $SS(\beta) = \sum_{j=1}^n \{y_j - \eta_j(\beta)\}^2$. The profile log likelihood for σ^2 is

$$\ell_p(\sigma^2) = \max_{\beta} \ell(\beta, \sigma^2) \equiv -\frac{1}{2} \{n \log \sigma^2 + SS(\hat{\beta})/\sigma^2\},$$

so the maximum likelihood estimator of σ^2 is $\hat{\sigma}^2 = SS(\hat{\beta})/n$. Although $S^2 = SS(\hat{\beta})/(n - p)$ is not unbiased when the model is nonlinear, it turns out to have smaller bias than $\hat{\sigma}^2$, and is preferable in applications.

In some cases the error variance depends on covariates, and we write the variance of the j th response as $\sigma_j^2 = \sigma^2(x_j, \gamma)$. Such models may be fitted by alternating iterative weighted least squares updates for β treating γ as fixed at a current value

with those for γ with β fixed, convergence being attained when neither estimates nor log likelihood change materially. ■

Iterative weighted least squares can be used for maximum likelihood estimation in linear models with non-normal errors and extends to situations with dependent responses.

Example 10.6 (Venice sea level data) In Section 5.1 the straight-line regression equation $y_j = \gamma_0 + \gamma_1(x_j - \bar{x}) + \varepsilon_j$ was fitted to data on annual maximum sea levels at Venice from 1931–1981. Fitting was by least squares, as is appropriate for normal responses, but the right-hand panel of Figure 5.2 suggests that the errors are non-normal. As the data are annual maxima, it is more appropriate to suppose that y_j has the Gumbel density

$$f(y_j; \eta_j, \tau) = \tau^{-1} \exp \left\{ -\frac{y_j - \eta_j}{\tau} - \exp \left(-\frac{y_j - \eta_j}{\tau} \right) \right\}, \quad (10.10)$$

where τ is a scale parameter and $\eta_j = \beta_0 + \beta_1(x_j - \bar{x})$; here we have replaced the γ s with β s for continuity with the general discussion above. Use of this density is justified by the arguments leading to (6.34).

In this case

$$\ell_j(\eta_j, \tau) = -\log \tau - \frac{y_j - \eta_j}{\tau} - \exp \left(-\frac{y_j - \eta_j}{\tau} \right), \quad (10.11)$$

and it is straightforward to establish that

$$\frac{\partial \ell_j(\eta_j, \tau)}{\partial \eta_j} = \tau^{-1} \left\{ 1 - \exp \left(-\frac{y_j - \eta_j}{\tau} \right) \right\}, \quad E \left\{ -\frac{\partial^2 \ell_j(\eta_j, \tau)}{\partial \eta_j^2} \right\} = \tau^{-2},$$

that $\partial \eta / \partial \beta^T = X$ is the $n \times 2$ matrix whose j th row is $(1, x_j - \bar{x})$, and $W = \tau^{-2} I_n$. Hence (10.8) becomes $\hat{\beta} \doteq (X^T X)^{-1}(X\beta + \tau^2 u)$, where the j th element of u is $\tau^{-1}[1 - \exp\{-(y_j - \eta_j)/\tau\}]$.

Here it is simplest to fix τ , to obtain $\hat{\beta}$ by iterating (10.8) for each fixed value of τ , and then to repeat this over a range of values of τ , giving the profile log likelihood $\ell_p(\tau)$ and hence confidence intervals for τ . Confidence intervals for β_0 and β_1 are obtained from the information matrix.

With starting value chosen to be the least squares estimates of β , and with $\tau = 5$, 19 iterations of (10.8) were required to give estimates and a maximized log likelihood whose relative change was less than 10^{-6} between successive iterations. We then took $\tau = 5.5, \dots, 40$, using $\hat{\beta}$ from the preceding iteration as starting-value for the next; in most cases just three iterations were needed. The left panel of Figure 10.3 shows a close-up of $\ell_p(\tau)$; its maximum is at $\hat{\tau} = 14.5$, and the 95% confidence interval for τ is (11.9, 18.1). The maximum likelihood estimates of β_0 and β_1 are 111.4 and 0.563, with standard errors 2.14 and 0.137; these compare with standard errors 2.61 and 0.177 for the least squares estimates. There is some gain in precision in using the more appropriate model. ■

Example 10.7 (ABO blood group system) In the usual model for the ABO blood-group system (Examples 4.38, 5.12), the probabilities for the observed blood groups

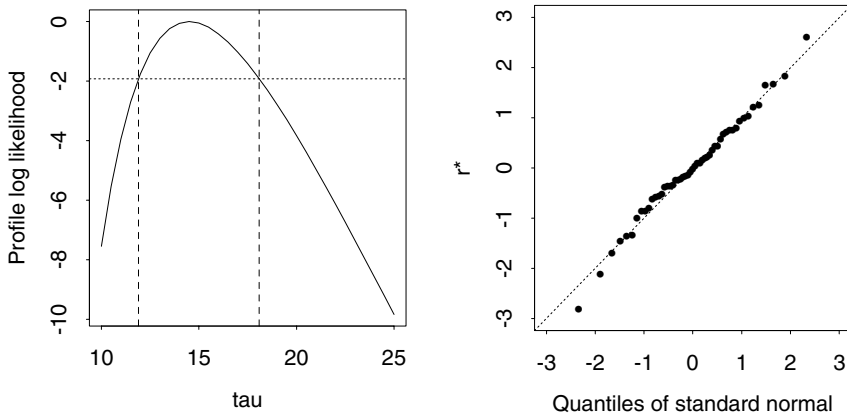


Figure 10.3 Gumbel analysis of Venice data. Left panel: profile log likelihood $\ell_p(\tau)$, with 95% confidence interval. Right panel: normal probability plot of residuals r_j^* .

A, B, AB and O are $\eta_A = \lambda_A^2 + 2\lambda_A\lambda_O$, $\eta_B = \lambda_B^2 + 2\lambda_B\lambda_O$, $\eta_{AB} = 2\lambda_A\lambda_B$, and $\eta_O = \lambda_O^2$, assuming random mating of a population in which λ_A , λ_B , and λ_O are the frequencies of alleles A, B, and O; here $\lambda_A + \lambda_B + \lambda_O = 1$. Given n independent individuals in which the groups appear with frequencies y_A , y_B , y_{AB} , and $y_O = n - y_A - y_B - y_{AB}$, the log likelihood is

$$y_A \log \eta_A + y_B \log \eta_B + y_{AB} \log \eta_{AB} + (n - y_A - y_B - y_{AB}) \log(1 - \eta_A - \eta_B - \eta_{AB}).$$

One of the η s is redundant and we have replaced η_O with $1 - \eta_A - \eta_B - \eta_{AB}$. If we set $\beta_1 = \log \lambda_A$ and $\beta_2 = \log \lambda_B$, we have

$$\frac{\partial \eta}{\partial \beta^T} = 2 \begin{pmatrix} \lambda_A \lambda_O & -\lambda_A \lambda_B \\ -\lambda_A \lambda_B & \lambda_B \lambda_O \\ \lambda_A \lambda_B & \lambda_A \lambda_B \end{pmatrix}, \quad \frac{\partial \ell}{\partial \eta} = \begin{pmatrix} y_A/\eta_A - y_O/\eta_O \\ y_B/\eta_B - y_O/\eta_O \\ y_{AB}/\eta_{AB} - y_O/\eta_O \end{pmatrix}$$

and

$$W = n \begin{pmatrix} 1/\eta_A - 1/\eta_O & -1/\eta_O & -1/\eta_O \\ -1/\eta_O & 1/\eta_B - 1/\eta_O & -1/\eta_O \\ -1/\eta_O & -1/\eta_O & 1/\eta_{AB} - 1/\eta_O \end{pmatrix}.$$

Once again iterative weighted least squares can be used for maximization, although the weight matrix is not diagonal because the log likelihood contribution from y_O depends on η_A , η_B and η_{AB} . ■

10.2.3 Model checking

The fit of a linear model is checked using residuals and other diagnostics and by embedding it into more complex models chosen to capture particular departures of interest. These ideas extend to nonlinear models through the components of the iterative weighted least squares algorithm. For example, the leverage of the j th case, h_{jj} , is defined as the j th diagonal element of the matrix $H = W^{1/2} X(X^T W X)^{-1} X^T W^{1/2}$,

evaluated at $\hat{\beta}$. As the weight matrix W generally depends on $\hat{\beta}$, the leverages depend on the responses as well as the covariates, but otherwise the h_{jj} have the same properties as in a linear model, where h_{jj} is the j th diagonal element of $X(X^T X)^{-1} X^T$.

There are several types of residual for nonlinear models, because no single definition plays all the roles of the standardized residual r_j defined at (8.26) for the linear model.

The residual sum of squares for a linear model can be written $\sum e_j^2$, so for more general models the analogy with the deviance suggests that we write $D = \sum d_j^2$, where d_j is the signed square root of the contribution y_j makes to the scaled deviance. This gives the *deviance residual*

$$d_j = \text{sign}(\tilde{\eta}_j - \hat{\eta}_j)[2\{\ell_j(\tilde{\eta}_j; \phi) - \ell_j(\hat{\eta}_j; \phi)\}]^{1/2}.$$

Analogy with the linear model suggests that the *standardized deviance residuals* $r_{Dj} = d_j/(1 - h_{jj})^{1/2}$ will be more homogeneous, and detailed calculations confirm that usually the r_{Dj} have roughly unit variances and distributions close to normal, though possibly with non-zero mean. Exceptions to this are binary data and Poisson data with small responses, whose residuals are essentially discrete.

A better general definition of residual combines the standardized deviance residual with the *standardized Pearson residual*

$$r_{Pj} = \frac{u_j(\hat{\beta})}{\{w_j(\hat{\beta})(1 - h_{jj})\}^{1/2}},$$

which is a standardized score statistic. Detailed calculations show that the distributions of the quantities $r_j^* = r_{Dj} + r_{Dj}^{-1} \log(r_{Pj}/r_{Dj})$ are close to normal for a wide range of models. For a normal linear model, $r_{Pj} = r_{Dj} = r_j^* = r_j$. The r_j^* or r_{Dj} may be used in the plots described in Section 8.6.1.

In a linear model, the influence of the j th case, (x_j, y_j) , is proportional to $(\hat{y} - X\hat{\beta}_{-j})^T(\hat{y} - X\hat{\beta}_{-j})$, where $\hat{\beta}_{-j}$ is the estimate when the case is deleted from the model. For more general models a better measure is

$$2p^{-1}\{\ell(\hat{\beta}) - \ell(\hat{\beta}_{-j})\}, \quad (10.12)$$

where p is the dimension of β . Calculation of all n of these requires n additional fits, and it is more convenient to use the *approximate Cook statistic*

$$C_j = \frac{h_{jj}}{p(1 - h_{jj})} r_{Pj}^2. \quad (10.13)$$

This is derived by Taylor series expansion of (10.12) and reduces to (8.30) in the case of the normal linear model.

Example 10.8 (Venice sea level data) Here the weight matrix W is proportional to I_n and the matrix $\partial\eta/\partial\beta^T = X$ is constant. It follows that leverages h_{jj} are simply the diagonal elements of $X(X^T X)^{-1} X^T$. If we set $z_j = (y_j - \hat{\eta}_j)/\hat{\tau}$, it is easy to check that the j th deviance residual is $\text{sign}(y_j - \hat{\eta}_j)[2\{z_j + \exp(-z_j) - 1\}]^{1/2}$. The r_j^* for the fitted model are shown in the right panel of Figure 10.3. They are close to standard normal, and cast no doubt on the adequacy of the model. ■

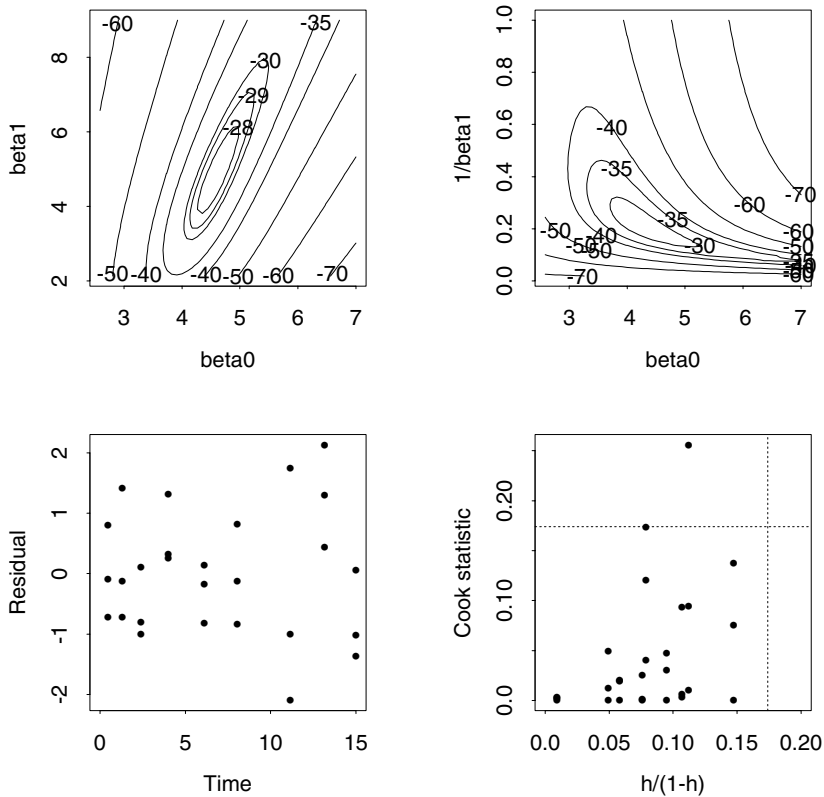


Figure 10.4 Fit of a nonlinear model to the calcium data. Upper left: contours for $\ell_p(\beta_0, \beta_1)$. Upper right: contours for $\ell_p(\beta_0, \gamma_1)$, where $\gamma_1 = 1/\beta_1$. Lower left: standardized residuals plotted against time. Lower right: plot of Cook statistics against $h/(1-h)$, where h is leverage.

The deviance can be used to check the fit of some types of models, but in moderate samples its distribution can be far from χ^2 and plots of deviances from simulated data can be useful.

Example 10.9 (Calcium data) The model for the calcium data of Example 10.1 sets $\eta_j(\beta) = \beta_0\{1 - \exp(-x_j/\beta_1)\}$, so $X = \partial\eta/\partial\beta^T$ has j th row

$$(\partial\eta_j/\partial\beta_0, \partial\eta_j/\partial\beta_1) = (1 - \exp(-x_j/\beta_1), -x_j\beta_0 \exp(-x_j/\beta_1)/\beta_1^2).$$

As the initial slope of η as a function of x is β_0/β_1 , and as η has asymptote β_0 for large x , we expect roughly that $\beta_0 \doteq 5$, and that $\beta_0/\beta_1 \doteq 1$. This suggests taking $\beta_0 = 5$ and $\beta_1 = 5$ as initial values for the iterative weighted least squares algorithm, which then converges rapidly to $\hat{\beta}_0 = 4.31$ and $\hat{\beta}_1 = 4.80$, with standard errors 0.303 and 0.905. The off-diagonal element of the inverse expected information matrix is 0.237, corresponding to $\text{corr}(\hat{\beta}_0, \hat{\beta}_1) \doteq 0.45$. The residual sum of squares is $SS(\hat{\beta}) = 7.465$, and the estimate of σ^2 is $s^2 = 7.465/(27 - 2) = 0.299$.

Large-sample likelihood theory suggests that $\hat{\beta}_0$ and $\hat{\beta}_1$ have approximately a bivariate normal distribution. This rests on quadratic approximation to the log likelihood, which seems reasonable from the upper left panel of Figure 10.4. The upper right

panel contains contours of $\ell_p(\beta_0, \gamma_1) = \max_{\sigma^2} \ell(\beta_0, \gamma_1, \sigma^2)$, where $\gamma_1 = 1/\beta_0$, for which quadratic approximation would be poor.

The standardized Pearson residuals in the lower left panel suggests possible poor fit at the three longest times, although the normal scores plot is good. The lower right panel shows that one of the approximate Cook statistics is somewhat large, but due to an outlier rather than a high leverage point.

To check model fit more formally, we allow unconnected means at each time, giving nine parameters and residual sum of squares 4.797 on 18 degrees of freedom. The F statistic that compares this and the fitted nonlinear model is equal to $\{(7.645 - 4.797)/7\}/(4.797/18) = 1.53$, and as the 0.95 quantile of the $F_{7,18}$ distribution is 2.58, there is no evidence of poor fit overall. Under the unconnected model the sums of squares between the three observations at each time have distribution $\sigma^2 \chi_2^2$, so a chi-squared probability plot of the ordered sums of squares should be straight. This plot suggests difficulties at $x = 11.15$ minutes, as was evident from the original data. ■

Exercises 10.2

- 1 Show that the scaled deviance contribution for a binomial response with probability density $\binom{m}{r} \pi^r (1 - \pi)^{m-r}$, $0 < \pi < 1$, $r = 0, \dots, m$, and $\eta = \pi$ is

$$2\{r \log\{r/(m\hat{\pi})\} + (m - r) \log[(m - r)/\{m(1 - \hat{\pi})\}]\}.$$

- 2 Show that the scaled deviance contribution for a Poisson response with density $\eta^y e^{-\eta}/y!$, $\eta > 0$, $y = 0, 1, \dots$, is $2\{y \log(y/\hat{\eta}) - y + \hat{\eta}\}$.
- 3 Consider a linear model with non-normal errors, in which the j th response is $y_j = \eta_j + \tau \varepsilon_j$, where $\eta_j = x_j^T \beta$, and ε_j has density $\exp u(\varepsilon)$, for $j = 1, \dots, n$.
 - (a) Show that the log likelihood contribution from y_j may be written as $u(z_j) - \log \tau$, and hence express in terms of $u(\cdot)$ and τ the quantities needed to obtain the maximum likelihood estimates of β by iterative weighted least squares.
 - (b) Let $x_j^T = (1, z_j^T)$, so that the covariate matrix equals $X = (1_n, Z)$ with $1_n^T Z = 0$. Show that the expected information matrix may be written

$$\tau^{-2} \begin{pmatrix} Z^T Z & 0 \\ 0 & nA \end{pmatrix},$$

where A is a 2×2 matrix that does not depend on the parameters. Show further that A is diagonal if the density of ε_j is symmetric about zero.

(c) Give the matrix A for the t density (3.11) and for the Gumbel density (10.10).

- 4 Suppose that n years of daily data are available, for each of which the r largest observations are known. Find the score and observed information when the model (6.37) is fitted to these, with $\eta = x_j^T \beta$ but constant τ and ξ . Hence write down the steps needed to apply weighted least squares to estimation of β , when τ and ξ are known. How would you estimate τ and ξ ?
- 5 Show that for a normal linear model in which ϕ is replaced by $\hat{\phi} = SS(\hat{\beta})/(n - p)$, the standardized deviance and Pearson residuals both equal the usual standardized residual r_j , and hence verify that (10.13) reduces to (8.30).
- 6 Verify the contents of the matrices X and W in Example 10.7.
- 7 In a nonlinear normal regression model, suppose that $\eta = \beta_0(1 - e^{-\beta_1 x})$. Let $SS(\beta_1)$ be the residual sum of squares when β_1 is known, that is, when the single covariate $1 - e^{-\beta_1 x}$

is fitted. Show that the profile log likelihood for β_1 can be written as

$$\ell_p(\beta_1) = \max_{\beta_0, \sigma^2} \ell(\beta_0, \beta_1, \sigma^2) \equiv -\frac{n}{2} \log s^2(\beta_1),$$

and give the form of a $(1 - 2\alpha)$ confidence interval for β_1 based on ℓ_p .

10.3 Generalized Linear Models

10.3.1 Density and link functions

Linear models play a central role in regression. In their simplest form they apply when the response variable is continuous, takes values on the real line, and has constant variance. Transformations and weighting broaden their applicability but can give fits that are awkward to interpret, as well as being unsatisfactory with discrete responses. In this section we describe how the key features of the linear model may be extended to situations where the response comes from any of a wide class of distributions. Such models are widely used in practice and the main ideas form the basis for much further development, in which the iterative weighted least squares algorithm plays an important role.

Three aspects of the normal linear model for a continuous response y are:

- the *linear predictor* $\eta = x^T \beta$ through which $\mu = E(y)$ depends on the $p \times 1$ vectors x of explanatory variables and β of parameters;
- the density of the response y , which is normal with mean μ and variance σ^2 ; and
- the fact that the mean response equals the linear predictor, $\mu = \eta$.

In a generalized linear model the second and third of these are extended to:

- the response y has density

$$f(y; \theta, \phi) = \exp \left\{ \frac{y\theta - b(\theta)}{\phi} + c(y; \phi) \right\}, \quad (10.14)$$

where θ depends on the linear predictor, and the *dispersion parameter* ϕ is often known; and

- the linear predictor and the mean of y are related by a monotone *link function* g , with

$$\eta = g(\mu). \quad (10.15)$$

If ϕ is known then (10.14) is a linear exponential family with natural parameter θ/ϕ . Thus it contains old friends such as the normal, gamma, binomial, and Poisson densities, but it includes also more casual acquaintances such as the inverse Gaussian and negative binomial models. This broadens the applicability of linear model ideas to data where the responses are positive, counts or proportions, without the need for transformations.

If Y has density (10.14), its moment-generating function is (Exercise 10.3.3)

$$M(t) = \exp\{b(\theta + t\phi) - b(\theta)\},$$

so the response has mean and variance

' denotes differentiation with respect to θ .

$$E(Y) = b'(\theta) = \mu, \quad \text{var}(Y) = \phi b''(\theta) = \phi b''\{b'^{-1}(\mu)\} = \phi V(\mu), \quad (10.16)$$

say, provided that a function inverse to $b'(\theta)$ exists. We met the variance function $V(\mu)$ in Section 5.2.1.

Example 10.10 (Poisson density) The Poisson density may be written as

$$f(y; \mu) = \exp(y \log \mu - \mu - \log y!), \quad y = 0, 1, \dots, \quad \mu > 0,$$

which has form (10.14) with $\theta = \log \mu$, $b(\theta) = e^\theta$, $\phi = 1$, and $c(y; \phi) = -\log y!$. The mean of y is $\mu = b'(\theta) = e^\theta = \mu$, and its variance is $b''(\theta) = e^\theta = \mu$, so the variance function is linear: $V(\mu) = \mu$; see Example 5.9. ■

Example 10.11 (Normal density) The normal density with mean μ and variance σ^2 may be written

$$f(y; \mu, \sigma^2) = \exp \left\{ -\frac{(y^2 - 2y\mu + \mu^2)}{2\sigma^2} - \frac{1}{2} \log(2\pi\sigma^2) \right\},$$

so

$$\theta = \mu, \quad \phi = \sigma^2, \quad b(\theta) = \frac{1}{2}\theta^2, \quad c(y; \phi) = -\frac{1}{2\phi}y^2 - \frac{1}{2} \log(2\pi\phi).$$

As the first and second derivatives of $b(\theta)$ are θ and 1, we have $V(\mu) = 1$; the variance function is constant. ■

Example 10.12 (Binomial density) We write the binomial density

$$f(r; \pi) = \binom{m}{r} \pi^r (1 - \pi)^{m-r}, \quad 0 < \pi < 1, \quad r = 0, \dots, m,$$

in the form

$$\exp \left[m \left\{ \frac{r}{m} \log \left(\frac{\pi}{1 - \pi} \right) + \log(1 - \pi) \right\} + \log \binom{m}{r} \right],$$

so

$$y = \frac{r}{m}, \quad \phi = \frac{1}{m}, \quad \theta = \log \left(\frac{\pi}{1 - \pi} \right), \quad b(\theta) = \log(1 + e^\theta), \quad c(y; \phi) = \log \binom{m}{r}.$$

The mean and variance of y are

$$\mu = b'(\theta) = \frac{e^\theta}{1 + e^\theta}, \quad \phi b''(\theta) = \frac{e^\theta}{m(1 + e^\theta)^2};$$

the variance function is $V(\mu) = \mu(1 - \mu)$. ■

By allowing nonlinear relations between the mean response and the covariates, the link function (10.15) permits the relationship between the linear part of the model and

the mean response to be chosen on subject-matter or statistical grounds. Restrictions on μ can be imposed through the choice of g , which maps the domain of μ to the set inhabited by η , usually the real line. One particular choice is the canonical link, which is obtained when $\eta = \theta = b'^{-1}(\mu)$. When ϕ is known and the canonical link is used, the model is a natural exponential family and there is a p -dimensional minimal sufficient statistic for β . This is attractive from the vantage of statistical theory, but substantive considerations are more important.

Example 10.13 (Poisson link functions) The Poisson mean is positive, so the most common link function is the log, for which $\log \mu = \eta$. As $\theta = \log \mu$, this is the canonical link.

Suppose that data n_1 and n_2 are gathered from two independent Poisson processes, with means $t_1\gamma_1$ and $t_2\gamma_2$, where t_1 and t_2 are known, but that the aggregated count $y = n_1 + n_2$ only is known. Then $E(y) = t_1\gamma_1 + t_2\gamma_2$ and the identity link function would be appropriate. ■

Example 10.14 (Normal link functions) The usual link function for the normal density is the identity, yielding the normal linear model. Suppose instead that $y = \mu + \varepsilon$, but

$$\mu = \alpha_0 \frac{x}{\alpha_1 + x}.$$

On rewriting μ^{-1} as $\eta = \beta_0 + \beta_1 z$, where $\beta_0 = \alpha_0^{-1}$, $\beta_1 = \alpha_1/\alpha_0$, and $z = x^{-1}$, we see that this model fits into our general setup with normal distribution for the response and the inverse link function, $\eta = \mu^{-1}$. ■

10.3.2 Estimation and inference

The log likelihood of independent responses $y_1, \dots, y_n$ from density (10.14) is

$$\ell(\beta) = \sum_{j=1}^n \left\{ \frac{y_j \theta_j - b(\theta_j)}{\phi_j} + c(y_j; \phi_j) \right\}, \quad (10.17)$$

where $\theta_j = \theta(\eta_j)$ with $\eta_j = x_j^\top \beta$. Maximum likelihood estimates $\hat{\beta}$ are obtained by solving the score equation (10.4) by iterative weighted least squares. Some differentiation shows that

$$\frac{\partial \ell(\beta)}{\partial \beta} = \frac{\partial \eta^\top}{\partial \beta} \frac{\partial \theta}{\partial \eta^\top} \frac{\partial \ell}{\partial \theta^\top} = X^\top u(\beta), \quad (10.18)$$

where the design matrix $\partial \eta / \partial \beta^\top = X$ does not depend on β . The components of the score statistic $u(\beta)$ and the weight matrix $W(\beta)$ may be expressed in terms of components μ_j of the mean vector μ as

$$\begin{aligned} u_j &= \frac{\partial \theta_j}{\partial \eta_j} \frac{\partial \ell_j(\theta_j)}{\partial \theta_j} = \frac{y_j - \mu_j}{g'(\mu_j) \phi_j V(\mu_j)}, \\ w_j &= \left(\frac{\partial \theta_j}{\partial \eta_j} \right)^2 \frac{\partial^2 \ell_j(\theta_j)}{\partial \theta_j^2} = \frac{1}{g'(\mu_j)^2 \phi_j V(\mu_j)}, \end{aligned} \quad (10.19)$$

where $g'(\mu_j) = dg(\mu_j)/d\mu_j$. Thus $\hat{\beta}$ is obtained by iterative weighted least squares regression of response $z = X\beta + g'(\mu)(y - \mu)$ on the columns of X using weights (10.19). By using y as an initial value for μ and $g(y)$ as an initial value for $\eta = X\beta$, we avoid needing an initial value for β .

Most generalized linear models in which the ϕ_j are unknown have $\phi_j = \phi a_j$, where the a_j are known and only ϕ must be estimated; this is analogous to weighted least squares, with a_j^{-1} playing the role of the weight attached to y_j . When ϕ is unknown, the scaled deviance is replaced by the *deviance*, defined as ϕ times the scaled deviance.

Under the usual regularity conditions, $\hat{\beta}$ has a large-sample normal distribution with mean β and variance matrix $X^T W X)^{-1}$.

The maximum likelihood estimator of ϕ can behave poorly, but another estimator is suggested by noting that if the regression parameters β were known, an unbiased estimator of $\phi = \text{var}(y_j)/\{a_j V(\mu_j)\}$ would be

$$\frac{1}{n} \sum_{j=1}^n \frac{(y_j - \mu_j)^2}{a_j V(\mu_j)}.$$

This motivates the use of

$$\hat{\phi} = \frac{1}{n-p} \sum_{j=1}^n \frac{(y_j - \hat{\mu}_j)^2}{a_j V(\hat{\mu}_j)}, \quad (10.20)$$

where the divisor $n - p$ allows for estimation of β , analogous to the unbiased estimate of variance in a normal linear model. When ϕ is known, fit can be measured using *Pearson's statistic*,

$$P = \frac{1}{\phi} \sum_{j=1}^n \frac{(y_j - \hat{\mu}_j)^2}{V(\hat{\mu}_j)/a_j}, \quad (10.21)$$

analogous to the scaled residual sum of squares in a normal model.

When the dispersion parameter is known, overall tests of fit are provided by Pearson's statistic and the scaled deviance, but their distributions depend on the situation. For gamma or binomial data with small dispersion, that is small ϕ , the distribution of D or P when the model fits is roughly χ^2_{n-p} . This corresponds to large v for gamma data and to large m for binomial data. For Poisson data χ^2 approximations are useful unless all the fitted means are small, $\hat{\mu} < 5$, say. Empirical evidence suggests that although such approximations are better for P than for D , they are poor if the data are sparse. The problem is most acute for the deviance of binary data, for which the large-sample approximation is useless (Exercise 10.4.1).

Example 10.15 (Jacamar data) For the data of Example 10.2, we treat the number of butterflies of species s painted the c th colour and eaten, r_{cs} , as binomial with denominator m_{cs} and probability

$$\pi_{cs} = \frac{\exp(\alpha_c + \gamma_s)}{1 + \exp(\alpha_c + \gamma_s)}, \quad c = 1, \dots, 8, s = 1, \dots, 6.$$

Terms	df	Deviance
1	43	134.24
1+Species	38	114.59
1+Colour	36	108.46
1+Species+Colour	31	67.28

Table 10.3 Deviances and analysis of deviance for models fitted to jacamar data. The lower part of the analysis of deviance table shows results for the reduced data, without two outliers.

Terms	df	Deviance reduction	Terms	df	Deviance reduction
Species (unadj. for Colour)	5	19.64	Species (adj. for Colour)	5	41.18
Colour (adj. for Species)	7	47.31	Colour (unadj. for Species)	7	25.78
Species (unadj. for Colour)	4	27.63	Species (adj. for Colour)	4	35.18
Colour (adj. for Species)	7	18.03	Colour (unadj. for Species)	7	10.48

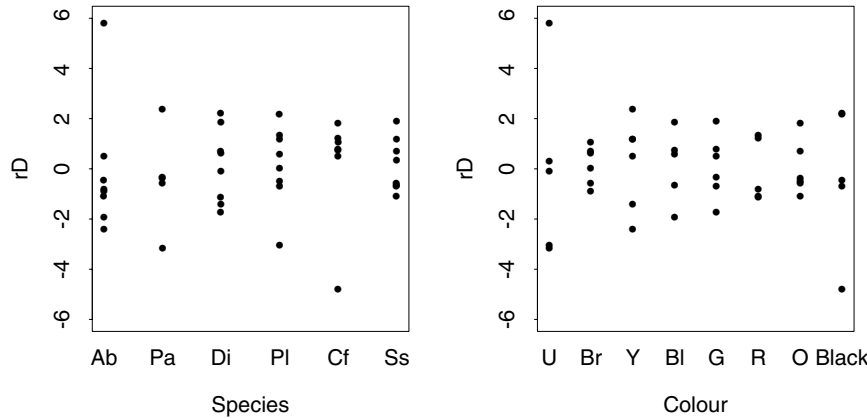


Figure 10.5 Standardized deviance residuals r_D for binomial two-way layout fitted to jacamar data.

From Example 10.12 we see that this is equivalent to a generalized linear model with binomial errors and response $y_{cs} = r_{cs}/m_{cs}$ whose mean $\mu_{cs} = \pi_{cs}$ is related to the linear predictor $\eta_{cs} = \alpha_c + \gamma_s$ by the *logit* link function

$$\eta = \log \left(\frac{\pi}{1 - \pi} \right).$$

Colour and species effects are represented by the α_c and γ_s respectively.

Table 10.3 contains the analysis of deviance. As there are four cells with zero counts there are 44 degrees of freedom in total. The reductions in deviance depend on the order of fitting, but are highly significant compared to the appropriate chi-squared distributions. The residual deviance, 67.28, is large compared to the χ^2_{31} distribution and suggests poor fit.

Figure 10.5 shows at least two outliers, corresponding to the first cell in the top row and the penultimate cell in the last row of Table 10.2. Deletion of the second of

Table 10.4 Estimated parameters and standard errors for the reduced jacamar data.

<i>Aphrissa boisduvalli</i>	<i>Phoebis argante</i>	<i>Dryas iulia</i>	<i>Pierella luna</i>	<i>Consul fabius</i>	<i>Siproeta stelenes</i>
−1.99 (0.79)	−2.22 (0.85)	−0.56 (0.67)	0.16 (0.54)	—	1.50 (0.78)

Brown	Yellow	Blue	Green	Red	Orange	Black
0.16 (0.73)	0.33 (0.68)	−0.53 (0.81)	−0.83 (0.75)	−1.93 (0.88)	−1.94 (0.85)	−1.26 (0.86)

Table 10.5 Times in minutes taken by four chimpanzees to learn ten words (Brown and Hollander, 1977, p. 257).

Chimpanzee	Word									
	1	2	3	4	5	6	7	8	9	10
1	178	60	177	36	225	345	40	2	287	14
2	78	14	80	15	10	115	10	12	129	80
3	99	18	20	25	15	54	25	10	476	55
4	297	20	195	18	24	420	40	15	372	190

Incidentally when only the last cell in this column was dropped, neither of the two packages used signalled a failure of $\hat{\gamma}_s$ to converge.

these necessitates dropping its entire column, because all the remaining *Consul fabius* butterflies were eaten and the corresponding $\hat{\gamma}_s$ is infinite. In this context a case is a single butterfly, so deletion of a cell involves deleting a number of cases.

The lower part of Table 10.3 shows the analysis of deviance for the reduced data. The overall deviance drops from 73.68 on 35 degrees of freedom to 28.02 on 24 degrees of freedom. The significance of the colours depends on whether or not species is fitted first. Overall significance for each term is assessed by its significance after adjusting for the other. Thus Colour is significant at about the 0.01 level, when treated as χ^2_7 . The residual deviance is not large compared to the χ^2_{24} distribution, and casts no doubt on the model.

Table 10.4 shows the estimates and standard errors. The odds that an unpainted *Aphrissa boisduvalli* is eaten are $e^{-1.99} = 0.14$ and the corresponding probability is $e^{-1.99}/(1 + e^{-1.99}) = 0.12$. Painting the underside of its wings green multiplies the odds of its being eaten by $e^{-0.83} = 0.44$. The most marked reductions are when red, orange, or black paint is used; locally these colours are associated with unpalatable butterflies.

Pearson's statistic is 25.58 on 24 degrees of freedom. The standardized deviance residuals lie between −2.03 and 1.96, and these and other case diagnostics show that the model fits the reduced data well. ■

Example 10.16 (Chimpanzee learning data) Table 10.5 gives times in minutes taken by four chimpanzees to learn each of ten words. The data are a two-way layout, but the responses are positive and vary over two orders of magnitude, suggesting that a linear model is inappropriate. When a linear model with mean $\alpha_c + \gamma_w$ is fitted,

Term	df	Deviance reduction	Term	df	Deviance reduction
Chimp (unadj. for Word)	3	6.95	Chimp (adj. for Word)	3	6.22
Word (adj. for Chimp)	9	38.46	Word (unadj. for Chimp)	9	39.19

Table 10.6 Analysis of deviance for models fitted to chimpanzee data.

where α_c and γ_w correspond to chimpanzee and words effect, the F statistic (8.27) for non-additivity strongly indicates a change of scale.

We fit a model with gamma errors and the log link function, in which the mean time taken by the c th chimpanzee to learn the w th word is μ_{cw} , where $\log \mu_{cw} = \eta_{cw} = \alpha_c + \gamma_w$. If the gamma density with mean μ and shape parameter ν is written in form

$$f(y; \mu, \nu) = \frac{1}{\Gamma(\nu)} y^{\nu-1} \left(\frac{\nu}{\mu}\right)^{\nu} \exp(-\nu y/\mu), \quad y > 0, \quad \nu, \mu > 0, \quad (10.22)$$

we see (Exercise 10.3.4) that the dispersion parameter $\phi = 1/\nu$, which must be estimated.

The deviances for models 1, 1+Chimp, 1+Word, and 1+Chimp+Word are 60.38, 53.43, 21.19, and 14.97, with 39, 36, 30, and 27 degrees of freedom. Table 10.6 shows the analysis of deviance. The two-way layout is balanced and the order of fitting matters less than in Example 10.15. Use of (10.20) gives $\hat{\phi} = 0.432$, and $\hat{\nu} = \hat{\phi}^{-1} = 2.31$. The significance of the deviance reductions for chimps and words is gauged by F tests using $\hat{\phi}$ as the denominator. For example, we test for differences between chimps by comparing $(6.22/3)/0.432 = 4.78$ with the $F_{3,27}$ distribution, giving significance level about 0.01.

An alternative is to fit a normal two-way layout to the log data, but the residuals suggest that the gamma model is preferable. The largest residual is for the first chimpanzee and fifth word, which is very large compared to the other times for that word. This is also the most influential value for the gamma model, but we shall not pursue this.

A different approach uses gamma errors and the inverse link $\eta_{cw} = 1/\mu_{cw}$, giving a linear model for the speed with which a word is learnt. The residual deviance for this model is 17.08. To test whether this link is suitable we extend the nonadditivity test outlined in Example 8.24 by adding the constructed variable $\hat{\eta}^2$ to the linear predictor, where $\hat{\eta}$ is the fitted linear predictor for the model with inverse link. The deviance drops by 1.26 on one degree of freedom, and since $\hat{\phi} = 0.47$ for the extended model, the test statistic is $(1.26/1)/0.47 = 2.68$, to be compared to the $F_{1,26}$ distribution. The significance level, 0.11, gives only weak evidence against the inverse link. ■

Exercises 10.3

- 1 Suppose that y is the number of events in a Poisson process of rate λ observed for a period of length T . Show that y has a generalized linear model density and give θ , $b(\theta)$, ϕ and $c(y; \phi)$.
- 2 Use the identities $E(\partial \ell / \partial \theta) = 0$ and $\text{var}(\partial \ell / \partial \theta) = E(-\partial^2 \ell / \partial \theta^2)$ to derive the mean and variance of the density (10.14).

- 3 Check that the moment-generating function that corresponds to (10.14) is $\exp\{b(\theta + t\phi) - b(\theta)\}$, give the corresponding cumulant-generating function, and hence verify the mean and variance in (10.16).
- 4 Show that the gamma density (10.22) can be put in form (10.14) with canonical parameter $\theta = -\mu^{-1}$, $b(\theta) = -\log(-\theta)$ and dispersion parameter $\phi = 1/\nu$. Give the canonical link function, and use $b(\theta)$ to show that (10.22) has mean μ , variance function $V(\mu) = \mu^2$, and variance μ^2/ν .
- 5 Verify that the inverse Gaussian density

$$f(y; \lambda, \mu) = \left(\frac{\lambda}{2\pi y^3} \right)^{1/2} \exp \left\{ -\frac{\lambda(y - \mu)^2}{2\mu^2 y} \right\}, \quad y > 0, \quad \lambda > 0, \mu > 0,$$

can be written in form (10.14) by giving θ , $b(\theta)$, ϕ , and $c(y; \phi)$, and show that its variance function is $V(\mu) = \mu^3$.

- 6 In (10.17), suppose that $\phi_j = \phi a_j$, where the a_j are known constants, and that ϕ is functionally independent of β . Show that the likelihood equations for β are independent of ϕ , and deduce that the profile log likelihood for ϕ is

$$\ell_p(\phi) = \phi^{-1} \sum_{j=1}^n \left\{ \frac{y_j \hat{\theta}_j - b(\hat{\theta}_j)}{a_j} + c(y_j; \phi a_j) \right\}.$$

Hence show that for gamma data the maximum likelihood estimate of ν solves the equation $\log \nu - \psi(\nu) = n^{-1} \sum (z_j - \log z_j - 1)$, where $z_j = y_j / \hat{\mu}_j$ and $\psi(\nu)$ is the digamma function $d \log \Gamma(\nu) / d\nu$.

- 7 Suppose that the canonical link is used with the log likelihood (10.17), so that $\theta_j = \eta_j$, and that it is required to check this link. Let $\theta_j = \theta(\eta_j)$, where $\theta(\cdot)$ is a potentially nonlinear function. By quadratic Taylor series expansion of $\theta(\cdot)$ about a suitable η^0 , show that provided η contains an intercept term, we can write $\theta(\eta) \doteq \eta' + \delta \eta^2$, where δ is proportional to $b''(\eta^0)$ and η' is a linear function of η . Hence verify that when the fit of a model with the canonical link has given linear predictor $\hat{\eta}$, a constructed variable test of the canonical link is based on the change in fit when $\hat{\eta}^2$ is added to the linear predictor. Discuss Example 8.24 in this light.
- 8 Show that in any generalized linear model, the Fisher information $E(-\partial^2 l / \partial \beta, \partial \phi)$ for the dispersion parameter ϕ and any regression parameter β , is zero; ϕ and β are said to be *orthogonal*. What is the implication for the maximum likelihood estimates $\hat{\phi}$ and $\hat{\beta}$? Prove also the stronger result that, whatever the value of ϕ , the joint observed information $-\sum \partial^2 \log f(y_j; \beta, \phi) / \partial \phi \partial \beta$ is zero, evaluated at $\hat{\beta}$.

10.4 Proportion Data

10.4.1 Binary data

A binary response Y takes values 1 and 0 with probabilities π and $1 - \pi$, denoting a dichotomous outcome such as success/failure, won/lost, or well/ill. Such data are common in applications. The simplest relation between $E(Y) = \pi$ and a linear predictor is $\pi = x^T \beta$. This is unsuitable for general use because π may then lie outside the unit interval. It is usually better to force $0 < \pi < 1$ by taking it to be a nonlinear monotone function of $x^T \beta$, whose inverse is the link function of the corresponding generalized linear model.

One way to derive link functions for dichotomous variables is to suppose that Y is a binary version of an underlying continuous response Z . Let $Z = x^T \gamma + \sigma \varepsilon$, and

Distribution		Link function	
Logistic	$e^u/(1 + e^u)$	Logit	$\eta = \log\{d\pi/(1 - \pi)\}$
Normal	$\Phi(u)$	Probit	$\eta = \Phi^{-1}(\pi)$
Log Weibull	$1 - \exp(-\exp(u))$	Log-log	$\eta = -\log\{-\log(\pi)\}$
Gumbel	$\exp\{-\exp(-u)\}$	Complementary log-log	$\eta = \log\{-\log(1 - \pi)\}$

Table 10.7 Tolerance distributions and corresponding link functions for binary data.

suppose that ε has continuous distribution function F . The mean of Z increases with $x^T\gamma$, and $Y = 1$ if $Z > 0$, with probability

$$\pi = \Pr(Y = 1) = 1 - F(-x^T\gamma/\sigma) = 1 - F(-x^T\beta),$$

say. The ratio $\beta = \gamma/\sigma$ is estimable from the binary data, but γ and σ are not. If F is symmetric about zero, then π equals $F(x^T\beta)$ and the corresponding link function (10.15) is $\eta = x^T\beta = F^{-1}(\pi)$. Some standard choices of the so-called *tolerance distribution* F and corresponding link functions are shown in Table 10.7. The logit and probit functions are symmetric and usually hard to distinguish in practice, while the log-log and complementary log-log functions are asymmetric in opposite directions. Numerous other links have been proposed, but those in the table usually suffice in applications.

Much information may be lost by splitting and it is generally better to work with the original responses if they are available. Otherwise less information is lost by taking several categories. Difficulties in the binary case are illustrated in the following example.

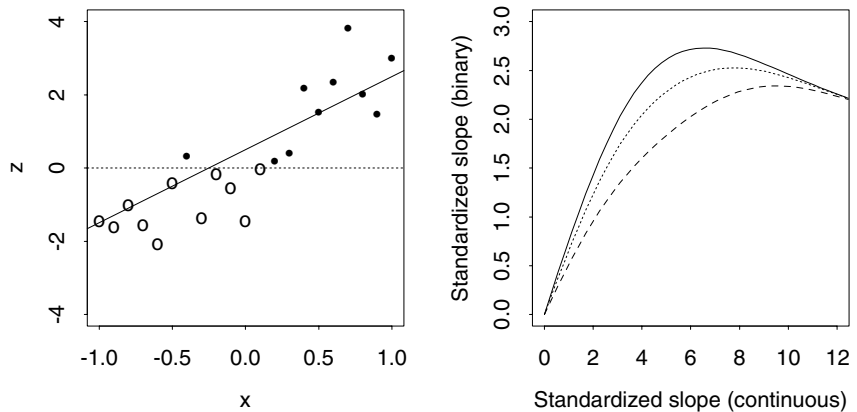
Example 10.17 (Dichotomization) Suppose independent observations $Z_j = x_j^T\beta + \varepsilon_j$ are dichotomized by setting $Y_j = 1$ if $Z_j > 0$ and $Y_j = 0$ otherwise, and let F and f denote the distribution and density of the ε_j . If the original Z_j were available, the j th log likelihood contribution would be $\log f(z_j - x_j^T\beta)$ and the expected information matrix would be (10.7), with X the constant matrix whose j th row is x_j^T and $W = kI_n$, where $k = -\int d^2 \log f(\varepsilon)/d\varepsilon^2 f(\varepsilon) d\varepsilon$. Thus if the Z_j are available the asymptotic covariance matrix of the maximum likelihood estimator $\hat{\beta}_Z$ is $k^{-1}(X^T X)^{-1}$.

Suppose now that only the binary variables $Y_1, \dots, Y_n$ are known. As Y_j has success probability $\pi_j = 1 - F(-\eta_j)$, where $\eta_j = x_j^T\beta$, its log likelihood contribution is $\ell_j = Y_j \log \pi_j + (1 - Y_j) \log(1 - \pi_j)$, and the Fisher information matrix is (10.7) with the same X as before but with W the diagonal matrix whose j th element is $E(-d^2 \ell_j / d\eta_j^2) = f(-\eta_j)^2 / [F(-\eta_j)\{1 - F(-\eta_j)\}]$. The asymptotic variance of the maximum likelihood estimator $\hat{\beta}_Y$ based on $Y_1, \dots, Y_n$ is thus $(X^T W X)^{-1}$.

The efficiency of large-sample inferences based on Z and Y may be compared through the asymptotic variance matrices $k^{-1}(X^T X)^{-1}$ and $(X^T W X)^{-1}$ of the corresponding maximum likelihood estimators. Rather than attempt a general discussion, we illustrate this numerically. Let $\eta_j = \beta_0 + \beta_1 x_j$, with x_j taking $n = 21$ values equally spaced from -1 to 1 . The left panel of Figure 10.6 shows data simulated

The asymptotics here arise if we imagine m replicate observations at each x_j , and let $m \rightarrow \infty$.

Figure 10.6 Efficiency loss due to reducing continuous variables to binary ones. Left panel: simulated data. Blobs above the dotted line are counted as successes, with zeros below it as failures; the solid line is $0.5 + 2x$. Right panel: Comparison of asymptotic t statistics when continuous data are dichotomized, for normal error distribution, when $\beta_0 = 0.5, 1, 1.5$ (solid, dots, dashes).



from this model, with $\beta_0 = 0.5$, $\beta_1 = 2$, and standard normal errors. The right panel plots $\beta_1/v_Y^{1/2}$ against $\beta_1/v_Z^{1/2}$, where v_Y and v_Z are the large-sample variances of the maximum likelihood estimates of β_1 based on the Y s and the Z s, for three values of β_0 . The quantity $\beta_1/v_Z^{1/2}$ is the limiting value of the t statistic for testing whether $\beta_1 = 0$, based on the full data, while $\beta_1/v_Y^{1/2}$ is the corresponding quantity for the binary data. The ratio v_Z/v_Y is the asymptotic efficiency for estimating β_1 from the binary data, relative to the original data, and in the graph this has largest value of about $(3/4)^2 \doteq 0.56$ when $\beta_1 = 0$, decreasing to about $(2/12)^2 \doteq 0.03$. For the data in the left panel, the t -statistic is $\hat{\beta}_1/v_Z^{1/2} = 2.39/0.36 = 6.6$, whereas the corresponding quantity for binary data is $3.15/1.20 = 2.6$, which is much weaker — though still strong — evidence of non-zero slope.

An argument analogous to that giving (7.28) shows that the power of a size α two-sided test of $\beta_1 = 0$ using the asymptotic normal distribution of $\hat{\beta}_1/v_Y^{1/2}$ is $\Phi(z_{\alpha/2} + \delta_Y) + \Phi(z_{\alpha/2} - \delta_Y)$, where $\delta_Y = \beta_1/v_Y^{1/2}$ is replaced by $\delta_Z = \beta_1/v_Z^{1/2}$ in the corresponding power from the full data. Use of the binary data can sharply reduce the power. When $\beta_1/v_Z^{1/2} \doteq 2$, for example, $\beta_1/v_Y^{1/2} \doteq 1.4$, and with $\alpha = 0.025$ the power is reduced from 0.52 to 0.29.

A peculiarity of binary regression is the decrease in $\beta_1/v_Y^{1/2}$ as $\beta_1 \rightarrow \infty$, because the information $f(-\eta)^2/[F(-\eta)\{1 - F(-\eta)\}]$ tends to zero so quickly that $v_Y^{1/2} \rightarrow \infty$ faster than $\beta_1 \rightarrow \infty$. Thus the reduced efficiency for estimating β_1 becomes extreme when $|\beta_1|$ is large; to put this another way, as $|\beta_1| \rightarrow \infty$, the power for testing for zero slope based on $\hat{\beta}_1$ tends to zero. The explanation for this is that most information in binary data is contributed by those responses whose variances are largest, for which π is not too close to zero or one, but as $\beta_1 \rightarrow \pm\infty$, the variances of all the observations tend to zero and β_1 cannot be reliably estimated.

Complete separation of successes and failures can occur. To see how, note that the estimate of β_1 from the binary data in the left panel of Figure 10.6 depends crucially on the value y_7 at $x_7 = -0.4$. If y_7 had equalled zero, then a perfect fit would have been obtained by setting $\hat{\beta}_1 = +\infty$ and choosing $\hat{\beta}_0$ so that $\hat{\pi} = 0$ for $x \leq 0.1$ and $\hat{\pi} = 1$ otherwise. This is harder to spot when there are several covariates, but a good

model-fitting routine will signal convergence problems as $|\hat{\beta}| \rightarrow \infty$. Near-complete separation will be indicated by regression diagnostics, which here suggest that the pair (y_7, x_7) is an outlier, as it has a large residual and is highly influential. ■

Logistic regression

The most common choice of function F is the logistic distribution, which gives the canonical, logit, link function. Then

$$\Pr(Y = 1) = \pi = \frac{\exp(x^\top \beta)}{1 + \exp(x^\top \beta)}, \quad \Pr(Y = 0) = 1 - \pi = \frac{1}{1 + \exp(x^\top \beta)},$$

and the resulting logistic regression model is a linear model for the logarithm of the odds of success,

$$\frac{\Pr(Y = 1)}{\Pr(Y = 0)} = \frac{\pi}{1 - \pi} = \exp(x^\top \beta).$$

The likelihood for independent binary observations $y_1, \dots, y_n$ with covariate vectors $x_1, \dots, x_n$ is

$$L(\beta) = \prod_{j=1}^n \left\{ \frac{\exp(x_j^\top \beta)}{1 + \exp(x_j^\top \beta)} \right\}^{y_j} \left\{ \frac{1}{1 + \exp(x_j^\top \beta)} \right\}^{1-y_j} = \frac{\exp(\sum_j y_j x_j^\top \beta)}{\prod_j \{1 + \exp(x_j^\top \beta)\}}. \quad (10.23)$$

This is a linear exponential family model in which $S = \sum Y_j x_j$ is minimal sufficient for β . If any of the covariate vectors are repeated, S may be written as $\sum_d x_d R_d$, where the distinct covariate vectors are labelled x_d and $R_d = \sum_j Y_{dj}$, the total number of successes for responses with covariates x_d , is a binomial variable. Apart from a constant, (10.23) is the same likelihood as would be obtained from responses aggregated by covariate vectors.

If R is binomial with denominator m , then the log odds may be estimated by the *empirical logistic transform* $\log\{(R + \frac{1}{2})/(m - R + \frac{1}{2})\}$, whose estimated variance is $(R + \frac{1}{2})^{-1} + (m - R + \frac{1}{2})^{-1}$, and this is sometimes useful for plotting.

Many model-checking procedures break down for unaggregated binary data. For a given fitted probability $\hat{\pi}$, a residual takes just two values, so comparison with a normal distribution is not useful. Moreover the deviance for a binary logistic model is a function of the data through $\hat{\beta}$ alone, and hence it provides no information about fit in any absolute sense (Exercise 10.4.1). Pearson's statistic is strongly correlated with the deviance and shares this difficulty.

Example 10.18 (Nodal involvement data) Table 10.8 summarizes data on 53 patients with prostate cancer. There are five binary explanatory variables: *age* in years (0 = less than 60, 1 = 60 or more); *stage*, a measure of the seriousness of the tumour (0 = less serious, 1 = more serious); *grade*, a measure of the pathology of the tumour (0 = less serious, 1 = more serious); *xray* (0 = less serious, 1 = more serious); and *acid*, the level of serum acid phosphatase (0 = less than 0.6, 1 = 0.6 or more). The response, nodal involvement, indicates whether the cancer has spread to neighbouring lymph nodes. The first row of the table shows that for five out of six patients

Table 10.8 Data on nodal involvement (Brown, 1980).

<i>m</i>	<i>r</i>	age	stage	grade	xray	acid
6	5	0	1	1	1	1
6	1	0	0	0	0	1
4	0	1	1	1	0	0
4	2	1	1	0	0	1
4	0	0	0	0	0	0
3	2	0	1	1	0	1
3	1	1	1	0	0	0
3	0	1	0	0	0	1
3	0	1	0	0	0	0
2	0	1	0	0	1	0
2	1	0	1	0	0	1
2	1	0	0	1	0	0
1	1	1	1	1	1	1
1	1	1	1	0	1	1
1	1	1	0	1	1	1
1	1	1	0	0	1	1
1	0	1	0	1	0	0
1	1	0	1	1	1	0
1	0	0	1	1	0	0
1	1	0	1	0	1	0
1	1	0	0	1	0	1
1	0	0	0	0	1	1
1	0	0	0	0	1	0

aged less than 60 and with high levels of the other explanatory variables, there was nodal involvement. A case is an individual patient rather than a row of the table. The explanatory variables are relatively easily collected and the aim of analysis was to predict nodal involvement from them.

Table 10.9 contains the deviances for all 2^5 combinations of explanatory variables when a binary logistic model is fitted to the data. The model with terms for *stage*, *xray*, and *acid* has deviance 19.64 on 49 degrees of freedom and the smallest AIC; it seems best overall, though it has several close competitors. The fitted linear predictor for this model is $-3.05 + 1.65I_{\text{stage}} + 1.91I_{\text{xray}} + 1.64I_{\text{acid}}$, where I_{stage} indicates that stage takes its higher level, and so forth. The fitted odds of nodal involvement when all the explanatory variables take their lower levels are a low $e^{-3.05} \doteq 0.047$, though this must be viewed with caution as there are no such cases in the data. The odds increase by a factor $e^{1.91} \doteq 6.75$ when *acid* takes its higher level, and are $e^{-3.05+1.91+1.65+1.64} \doteq 8.6$ at the higher levels of *stage*, *acid*, and *xray*.

The residual scaled deviance of 19.64 on 49 degrees of freedom suggests that the model fits well, but the binomial denominators are too small for confidence in χ^2 asymptotics. The deviance does not measure model fit for binary data: it is a function of $\hat{\beta}$ alone and hence does not contrast the data with the fitted model. If the data in Table 10.8 had been analyzed as written there, that is as 23 binomial rather than 53 binary observations, the degrees of freedom for the best-fitting model would be

age	stage	grade	xray	acid	df	Deviance	age	stage	grade	xray	acid	df	Deviance
					52	40.71	+	+	+			49	29.76
+					51	39.32	+	+		+		49	23.67
	+				51	33.01	+	+			+	49	25.54
		+			51	35.13	+		+	+		49	27.50
			+		51	31.39	+		+		+	49	26.70
				+	51	33.17	+			+	+	49	24.92
+	+				50	30.90		+	+	+		49	23.98
+		+			50	34.54		+	+		+	49	23.62
+			+		50	30.48		+		+	+	49	19.64
+				+	50	32.67			+	+	+	49	21.28
	+	+			50	31.00	+	+	+	+		48	23.12
	+		+		50	24.92	+	+	+		+	48	23.38
	+			+	50	26.37	+	+		+	+	48	19.22
		+	+		50	27.91	+		+	+	+	48	21.27
		+		+	50	26.72		+	+	+	+	48	18.22
			+	+	50	25.25	+	+	+	+	+	47	18.07

Table 10.9 Scaled deviances for 32 logistic regression models for nodal involvement data. A plus denotes a term included in the model.

23 − 4 = 19 rather than 49, but the deviance of 19.64 would be unchanged. This ambiguity is another reason not to rely on the deviance to measure fit for binary data.

Figure 10.7 illustrates difficulties with binary residuals. The left panel shows the 53 residuals for the unaggregated data. The linear predictors are slightly jittered to prevent over-plotting. The upper and lower bands, corresponding to ones and zeros respectively, are typical of data with only a few response values. The right panel shows the 23 residuals for the aggregated data; the fitted values are the same as on the left. Banding remains but is much less obvious, and the apparent outliers are gone. There is little useful information in either plot.

We reconsider these data in Example 12.18. ■

10.4.2 2 × 2 table

A very common data structure classifies individuals by two sets of binary categories. In a medical setting, for example, we may observe success or failure for patients randomly allocated to be either a case — receiving some treatment — or a control. The resulting data may be laid out as in Table 10.10. The simplest model for this regards the numbers of successes R_1 and R_0 as independent binomial variables with probabilities

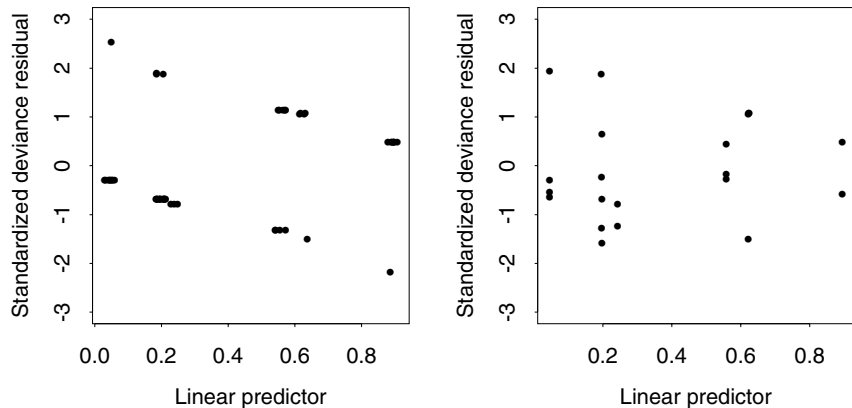
$$\pi_1 = \frac{e^{\lambda+\psi}}{1 + e^{\lambda+\psi}}, \quad \pi_0 = \frac{e^{\lambda}}{1 + e^{\lambda}}$$

and denominators m_1 and m_0 . Then the joint density of R_1 and R_0 is

$$\Pr(R_1 = r_1, R_0 = r_0; \psi, \lambda) = \binom{m_1}{r_1} \binom{m_0}{r_0} \frac{e^{(\lambda+\psi)r_1}}{(1 + e^{\lambda+\psi})^{m_1}} \frac{e^{\lambda r_0}}{(1 + e^{\lambda})^{m_0}}, \quad (10.24)$$

Table 10.10 Notation for 2×2 table.

	Success	Failure	Total
Case	R_1	$m_1 - R_1$	m_1
Control	R_0	$m_0 - R_0$	m_0
Total	$R_1 + R_0$	$m_1 + m_0 - R_1 - R_0$	$m_1 + m_0$

Figure 10.7 Standardized deviance residuals for nodal involvement data, for ungrouped responses (left) and grouped responses (right).

which is an exponential family of order two with natural parameter (ψ, λ) and natural observation $(R_1, R_1 + R_0)$ (Section 5.2.2). This is a generalized linear model with binomial errors and logit link function.

The usual purpose of analysis is to compare π_1 with π_0 . Although quantities such as the difference $\pi_1 - \pi_0$ are sometimes of interest, we focus here on the difference in log odds,

$$\psi = \log\left(\frac{\pi_1}{1 - \pi_1}\right) - \log\left(\frac{\pi_0}{1 - \pi_0}\right).$$

This is a natural parameter of the exponential family, but more importantly its interpretation does not depend on whether the data are obtained prospectively or retrospectively. To appreciate this, suppose that a prospective study is performed: an individual is allocated randomly to cases ($T = 1$) or controls ($T = 0$) and then followed until a binary outcome Y is observed. Then

$$\Pr(Y = 1 \mid T = 1) = \frac{e^{\lambda + \psi}}{1 + e^{\lambda + \psi}}, \quad \Pr(Y = 1 \mid T = 0) = \frac{e^{\lambda}}{1 + e^{\lambda}} \quad (10.25)$$

and as T is allocated and then Y observed, the scheme fixes the vertical margin in Table 10.10. The drawback is that it may be costly and difficult to follow up enough individuals to obtain a precise estimate of ψ .

In a retrospective study the treatment status T is determined only after the outcomes Y are known; the scheme fixes the horizontal margin in Table 10.10. Often the treatment undergone can be ascertained from medical records, so large samples can

be assembled more easily and cheaply than a prospective study, though the lack of randomization weakens subsequent inferences. Let $Z = 1$ indicate that an individual is chosen for the retrospective study, and suppose that this occurs with probabilities

$$\Pr(Z = 1 \mid Y = 1) = p_1, \quad \Pr(Z = 1 \mid Y = 0) = p_0,$$

independent of treatment status T . Then the success probability for an individual who was treated, conditional on their being chosen for inclusion in the study is $\Pr(Y = 1 \mid Z = 1, T = 1)$. This equals

$$\frac{\Pr(Z = 1 \mid Y = 1)\Pr(Y = 1 \mid T = 1)}{\Pr(Z = 1 \mid Y = 1)\Pr(Y = 1 \mid T = 1) + \Pr(Z = 1 \mid Y = 0)\Pr(Y = 0 \mid T = 1)}$$

by Bayes' theorem, so

$$\Pr(Y = 1 \mid Z = 1, T = 1) = \frac{p_1 e^{\lambda + \psi}}{p_1 e^{\lambda + \psi} + p_0} = \frac{e^{\lambda' + \psi}}{1 + e^{\lambda' + \psi}},$$

where $\lambda' = \lambda + \log(p_1/p_0)$. A similar argument gives

$$\Pr(Y = 1 \mid Z = 1, T = 0) = \frac{e^{\lambda'}}{1 + e^{\lambda'}},$$

so although retrospective sampling alters λ , the difference of log odds ψ is unchanged. This gives a strong motivation for using ψ to summarize the treatment effect, particularly if estimates from both types of study will ultimately be combined.

This argument applies also if ψ is replaced by $x^T \beta$, where x contains covariates as well as an indicator of treatment status. The key point is that the selection probabilities p_1 and p_0 must be independent of x .

Example 10.19 (Smoking and the Grim Reaper) Table 6.8 contains seven 2×2 tables, containing a prospective observational, that is, non-randomized, study on outcomes for women smokers and non-smokers. The simplest model ignores age by using only the overall data in the first line of Table 6.8, and gives parameter estimates for (10.25) of $\hat{\lambda} = 0.78$ (0.08) and $\hat{\psi} = 0.38$ (0.13). The significant positive value of $\hat{\psi}$ shows an unlikely preservative effect of smoking. The deviance is 632.3 on 12 degrees of freedom, however, so the model is evidently inadequate.

When different values of λ are fitted to each table, the deviance drops to 2.38 on 6 degrees of freedom, and $\hat{\psi} = -0.43$ (0.18): smoking significantly increases the death rate. There are 14 residuals, but as they arise in negatively correlated pairs and have only 6 degrees of freedom, it is better to examine one residual for each 2×2 table. They show nothing untoward. ■

Small sample analysis

The discussion above relies on large-sample likelihood results. Special techniques are needed for 2×2 tables with small counts. As (10.24) is an exponential family, the

nuisance parameter λ may be eliminated by conditioning on its associated statistic $A = R_1 + R_0$, whose density is

$$\sum_{u=r_-}^{r_+} \binom{m_1}{u} \binom{m_0}{a-u} \frac{e^{\lambda a}}{(1+e^\lambda)^{m_0}} \frac{e^{\psi u}}{(1+e^{\lambda+\psi})^{m_1}}, \quad a = 0, \dots, m_1 + m_0,$$

where $r_- = \max(0, a - m_0)$, $r_+ = \min(m_1, a)$. The conditional density of R_1 given $A = a$ is the *non-central hypergeometric density*

$$f(r | a; \psi) = \frac{\binom{m_1}{r} \binom{m_0}{a-r} e^{\psi r}}{\sum_{u=r_-}^{r_+} \binom{m_1}{u} \binom{m_0}{a-u} e^{\psi u}}, \quad r = r_-, \dots, r_+, \quad (10.26)$$

on which exact inferences for ψ may be based; this amounts to conditioning on both margins of Table 10.10. Tests of $\pi_1 = \pi_0$ compare the observed value of R_1 with its null distribution, obtained by setting $\psi = 0$ in (10.26). To test $\psi = 0$ against the one-sided alternative $\psi > 0$ we use the P-value

$$\Pr(R_1 \geq r_1 | A = a; 0) = \sum_{r=r_1}^{r_+} f(r | a; 0), \quad (10.27)$$

and take $\Pr(R_1 \leq r_1 | A = a; 0)$ when testing $\psi < 0$. Exact confidence intervals are obtained by inverting these tests, solving for ψ_α , ψ^α the equations

$$\Pr(R_1 \geq r_1 | A = a; \psi_\alpha) = \alpha, \quad \Pr(R_1 \leq r_1 | A = a; \psi^\alpha) = \alpha.$$

When the margins of the table are small, the conditional distribution of R_1 is very discrete and the difficulties seen in Example 7.38 arise: exact conditional confidence intervals are quite conservative and it is preferable to replace (10.27) by the *mid-p significance level*

$$p_{+, \text{mid}} = \frac{1}{2} \Pr(R_1 = r_1 | a; 0) + \Pr(R_1 > r_1 | a; 0). \quad (10.28)$$

When exact significance levels for testing $\psi = 0$ are unavailable, approximate ones may be obtained by treating

$$Z = \frac{R_1 - \frac{1}{2} - E(R_1 | A = a; 0)}{\text{var}(R_1 | A = a; 0)^{1/2}}$$

as standard normal, where the $\frac{1}{2}$ is a continuity correction, and

$$E(R_1 | A = a; 0) = \frac{m_1 a}{m_0 + m_1}, \quad \text{var}(R_1 | A = a; 0) = \frac{m_0 m_1 a (m_0 + m_1 - a)}{(m_0 + m_1)^2 (m_0 + m_1 - 1)}.$$

Example 10.20 (Ulcer data) In a trial to compare two treatments for stomach ulcer, 28 persons with ulcers were divided randomly into two groups, one of size $m_1 = 15$ who were given a new surgical treatment, and the other of size $m_0 = 13$ who were given an existing one; see Table 10.11, which also contains data from other trials. The numbers in these groups without an adverse outcome, recurrent bleeding, were $r_1 = 8$ and $r_0 = 2$. Does the new treatment reduce the number of adverse outcomes? Here the null hypothesis is $\psi = 0$, with alternative $\psi > 0$. The attainable significance levels for the conditional test are in Table 10.12, and $p_+ = 0.0434$ and $p_{+, \text{mid}} = 0.0243$. There is some evidence that the new treatment improves on the old.

	r_1	m_1	r_0	m_0		r_1	m_1	r_0	m_0
1	8	15	2	13	21	34	40	8	21
2	11	19	8	16	22	14	18	34	39
3	29	34	35	39	23	54	68	61	74
4	14	20	13	21	24	20	24	19	27
5	9	12	12	12	25	6	6	0	6
6	3	7	0	4	26	9	10	10	15
7	13	17	11	24	27	12	17	10	15
8	15	16	3	16	28	10	10	2	14
9	11	14	15	22	29	22	22	16	24
10	36	38	20	32	30	16	18	11	21
11	6	12	0	8	31	14	15	6	13
12	5	7	2	9	32	9	12	2	9
13	12	21	17	24	33	20	20	18	23
14	14	21	20	25	34	13	17	14	16
15	22	25	21	32	35	30	40	8	20
16	7	11	4	10	36	13	16	14	16
17	8	10	2	10	37	30	34	14	19
18	30	31	23	27	38	31	38	22	37
19	24	28	16	31	39	34	34	0	34
20	36	43	27	43	40	9	9	16	16

Table 10.11 Data from 40 independent experiments to compare a new surgery for stomach ulcer with an older surgery; data from Efron (1996) corrected from original articles. Shown are the number of persons given the new treatment, m_1 , of whom r_1 did not have recurrent bleeding, and the number given the old treatment, m_0 , of whom r_0 did not have recurrent bleeding.

r_1	0	1	2	3	4	5	6	7	8	9	10
p_+	1	1	0.999	0.989	0.929	0.751	0.456	0.184	0.043	0.005	0
$1 - \Phi(z)$	1	1	0.999	0.987	0.925	0.747	0.456	0.187	0.048	0.007	0
$p_{+,mid}$	1	1	0.994	0.959	0.840	0.604	0.320	0.114	0.024	0.003	0
$1 - \Phi(z')$	1	1	0.995	0.966	0.854	0.609	0.309	0.101	0.020	0.002	0

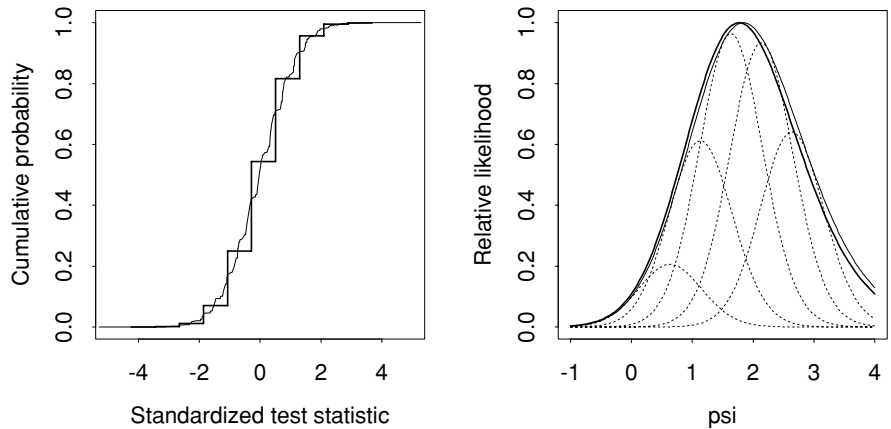
Table 10.12 Significance probabilities for a test of no treatment effect in the first 2×2 table of the ulcer data. Here $p_+ = \Pr(R_1 \geq r_1 \mid A = a; 0)$, and z' and z are the standardized forms of R_1 without and with continuity correction. Note how closely $1 - \Phi(z)$ and $1 - \Phi(z')$ match p_+ and $p_{+,mid}$ respectively.

The left panel of Figure 10.8 shows the conditional and unconditional distributions of Z ; for the unconditional distribution $\lambda = 0$. Though both are discrete, the unconditional distribution is much more nearly continuous.

The right panel shows summaries for likelihood analysis of the data. The difference between the conditional likelihood based on (10.26) and the profile likelihood for ψ is small, but we should be wary of using large-sample likelihood approximations, because the sample is rather small. The panel also shows slices through the likelihood corresponding to various values of λ ; evidently the likelihood depends strongly on both parameters. ■

The history of the 2×2 table has been dogged by controversy, partly because of the effect of the discreteness of the conditional distribution of R_1 on confidence intervals for ψ . The unconditional distribution is more nearly continuous, so it yields shorter confidence intervals and more powerful tests. Hence some authors believe that inference should be based on the unconditional rather than on the conditional distribution. The drawback is that as the unconditional distribution depends on λ it does not give exact tests and confidence intervals, whereas the conditional approach does.

Figure 10.8 Analysis for first 2×2 table of ulcer data. Left: conditional (bold) and unconditional distribution of standardized R_1 . Right: relative likelihoods based on conditional distribution of R_1 given A (heavy), profile likelihood (solid), and slices through likelihood based on R_1 and R_2 for fixed values of λ , equal to $-0.5, -1, -1.5, 2, -2.5$, from left to right (dots).



Exercises 10.4

- Data $y_1, \dots, y_n$ are assumed to follow a binary logistic model in which y_j takes value 1 with probability $\pi_j = \exp(x_j^T \beta) / \{1 + \exp(x_j^T \beta)\}$ and value 0 otherwise, for $j = 1, \dots, n$. (a) Show that the deviance for a model with fitted probabilities $\hat{\pi}_j$ can be written as

$$D = -2 \left\{ y^T X \hat{\beta} + \sum_{j=1}^n \log(1 - \hat{\pi}_j) \right\}$$

and that the likelihood equation is $X^T y = X^T \hat{\pi}$. Hence show that the deviance is a function of the $\hat{\pi}_j$ alone.

- If $\pi_1 = \dots = \pi_n = \pi$, then show that $\hat{\pi} = \bar{y}$, and verify that

$$D = -2n \{ \bar{y} \log \bar{y} + (1 - \bar{y}) \log(1 - \bar{y}) \}.$$

Comment on the implications for using D to measure the discrepancy between the data and fitted model.

- In (b), show that Pearson's statistic (10.21) is identically equal to n . Comment.

- (a) Show that the parametric link function

$$g(\pi; \gamma) = \log [\gamma^{-1} \{ (1 - \pi)^{-\gamma} - 1 \}], \quad \gamma \neq 0,$$

gives the logit and complementary log-log links when $\gamma = 1$ and when $\gamma \rightarrow 0$.

Give a similar function containing the logit and log-log link functions.

- Show that the link function

$$g(\pi; \gamma) = 2\gamma^{-1} \frac{\pi^\gamma - (1 - \pi)^\gamma}{\pi^\gamma + (1 - \pi)^\gamma}, \quad \gamma \neq 0,$$

is symmetric for all γ and gives the logit and identity functions when $\gamma \rightarrow 0$ and when $\gamma = 1$.

- If X is a Poisson variable with mean $\mu = \exp(x^T \beta)$ and Y is a binary variable indicating the event $X > 0$, find the link function between $E(Y)$ and $x^T \beta$.

10.5 Count Data

10.5.1 Log-linear models

The basic model for count data treats the response Y as a Poisson variable with mean μ . With the canonical, log, link, $\mu = \exp(x^T \beta)$; this is a log-linear model. In certain applications Y may be thought of as the number of events in a Poisson process of rate $\exp(x^T \beta)$ observed for a period T , in which case $\mu = T \exp(x^T \beta) = \exp(x^T \beta + \log T)$. This is a log-linear model with linear predictor $\eta' = x^T \beta + \log T$; the *offset* term $\log T$ is a fixed part of the linear predictor.

The connection between the Poisson and binomial distributions induces a relationship between log-linear and logistic models. Let Y_1 and Y_2 be independent Poisson variables with means μ_1 and μ_2 . Then the conditional distribution of Y_2 given that $Y_1 + Y_2 = m$ is binomial with probability $\pi = \mu_2 / (\mu_1 + \mu_2)$ and denominator m . If $\mu_1 = \exp(\gamma + x_1^T \beta)$ and $\mu_2 = \exp(\gamma + x_2^T \beta)$, then $\pi = \exp\{(x_2 - x_1)^T \beta\} / [1 + \exp\{(x_2 - x_1)^T \beta\}]$, so β may be estimated either by a log-linear model based on both observations, or by a logistic model using the conditional distribution of the second given their sum; in this second case γ cannot be estimated.

Example 10.21 (Premier League data) We consider the numbers of goals scored in the 380 soccer matches played in the English Premier League in the 2000–2001 season. The data are the home and away scores, y_{ij}^h and y_{ij}^a , when team i is at home to team j , treated as independent Poisson variables with means

$$\mu_{ij}^h = \exp(\Delta + \alpha_i - \beta_j), \quad \mu_{ij}^a = \exp(\alpha_j - \beta_i),$$

where Δ represents the home advantage and α_i and β_i the offensive and defensive strengths of team i . We expect to find $\Delta > 0$, corresponding to better performance for teams playing at home.

Table 10.13 contains the analysis of deviance for this log-linear model. There are large home and offensive effects and weaker but still very significant defensive effects. Although the residual deviance is substantially larger than its degrees of freedom, only 36 of the individual scores exceed three goals, so asymptotics based on large counts are suspect. For the same reason residual analysis is not very useful, and we assess model fit by Monte Carlo methods. Simulation from the fitted model gave 999 deviances with average value of 826, of which 748 exceeded the observed value. Thus the observed residual deviance is not unusual, suggesting that the model is broadly adequate. Under this model, $\hat{\Delta} = 0.37$ (0.07), so the mean score of a team playing at home is increased by a substantial multiplier of $\exp(\hat{\Delta}) = 1.45$. Estimates of the other parameters are given in the lower part of Table 10.13. The fitted mean scores are readily computed; for example when Manchester United is at home to Coventry the fitted means are $\exp\{0.37 + 0.22 - (-0.52)\} = 3.03$ and $\exp\{-0.53 - 0.15\} = 0.51$. In fact this match was a 4–2 win for the home team, Coventry doing better than expected but losing anyway.

A different analysis models the home score, given the total score m for each match. The paragraph preceding this example shows that if the log-linear model is correct,

Table 10.13 Log-linear and logistic models fitted to Premier League data. The upper part shows the analysis of deviance for log-linear models with parameters for home advantage, offense and defense. The lower part shows a league table based on the overall strengths estimated from the binomial model, with estimated offensive and defensive capabilities from the log-linear model. The baseline team is Arsenal, some of whose parameters are aliased. Individual standard errors are not shown, but they are within ± 0.02 of the values at the foot of the table.

Log-linear model			Logistic model		
Terms	df	Deviance reduction	Terms	df	Deviance reduction
Home	1	33.58	Home	1	33.58
Defense	19	39.21	Team	19	79.63
Offense	19	58.85			
Residual	720	801.08	Residual	332	410.65
			Overall (δ)	Offensive (α)	Defensive (β)
Manchester United			0.39	0.22	0.15
Liverpool			0.13	0.12	−0.08
Arsenal			—	0.04	—
Chelsea			−0.09	0.08	−0.22
Leeds			−0.10	0.02	−0.17
Ipswich			−0.16	−0.10	−0.13
Sunderland			−0.33	−0.31	−0.10
Aston Villa			−0.48	−0.31	−0.15
West Ham			−0.53	−0.33	−0.30
Middlesbrough			−0.53	−0.35	−0.17
Charlton			−0.55	−0.21	−0.43
Tottenham			−0.58	−0.28	−0.38
Newcastle			−0.59	−0.35	−0.30
Southampton			−0.60	−0.45	−0.25
Everton			−0.75	−0.32	−0.46
Leicester			−0.77	−0.47	−0.31
Manchester City			−0.90	−0.40	−0.56
Coventry			−0.93	−0.53	−0.52
Derby			−0.93	−0.51	−0.45
Bradford			−1.29	−0.71	−0.62
SEs			0.29	0.20	0.20

the distribution of the number of goals scored when team i plays at home to team j is binomial with denominator m and probability

$$\frac{\mu_{ij}^h}{\mu_{ij}^h + \mu_{ij}^a} = \frac{\exp(\Delta + \alpha_i - \beta_j)}{\exp(\Delta + \alpha_i - \beta_j) + \exp(\alpha_j - \beta_i)} = \frac{\exp(\Delta + \delta_i - \delta_j)}{1 + \exp(\Delta + \delta_i - \delta_j)}, \quad (10.29)$$

where $\delta_i = \alpha_i + \beta_i$ represents the overall strength of team i . Under this logistic model, no-score draws contribute no information, as the conditional distribution of Y_2 is degenerate when $m = 0$, and if there was no home advantage and no differences among the teams, the number of goals scored by the home side would be binomial with denominator m and probability $\frac{1}{2}$. This analysis will give no information on the

absolute goal-scoring abilities of the teams, merely their relative strengths. As an arbitrary constant may be added to the δ_i , they cannot all be estimated; we deal with this by declaring that $\delta = 0$ for Arsenal.

When this model is fitted to the 352 matches with at least one goal scored, we obtain the deviances in the right part of Table 10.13. There are strong differences among the teams. The home advantage remains $\hat{\Delta} = 0.37$ (0.07), and the estimated δ s are given in the lower part of the table. The broad pattern is the same as in the log-linear model, though the ordering of clubs in the middle of the league is different. However, the standard errors for the team effects are larger than those for the log-linear model, because information is lost when the logistic model is fitted; see Exercise 10.5.2.

The logistic model gives an overall ranking of the teams similar to the official ranking, though with differences of detail: Arsenal and Liverpool are interchanged, and so are Derby and Manchester City. The centre of the table has further differences, but as the standard error for each $\hat{\delta}_j - \hat{\delta}_i$ is about 0.3, it is dangerous to read much into them. One reason for the differences is that the ranking here is based on numbers of goals, while the official ranking gives 2 for a win, 1 for a draw, and 0 for a loss. ■

As the lowest three sides were relegated to the first division, their supporters might not regard this as a matter of detail!

10.5.2 Contingency tables

Count data often arise in the form of contingency tables that cross-classify individuals according to their attributes. The appropriate class of models for such a table depends on the sampling scheme. Suppose that an $R \times C$ table arises by randomly sampling a population over a fixed period and then classifying the resulting individuals. For example, a researcher interested in the association of gender (rows) and voting intentions (columns) might stand on a street corner for an hour recording data from anyone willing to talk to him. There are then no constraints on the row and column totals, and a simple model is that the count in the (r, c) cell, y_{rc} , has a Poisson distribution with mean μ_{rc} . The resulting likelihood is

This would not be a good way to proceed because of likely bias due to non-random sampling.

$$\prod_{r,c} \left\{ \frac{\mu_{rc}^{y_{rc}}}{y_{rc}!} e^{-\mu_{rc}} \right\};$$

this is simply the Poisson likelihood for the counts in the RC groups.

Our hapless researcher may set out with the intention of interviewing a fixed number m of individuals, stopping only when $\sum_{rc} y_{rc} = m$. In this case the data are multinomially distributed, with likelihood

$$\frac{m!}{\prod_{r,c} y_{rc}!} \prod_{r,c} \pi_{rc}^{y_{rc}}, \quad \sum_{r,c} \pi_{rc} = 1,$$

with $\pi_{rc} = \mu_{rc} / \sum_{s,t} \mu_{st}$ the probability of falling into the (r, c) cell.

A third scheme is to interview fixed numbers of men and of women, thus fixing the row totals $m_r = \sum_c y_{rc}$ in advance. In effect this treats the row categories as subpopulations, and the column categories as the response. This yields independent

multinomial distributions for each row, and product multinomial likelihood

$$\prod_r \left\{ \frac{m_r!}{\prod_c y_{rc}!} \prod_c \pi_{rc}^{y_{rc}} \right\}, \quad \sum_c \pi_{1c} = \cdots = \sum_c \pi_{Rc} = 1,$$

in which $\pi_{rc} = \mu_{rc} / \sum_t \mu_{rt}$. See Table 10.2, in which the response is the fate of a fixed number of butterflies for each combination of species and colour; the appropriate product multinomial model fixes the total for each triplet.

These three set-ups can all be fitted as log-linear models, provided the appropriate baseline terms are included in the linear predictor. To see this, we arrange our data as a two-way layout, with row totals fixed: the multinomial sampling scheme gives just one row, whereas we would arrange the data in Table 10.2 as a 48×3 table. Suppose that the cell counts y_{rc} are independent Poisson variables with means $\mu_{rc} = \exp(\gamma_r + x_{rc}^T \beta)$, where γ_r corresponds to the overall count in the r th row; interest focuses on the parameter β . The multinomial model has fixed row totals $\sum_c y_{rc} = m_r$ and probabilities

$$\pi_{rc} = \frac{\mu_{rc}}{\sum_d \mu_{rd}} = \frac{\exp(\gamma_r + x_{rc}^T \beta)}{\sum_d \exp(\gamma_r + x_{rd}^T \beta)} = \frac{\exp(x_{rc}^T \beta)}{\sum_d \exp(x_{rd}^T \beta)},$$

so the corresponding log likelihood is

$$\begin{aligned} \ell_{\text{Mult}}(\beta; y \mid m) &\equiv \sum_{rc} y_{rc} \log \pi_{rc} \\ &= \sum_r \left\{ \sum_c y_{rc} x_{rc}^T \beta - m_r \log \left(\sum_c e^{x_{rc}^T \beta} \right) \right\}, \end{aligned} \quad (10.30)$$

where we have emphasized the fact that the likelihood is based on the conditional distribution of the counts y given the row totals m .

For the Poisson model there is no conditioning, so the log likelihood is

$$\begin{aligned} \ell_{\text{Pois}}(\beta, \gamma) &\equiv \sum_{r,c} (y_{rc} \log \mu_{rc} - \mu_{rc}) \\ &= \sum_r \left(m_r \gamma_r + \sum_c y_{rc} x_{rc}^T \beta - e^{\gamma_r} \sum_c e^{x_{rc}^T \beta} \right). \end{aligned}$$

As the γ_r are not of central concern, we express this log likelihood as a function of the row totals $\tau_r = \sum_c \mu_{rc} = e^{\gamma_r} \sum_c e^{x_{rc}^T \beta}$ and the parameter of interest, β . In terms of β and the τ_r , we have $\gamma_r = \log \tau_r - \log \{ \sum_c \exp(x_{rc}^T \beta) \}$, giving

$$\begin{aligned} \ell_{\text{Pois}}(\beta, \tau) &\equiv \sum_r (m_r \log \tau_r - \tau_r) + \sum_r \left\{ \sum_c y_{rc} x_{rc}^T \beta - m_r \log \left(\sum_c e^{x_{rc}^T \beta} \right) \right\}, \\ &= \ell_{\text{Pois}}(\tau; m) + \ell_{\text{Mult}}(\beta; y \mid m), \end{aligned}$$

say. The first term on the right of this decomposition is the log likelihood that corresponds to the Poisson distribution of the row total m_r — a sum of independent Poisson variables — while the second is the multinomial log likelihood (10.30). Thus the m_r form a cut, and the maximum likelihood estimates of β and τ based

on $\ell_{\text{Pois}}(\beta, \tau)$ are the same as those based on separate maximizations of $\ell_{\text{Pois}}(\tau; m)$ and $\ell_{\text{Mult}}(\beta; y \mid m)$; see (5.21). Hence $\hat{\beta}$ equals the maximum likelihood estimate for the multinomial log likelihood (10.30), and $\hat{\tau}_r = m_r$. Moreover, the observed and expected information matrices for the model are block diagonal, with blocks corresponding to $-\partial^2 \ell_{\text{Pois}}(\tau; m) / \partial \tau \partial \tau^\top$ and $-\partial^2 \ell_{\text{Mult}}(\beta; y \mid m) / \partial \beta \partial \beta^\top$.

To see that the standard errors for $\hat{\beta}$ based on the multinomial and Poisson models are equal, note that $\partial^2 \ell_{\text{Pois}}(\beta, \tau) / \partial \beta \partial \beta^\top$ depends on the data only through the m_r . Therefore the expected information for β under the multinomial model, in which the m_r are fixed, equals the observed information for β under the Poisson model. Under the Poisson model, the expected information for β is

$$\sum_r E(m_r) \frac{\partial^2 \log(\sum_c e^{x_{rc}^\top \beta})}{\partial \beta \partial \beta^\top} = \sum_r \tau_r \frac{\partial^2 \log(\sum_c e^{x_{rc}^\top \beta})}{\partial \beta \partial \beta^\top},$$

and the standard errors for $\hat{\beta}$ are obtained by replacing τ_r and β with their estimates, and inverting the resulting matrix. But as $\hat{\tau}_r = m_r$, the resulting standard errors will equal those obtained by inverting the expected information matrix obtained from (10.30). It follows that the numerical values of standard errors and maximum likelihood estimates for β under the Poisson model are the same as those under the multinomial model, provided that the parameters associated with the margin fixed under the multinomial model, the γ_r , are included in the fit. The log linearity is important here, as it ensures that second derivatives of both log likelihoods with respect to β involve the counts y_{rc} only through their row totals m_r .

Example 10.22 (Jacamar data) Let y_{csf} denote the number of butterflies of the c th colour and s th species suffering the f th fate, where $c = 1, \dots, 8$, $s = 1, \dots, 6$, and $f = 1, 2, 3$. If we treat fate as the response, any model should fix the total count for each of the 48 combinations of species and colour, giving 48 trinomial variables. Any Poisson model should have a term α_{cs} in the linear predictor. For example, $\log \mu_{csf} = \alpha_{cs}$ corresponds to equal probability of each of the three fates, whatever the colour and species, because

$$(\pi_{cs1}, \pi_{cs2}, \pi_{cs3}) = \left(\frac{\mu_{cs1}}{\sum_f \mu_{csf}}, \frac{\mu_{cs2}}{\sum_f \mu_{csf}}, \frac{\mu_{cs3}}{\sum_f \mu_{csf}} \right) = \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3} \right),$$

and this is independent of c and s , while $\log \mu_{csf} = \alpha_{cs} + \gamma_f$ corresponds to probabilities

$$\begin{aligned} (\pi_{cs1}, \pi_{cs2}, \pi_{cs3}) &= \left(\frac{\mu_{cs1}}{\sum_f \mu_{csf}}, \frac{\mu_{cs2}}{\sum_f \mu_{csf}}, \frac{\mu_{cs3}}{\sum_f \mu_{csf}} \right) \\ &= \frac{1}{e^{\gamma_1} + e^{\gamma_2} + e^{\gamma_3}} (e^{\gamma_1}, e^{\gamma_2}, e^{\gamma_3}), \end{aligned}$$

also independent of colour and species. Linear predictor $\log \mu_{csf} = \alpha_{cs} + \gamma_{cf}$

Table 10.14 Deviances for log-linear models fitted to jacamar data.

Terms	df	Deviance
C★S	22	259.42
C★S+F	86	173.86
C★S+C★F	72	139.62
C★S+S★F	76	148.23
C★S+C★F+S★F	62	90.66
C★S★F	0	0

corresponds to probability vector

$$\begin{aligned}
 (\pi_{cs1}, \pi_{cs2}, \pi_{cs3}) &= \left(\frac{\mu_{cs1}}{\sum_f \mu_{csf}}, \frac{\mu_{cs2}}{\sum_f \mu_{csf}}, \frac{\mu_{cs3}}{\sum_f \mu_{csf}} \right) \\
 &= \frac{1}{e^{\gamma_{c1}} + e^{\gamma_{c2}} + e^{\gamma_{c3}}} (e^{\gamma_{c1}}, e^{\gamma_{c2}}, e^{\gamma_{c3}}),
 \end{aligned}$$

in which the probabilities of the different fates depend on colour, but not on species. Let C★S+F denote the terms of the linear predictor $\alpha_{cs} + \gamma_f$. Then the terms for the three models above are C★S, C★S+F, and C★S+C★F. Any model that treats F as the response must contain a term C★S, which fixes the row totals. The term C★F indicates that the response probabilities depend on colour.

Table 10.14 contains the deviances for the models with C★S, with degrees of freedom adjusted for triplets with zero totals. The best-fitting model is the full model C★S★F. The best reasonable model is C★S+C★F+S★F, which extends to trinomial responses the binomial two-way layout model of Example 10.15, but its deviance is large compared to its asymptotic χ^2_{62} distribution.

If categories N and S were merged there would be 96 observations and two possible fates. In this case the linear predictor $\log \mu_{csf} = \alpha_{cs} + \gamma_{cf}$ corresponds to probabilities

$$(\pi_{cs1}, \pi_{cs2}) = \frac{1}{e^{\gamma_{c1}} + e^{\gamma_{c2}}} (e^{\gamma_{c1}}, e^{\gamma_{c2}}) = \frac{1}{1 + e^{\gamma_{c2} - \gamma_{c1}}} (1, e^{\gamma_{c2} - \gamma_{c1}}),$$

which is the binomial logistic model with terms 1+Colour fitted in Example 10.15. When the response classification has two categories, it simplifies matters to fit the model as binomial rather than Poisson, although identical inferences are drawn about its parameters. ■

Example 10.23 (Lung cancer data) Example 1.4 gives data on the lung cancer mortality of cigarette smokers among British male physicians. The response is the number of deaths in each cell of the table, which also gives the total number of man-years of exposure T in each category.

We initially fit a log-linear model with factors for both margins and offset log man-years at risk in each cell. Thus $T_{rc} \exp(\alpha_r + \beta_c)$ is the mean number of deaths in the (r, c) cell. This model has deviance 51.47 on 48 degrees of freedom and appears to fit well. Figure 10.9 shows the coefficients for this model; the first level of each factor is taken to have coefficient zero. The figure suggests that there is a linear effect of dose d on the cancer rate, but that the increase with age is faster. However the standard

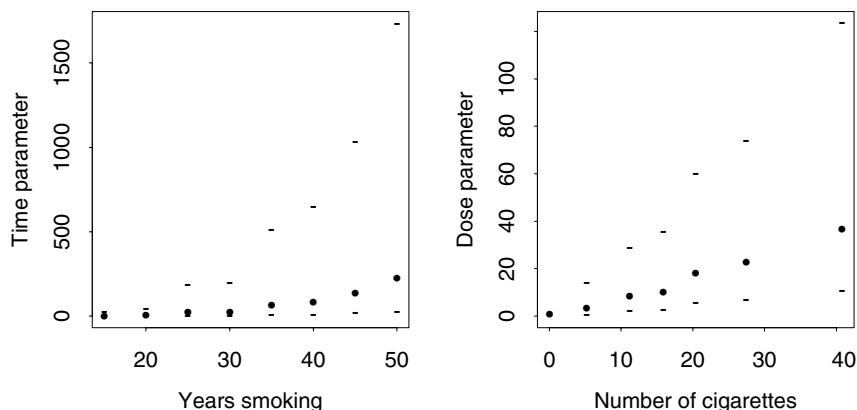


Figure 10.9 Results for two-way layout model, plotted against age and cigarette consumption. Shown are exponentials of coefficients, plus/minus two standard errors.

errors for the individual parameters are very large, reflecting the small numbers of deaths in most cells.

For a more concise model that is not log-linear, let the death rate for those smoking d cigarettes per day after t years of smoking be

$$\lambda(d, t) = (\beta_0 + \beta_1 d^{\beta_2}) t^{\beta_3}, \quad (10.31)$$

deaths per 100,000 man-years at risk; here β_0 and β_1 are non-negative, and β_2 and β_3 are real. We take t to be the midpoint of each group, divided by 42.5, so that β_0 represents the background rate of cancer for non-smokers aged 62.5 years, for whom the rescaled $t = 1$. The broadly exponential pattern in the left panel of Figure 10.9 suggests that $\beta_3 > 1$. The term $\beta_1 d^{\beta_2}$ describes the effect of smoking on death rates; we expect $\beta_2 \doteq 1$, corresponding to the linear increase seen in the right panel of Figure 10.9.

A likelihood ratio test of the effect of smoking on death rate would be non-regular, because setting either $\beta_1 = 0$ or $\beta_2 = 0$ eliminates both of these parameters; moreover $\beta_1 = 0$ is a boundary hypothesis. In either case the resulting model is log-linear, with deviance 180.8 on 61 degrees of freedom; the fit seems poor, despite the low counts and hence the likely inapplicability of chi-squared deviance asymptotics.

To fit the full model it is better to recast (10.31) as

$$\lambda(d, t) = \{e^{\gamma_0} + \exp(\gamma_1 + \beta_2 \log d)\} \exp(\beta_3 \log t),$$

so that all the parameters are unconstrained; the term $\exp(\gamma_1 + \beta_2 \log d)$ is omitted for the non-smokers. In this form it is straightforward to maximize the log likelihood by iterative weighted least squares, giving deviance 59.58 on 59 degrees of freedom, so marked an improvement on the model without smoking that the non-regularity of the asymptotics is immaterial.

Table 10.15 shows that the precision of $\hat{\gamma}_0$ depends heavily on the data for non-smokers. The background non-smoker death-rate from cancer at age 62.5 is $e^{\hat{\gamma}_0} = 18.9$ per 100,000 years at risk.

With the restriction $\beta_2 = 1$ the deviance increases to 61.84 on 60 degrees of freedom, a deviance difference of 2.26 on 1 degree of freedom. Linear dependence of

Table 10.15 Parameter estimates (standard errors) for lung cancer data.

	γ_0	γ_1	β_2	β_3
Smokers only	0.96 (25.4)	2.15 (1.45)	1.20 (0.40)	4.50 (0.34)
All data	2.94 (0.58)	1.82 (0.66)	1.29 (0.20)	4.46 (0.33)
All data ($\beta_2 = 1$)	2.75 (0.56)	2.72 (0.09)	—	4.43 (0.33)

Table 10.16 Joint distribution of visual impairment on both eyes by race and age Liang *et al.* (1992). Combination (0, 0) means neither eye is visually impaired.

Eye		Prevalence for whites aged				Prevalence for blacks aged			
Left	Right	40–50	51–60	61–70	70+	40–50	51–60	61–70	70+
0	0	602	541	752	606	729	551	452	307
1	0	11	15	31	60	19	24	22	29
0	1	15	16	37	67	21	23	21	37
1	1	4	9	11	79	10	14	28	56

death rate on d appears plausible, in which case the background death rate drops somewhat to 15.6 deaths per 100,000 man-years at risk, but rises by an additional $e^{\hat{\gamma}_1} \doteq 15.2$ for every cigarette smoked daily. Case analysis shows no residuals out of line, and the model appears to fit well. It is both more parsimonious than the log-linear model and motivated by substantive considerations, so it seems preferable. ■

Marginal models

Although mathematically elegant and simple to fit, log-linear models have some awkward statistical properties because their parameters have interpretations that depend on other terms in the model, as we shall now see.

Example 10.24 (Eye data) Table 10.16 gives data from the Baltimore Eye Study Survey. Drivers are classified by age, race and visual impairment, defined as vision less than 20/60; in the original data their level of education is also available, and is treated as a surrogate for socioeconomic status. The aim of the original analysis was to see how visual impairment depends on age and race, controlling for education, but we shall simply consider dependence on age and race.

We treat the data as eight 2×2 tables corresponding to columns 3–10 of the table, that is one for each different combination of race and age, given by the covariate vector x . Each table has elements $(y_{00}, y_{01}; y_{10}, y_{11})$, where y_{00} is the number of men without visual impairment, y_{01} is the number whose right eye only is poor, and so forth. The total number of men with covariate combination x is $m = y_{00} + y_{01} + y_{10} + y_{11}$, which we treat as fixed. The corresponding probabilities $(\pi_{00}, \pi_{01}; \pi_{10}, \pi_{11})$ depend on x .

A natural preliminary to joint analysis of data for both eyes is to fit logistic regression models and estimate the probability of impairment in each eye separately. For the left eye we would treat $r_L = y_{10} + y_{11}$ as a binomial response with denominator m and probability

$$\pi_{10} + \pi_{11} = \pi_L = \exp(x^T \beta_L) / \{1 + \exp(x^T \beta_L)\},$$

say. For the right eye the response would be $r_R = y_{01} + y_{11}$ with denominator m and probability $\pi_{01} + \pi_{11} = \pi_R = \exp(x^T \beta_R) / \{1 + \exp(x^T \beta_R)\}$. Here β_L and β_R summarize the effect of x on the marginal distributions of r_L and r_R . Our earlier arguments show that these logistic models are also log-linear.

In generalizing these marginal models to allow for the anticipated dependence between the eyes, it is natural to augment π_L and π_R by adding further parameters. One possibility is to write the odds ratio as

$$\frac{\pi_{11}\pi_{00}}{\pi_{10}\pi_{01}} = \frac{\pi_{11}(1 - \pi_L - \pi_R + \pi_{11})}{(\pi_L - \pi_{11})(\pi_R - \pi_{11})} = \exp(x^T \beta_{LR}).$$

If $x^T \beta_{LR} = \gamma$ was independent of x , there would be constant association between the eyes after adjusting for marginal effects of age and race, with more complicated models indicating more complex patterns of association. As $0 < \pi_{00}, \pi_{01}, \pi_{10}, \pi_{11} < 1$, the probability π_{11} must lie in the interval $(\max(0, \pi_L + \pi_R - 1), \min(\pi_L, \pi_R))$, and a little algebra shows that π_{11} may be expressed as the root of a quadratic equation whose coefficients depends on π_L, π_R , and $x^T \beta_{LR}$, thereby enabling us to express the probabilities in each 2×2 table in terms of the marginal probabilities and the odds ratio.

The log-linear model for the joint density of $(y_{00}, y_{01}; y_{10}, y_{11})$ has probabilities

$$(\pi_{00}, \pi_{01}; \pi_{10}, \pi_{11}) = \frac{1}{1 + e^{\gamma_R} + e^{\gamma_L} + e^{\gamma_R + \gamma_L + \gamma_{LR}}} (1, e^{\gamma_R}, e^{\gamma_L}, e^{\gamma_R + \gamma_L + \gamma_{LR}}),$$

where $\gamma_L = x^T \delta_L$, $\gamma_R = x^T \delta_R$, $\gamma_{LR} = x^T \delta_{LR}$. Under this model the marginal probability of an unimpaired left eye is

$$\pi'_L = \frac{e^{\gamma_L} + e^{\gamma_R + \gamma_L + \gamma_{LR}}}{1 + e^{\gamma_R} + e^{\gamma_L} + e^{\gamma_R + \gamma_L + \gamma_{LR}}},$$

which has logistic form $e^{\gamma_L} / (1 + e^{\gamma_L})$ only when $\gamma_{LR} = 0$, that is conditional on x visual impairment occurs independently in each eye. Otherwise the marginal probability of an impaired left eye depends on γ_R and γ_{LR} , implying that the initial logistic fits shed no light on γ_L .

To put this another way, note that γ_L may be written as

$$\log \left(\frac{\pi_{10}}{\pi_{00}} \right) = \log \frac{\Pr(L = 1 \mid R = 0, x)}{\Pr(L = 0 \mid R = 0, x)},$$

$L = 1$ denotes visual impairment in the left eye, etc.

with a similar expression for γ_R , and that

$$\gamma_{LR} = \log \left\{ \frac{\Pr(L = 1 \mid R = 1, x)}{\Pr(L = 0 \mid R = 1, x)} \right\} - \log \left\{ \frac{\Pr(L = 1 \mid R = 0, x)}{\Pr(L = 0 \mid R = 0, x)} \right\}.$$

Thus the parameters of the log-linear model have interpretations in terms of contrasts of log odds for one eye *conditional* on the state of the other, and these do not yield marginal probabilities with simple interpretations. Therefore the log-linear model for the joint outcomes is not upwardly compatible with the logistic models for the marginal outcomes. This poses problems in applications where marginal properties of the variables are of interest. ■

Inference for marginal models is awkward because complete specification of their likelihoods is ordinarily neither possible nor desirable. An alternative is to base inference on systems of estimating equations, and we now sketch how this is done.

Suppose that the j th of n individuals contributes a $q \times 1$ response vector Y_j and a $p \times 1$ vector of explanatory variables x_j , and let Z_j denote the $q(q-1)/2 \times 1$ vector containing the distinct products of pairs of elements of Y_j . Now $E(Y_j) = \mu(x_j^T \beta)$ is specified by the marginal model, while the covariance structure among the responses is given by $E(Z_j) = \xi(x_j^T \beta, \gamma)$, where β represents the parameters of the marginal model, and γ additional parameters that account for association among elements of Y_j .

In the preceding example $q = 2$ and Y_j^T equals $(0, 0)$, $(0, 1)$, $(1, 0)$, or $(1, 1)$, indicating the state of the left and right eyes, while

$$E(Y_j^T) = \mu(x_j^T \beta)^T = (\pi_L, \pi_R) = \left(\frac{\exp(x^T \beta_L)}{1 + \exp(x^T \beta_L)}, \frac{\exp(x^T \beta_R)}{1 + \exp(x^T \beta_R)} \right),$$

and $\xi(x_j^T \beta, \gamma)$, the probability of visual impairment in both eyes for individual j , depends both on x_j and on the degree of association between the eyes.

Ideas from Section 7.2 suggest that consistent estimators of β and γ may be obtained by combining the unbiased estimating functions $Y_j - \mu(x_j^T \beta)$ and $Z_j - \xi(x_j^T \beta, \gamma)$, and the form of (7.21) suggests that the estimators that solve the *generalized estimating equations*

$$\sum_{j=1}^n \frac{\partial \{\mu(x_j^T \beta)^T, \xi(x_j^T \beta, \gamma)^T\}}{\partial(\beta, \gamma)} \begin{pmatrix} \text{var}(Y_j) & \text{cov}(Y_j, Z_j) \\ \text{cov}(Z_j, Y_j) & \text{var}(Z_j) \end{pmatrix}^{-1} \times \begin{pmatrix} Y_j - \mu(x_j^T \beta) \\ Z_j - \xi(x_j^T \beta, \gamma) \end{pmatrix} = 0$$

will have smallest asymptotic variance. The presence of $\text{cov}(Y_j, Z_j)$ means that third-order moments of Y_j must in principle be specified, and one way to avoid this is to replace this term by a zero matrix. The resulting estimators $\hat{\beta}$ and $\hat{\gamma}$ are consistent but the variance of $\hat{\gamma}$ can be much larger than when the correct covariance matrix is used. If γ is of interest then some of the lost efficiency can be retrieved by assuming a simple form for $\text{cov}(Y_j, Z_j)$. Standard errors for $\hat{\beta}$ and $\hat{\gamma}$ are based on a sandwich covariance matrix; see Section 7.2. In many applications it is important to be able to accommodate missing data, and this is achieved by allowing the length of Y_j to vary with the individual; no essentially new points arise.

10.5.3 Ordinal responses

Discrete data often arise in which the response comprises numbers in ordered categories that may be labelled $1, \dots, k$. Examples are individuals undergoing some treatment and asked to say if they experience one of {no pain, slight pain, moderate pain, extreme pain}, or where curries are classified as {bland, mild, ..., volcanic}. The goal is then typically to assess how these *ordinal responses* depend on explanatory variables x . Sometimes the response is a discretized version of an underlying continuous variable, though this interpretation is not always plausible. In either case suitable models are based on the multinomial distribution. If there are n independent

individuals whose responses are $I_1, \dots, I_n$, and $I_j = l$ indicates that the j th response falls in category l , then $\Pr(I_j = l) = \pi_l$ for $l = 1, \dots, k$, and the corresponding cumulative probabilities are $\gamma_l = \Pr(I_j \leq l) = \pi_1 + \dots + \pi_l$ for $l = 1, \dots, k$; of course $\gamma_k = 1$. Individual responses with common explanatory variables x can be merged to give a multinomial variable $(Y_1, \dots, Y_k)$, where Y_l represents the number in category l ; thus $Y_1 + \dots + Y_k = n$. Typically the joint distribution of $(Y_1, \dots, Y_k)$ depends on x through a linear predictor $x^T \beta$.

In many applications it is appropriate to require that the interpretation of the model parameters remains unchanged when adjacent categories are merged. One class of models with this property may be motivated by positing the existence of an underlying continuous variable ε with distribution function F , with I indicating into which of the k intervals

$$(-\infty, \zeta_1], (\zeta_1, \zeta_2], \dots, (\zeta_{k-2}, \zeta_{k-1}], (\zeta_{k-1}, \infty), \quad \zeta_1 < \dots < \zeta_{k-1},$$

$x^T \beta + \varepsilon$ falls. For convenience let $\zeta_0 = -\infty$ and $\zeta_k = \infty$. Then

$$\begin{aligned} \pi_l(x^T \beta) &= \Pr(I = l; x^T \beta) = \Pr(\zeta_{l-1} < x^T \beta + \varepsilon \leq \zeta_l) \\ &= F(\zeta_l - x^T \beta) - F(\zeta_{l-1} - x^T \beta), \end{aligned}$$

and $\gamma_l(x^T \beta) = F(\zeta_l - x^T \beta)$, for $l = 1, \dots, k$. Thus large $x^T \beta$ leads to higher probabilities for the higher categories. A natural choice is the logistic distribution function $F(u) = \exp(u)/\{1 + \exp(u)\}$, which leads to the *proportional odds model*, so-called because the odds ratio of appearing in category l or lower for two individuals with explanatory variables x_1 and x_2 ,

$$\frac{\Pr(I \leq l; x_2)/\Pr(I > l; x_2)}{\Pr(I \leq l; x_1)/\Pr(I > l; x_1)} = \frac{\exp(\zeta_l - x_2^T \beta)}{\exp(\zeta_l - x_1^T \beta)} = \exp\{-(x_2 - x_1)^T \beta\},$$

is independent of l . Another possibility that often works well in practice is $F(u) = 1 - \exp\{-\exp(u)\}$. Whatever the choice of F , interest focuses on how the response depends on the covariates, summarized in β ; typically $\zeta_1, \dots, \zeta_{k-1}$ are of little concern. Any overall intercept term in $x^T \beta$ is aliased with the ζ_l .

Although this model is motivated by arguing as if an underlying continuous variable exists, this is not essential in order for it to be applied — the model may be useful even when ε clearly does not exist. As Examples 4.21 and 10.17 show, the loss of information due to categorizing continuous data can be substantial if the number of categories is very small.

The log likelihood based on independent responses $i_1, \dots, i_n$ with corresponding vectors of explanatory variables $x_1, \dots, x_n$ may be written as

$$\ell(\beta, \zeta) = \sum_{j=1}^n \sum_{l=1}^k I(i_j = l) \log \pi_l(x_j^T \beta),$$

a multinomial log likelihood to which by now familiar methods can be applied.

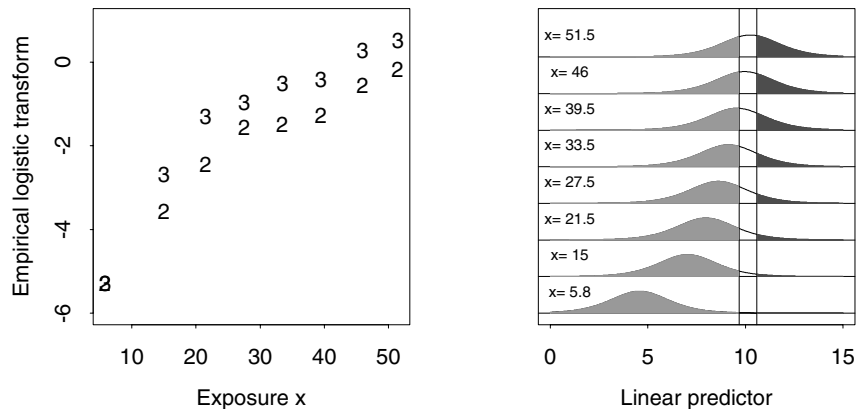
Example 10.25 (Pneumoconiosis data) The data in Table 10.17 concern the period x in years of work at a coalface and the degree of pneumoconiosis in a group of miners. The response consists of counts $\{y_1, y_2, y_3\}$ in $k = 3$ categories {Normal,

Table 10.17 Period of exposure x and prevalence of pneumoconiosis amongst coalminers (Ashford, 1959).

	Period of exposure (years)							
	5.8	15	21.5	27.5	33.5	39.5	46	51.5
Normal	98	51	34	35	32	23	12	4
Present	0	2	6	5	10	7	6	2
Severe	0	1	3	8	9	8	10	5

Figure 10.10

Pneumoconiosis data analysis. The left panel shows how empirical logistic transformations z_2 and z_3 depend on exposure x . The right panel shows how the implied fitted logistic distributions depend on x . The vertical lines show $\hat{\zeta}_1$ and $\hat{\zeta}_2$; the areas lying left of, between, and right of them equal the fitted probabilities $\hat{\pi}_1(x)$, $\hat{\pi}_2(x)$, and $\hat{\pi}_3(x)$.



Present, Severe} assessed radiologically and is qualitative. As the period of exposure increases, the proportion of miners with the disease present or in severe form increases sharply.

A simple analysis starts by combining categories, either as {Normal or Present, Severe} or as {Normal, Present or Severe}, to which models for binomial responses may be fitted. The plot of the empirical logistic transforms

$$z_2 = \log \left(\frac{y_3 + \frac{1}{2}}{y_1 + y_2 + \frac{1}{2}} \right), \quad z_3 = \log \left(\frac{y_2 + y_3 + \frac{1}{2}}{y_1 + \frac{1}{2}} \right),$$

in the left panel of Figure 10.10 shows that the linear predictor should contain $\log x$ rather than x . The logistic regression model with linear predictor $\beta_0 + \beta_1 \log x$ and response $y_2 + y_3$ gives $\hat{\beta}_0 = -9.6$ and $\hat{\beta}_1 = 2.58$. The corresponding model with response y_3 yields $\hat{\beta}_0 = -10.9$ and $\hat{\beta}_1 = 2.69$. Both models fit well, and the similarity of the slope estimates suggests that fitting the proportional odds model will be worthwhile.

Maximum likelihood fitting of the proportional odds model with linear predictor $\beta_1 \log x$ gives $\hat{\beta}_1 = 2.60$ (0.38), $\hat{\zeta}_1 = 9.68$ (1.32), and $\hat{\zeta}_2 = 10.58$ (1.34), entirely consistent with the binomial fits. Pearson's statistic is 4.7 on 13 degrees of freedom, so the fit seems good. The interpretation of $\hat{\beta}_1$ is that every doubling of exposure increases the linear predictor by $2.6 \times \log 2 \doteq 1.8$ and hence the odds of having the disease by a factor 6 or so. The same increase applies to the odds of having the disease in severe form. The right panel of Figure 10.10 illustrates how the fitted logistic distribution implied by the model changes with x . ■

Such models can be broadened by taking an underlying variable $x^T\beta + \sigma\varepsilon$, with σ dependent on explanatory variables.

Continuation ratio models may be based on the decomposition of the multinomial distribution of $(Y_1, \dots, Y_k)$ as

$$\begin{aligned} Y_1 &\sim B(n, \pi_1), \\ Y_2 | Y_1 &\sim B\left(n - y_1, \frac{\pi_2}{1 - \pi_1}\right), \\ &\vdots \\ Y_{k-1} | Y_1, \dots, Y_{k-2} &\sim B\left(n - y_1 - \dots - y_{k-2}, \frac{\pi_{k-1}}{1 - \pi_1 - \dots - \pi_{k-2}}\right); \end{aligned} \quad (10.32)$$

of course Y_k is constant conditional on $Y_1, \dots, Y_{k-1}$. At each stage the number of individuals in category l , given the numbers in categories $1, \dots, l-1$, is treated as a binomial variable with response probability $\pi_l/(1 - \pi_{l-1})$, to which a logistic regression or other suitable binomial response model may be fitted. Thus the original k -nomial response is broken into $k-1$ separate binomial responses. Unlike in the proportional odds model there is no necessity that the same explanatory variables be used in each of the $k-1$ fits, nor that their link functions be the same; this would depend on the scientific context.

Exercises 10.5

- 1 Consider the $2 \times n$ table of independent Poisson variables

$$\begin{array}{cccc} Y_{11} & \cdots & Y_{1j} & \cdots & Y_{1n} \\ Y_{21} & \cdots & Y_{2j} & \cdots & Y_{2n} \end{array},$$

where

$$\eta_{1j} = \log E(Y_{1j}) = x_{1j}^T \beta, \quad \eta_{2j} = \log E(Y_{2j}) = x_{2j}^T \beta.$$

Show that the conditional density of Y_{1j} given that $Y_{1j} + Y_{2j} = m_j$ is binomial with denominator m_j and probability π_j satisfying $\log\{\pi_j/(1 - \pi_j)\} = x_j^T \beta$, where $x_j = x_{1j} - x_{2j}$. This implies that a contingency table in which a single, binary, classification is regarded as the response can be analyzed using logistic regression. What advantages are there to doing so in terms of model-fitting and the examination of residuals?

- 2 In light of the preceding exercise and the discussion on page 501, reconsider the models fitted in Example 10.21. Say why Table 10.13 contains much larger standard errors for the logistic than for the log-linear model.
- 3 For a 2×2 contingency table with probabilities

$$\begin{array}{cc} \pi_{00} & \pi_{01} \\ \pi_{10} & \pi_{11} \end{array},$$

the maximal log-linear model may be written as

$$\begin{aligned} \eta_{00} &= \alpha + \beta + \gamma + (\beta\gamma), & \eta_{01} &= \alpha + \beta - \gamma - (\beta\gamma), \\ \eta_{10} &= \alpha - \beta + \gamma - (\beta\gamma), & \eta_{11} &= \alpha - \beta - \gamma + (\beta\gamma), \end{aligned}$$

where $\eta_{jk} = \log E(Y_{jk}) = \log(m\pi_{jk})$ and $m = \sum_{j,k} y_{jk}$. Show that the ‘interaction’ term $(\beta\gamma)$ may be written $(\beta\gamma) = \frac{1}{4} \log \Delta$, where Δ is the odds ratio $(\pi_{00}\pi_{11})/(\pi_{01}\pi_{10})$, so that $(\beta\gamma) = 0$ is equivalent to $\Delta = 1$.

- 4 Give the matrices needed for iterative weighted least squares for the nonlinear model (10.31) in Example 10.23. How might starting-values be obtained?

- 5 In Example 10.24, discuss whether a marginal model or a log-linear model is preferable for (a) a white man aged 43 with a visually impaired left eye, who wants to assess his probability of having visual impairment in the other eye at the age of 65, and (b) a scientist comparing how visual impairment develops with age for men of different races.
- 6 Give the form of the proportional odds model obtained when an underlying continuous variable $x^T\beta + \exp(x^T\gamma)\varepsilon$ is categorized; ε has the logistic density $e^u/(1+e^u)^2$, $-\infty < u < \infty$. Derive the iterative weighted least squares algorithm for estimation of β when it is known that $\gamma = 0$. Explain how you would need to change your algorithm to deal with $\gamma \neq 0$.
- 7 Establish (10.32).

10.6 Overdispersion

Thus far we have supposed that our data are well-described by a model with a simple error distribution. Nature is not usually so obliging, however, and in practice it is common to find that count and proportion data are more variable than would be expected under the Poisson and binomial models. Other types of data may also exhibit such *overdispersion*, manifested by models with over-large deviances and residuals, but otherwise showing no systematic lack of fit. Structure in the data is obscured by additional noise, so overdispersion increases uncertainty. Underdispersion also arises but is much rarer.

Two approaches to dealing with overdispersion are explicit parametric modelling of the heterogeneity, and the use of quasi-likelihood and associated estimating functions.

Parametric models

Suppose that the response Y has a standard distribution conditional on the unobserved variable ε , but that ε induces extra variation in Y . Here ε might represent unobserved — perhaps unobservable — covariates that affect the response. Let ε have unit mean and variance $\xi > 0$, and to be concrete suppose that conditional on ε , Y has the Poisson distribution with mean $\mu\varepsilon$. Then (3.12) and (3.13) give

$$E(Y) = E_{\varepsilon} \{E(Y | \varepsilon)\}, \quad \text{var}(Y) = \text{var}_{\varepsilon} \{E(Y | \varepsilon)\} + E_{\varepsilon} \{\text{var}(Y | \varepsilon)\},$$

so the response has mean and variance

$$E(Y) = E_{\varepsilon}(\mu\varepsilon) = \mu, \quad \text{var}(Y) = \text{var}_{\varepsilon}(\mu\varepsilon) + E_{\varepsilon}(\mu\varepsilon) = \mu(1 + \xi\mu).$$

If on the other hand the variance of ε is ξ/μ , then $\text{var}(Y) = (1 + \xi)\mu$. In both cases the variance of Y is greater than its value under the standard Poisson model, for which $\xi = 0$. In the first case the variance function is quadratic, and in the second it is linear.

Table 10.18 illustrates the difference between these variance functions under modest overdispersion. Large amounts of data will be needed to detect overdispersion when the counts are small. The variances are equal when $\mu = 15$, but evidently a lot of data over a limited range of values of μ or alternatively a large range of mean responses would be needed to discriminate well between the two variance functions. This is one reason to consider a more robust approach, rather than to model the

μ	1	2	5	10	15	20	30	40	60
Linear	1.5	3.0	7.5	15.0	22.5	30	45	60	90
Quadratic	1.0	2.1	5.8	13.3	22.5	33	60	93	180

Table 10.18

Comparison of variance functions for overdispersed count data. The linear and quadratic variance functions are $V_L(\mu) = (1 + \xi_L)\mu$ and $V_Q(\mu) = \mu(1 + \xi_Q\mu)$, with $\xi_L = 0.5$ and ξ_Q chosen so that $V_L(15) = V_Q(15)$.

overdispersion in detail. If a full likelihood analysis is desired regardless, one can proceed as in the following example.

Example 10.26 (Negative binomial model) In the discussion above, suppose that ε has the gamma distribution with unit mean and variance $1/\nu$. Then Y has the negative binomial density (Exercise 10.6.1)

$$f(y; \mu, \nu) = \frac{\Gamma(y + \nu)}{\Gamma(\nu)y!} \frac{\nu^\nu \mu^y}{(\nu + \mu)^{\nu+y}}, \quad y = 0, 1, \dots, \quad \mu, \nu > 0, \quad (10.33)$$

and quadratic and linear variance functions are obtained on setting $\nu = 1/\xi$ and $\nu = \mu/\xi$ respectively. The first leads to simpler likelihood equations and so is preferable in purely numerical terms. When independent responses y_j have associated covariates x_j , it is natural to take the log link, giving means $\mu_j = \exp(x_j^\top \beta)$. The value of ξ may be estimated from its profile log likelihood or by equating the Pearson statistic and its expected value; see Example 10.28. ■

A similar analysis applies to proportions. Suppose that conditional on ε , $R = mY$ is binomial with denominator m and success probability $\pi\varepsilon$, and that ε has unit mean and variance ξ . Then calculations like those above give

$$E(Y) = \pi, \quad \text{var}(Y) = m^{-1} \{ \pi(1 - \pi) + \xi \pi^2(m - 1) \}. \quad (10.34)$$

Hence overdispersion increases with m if ξ is constant. Heterogeneity is undetectable in pure binary data, for which $m = 1$. When $m > 1$ and $\gamma > 0$, the choice $\xi = \gamma(1 - \pi)/\{\pi(m - 1)\}$ gives $\text{var}(Y) = (1 + \gamma)\pi(1 - \pi)/m$, corresponding to uniform overdispersion. This is explored further in Exercise 10.6.4.

Quasi-likelihood

In all but the simplest cases the modelling of overdispersion by integrating out an unobserved variable leads to use of numerical integration. This can be awkward, but a more serious difficulty is that inferences might depend strongly on the unobserved component, which can be validated only indirectly. Hence it is often preferable to modify standard methods to accommodate overdispersion, in analogy with the use of least squares estimation when responses are non-normal (Section 8.4). We shall see below that provided the mean and variance functions are correctly specified, the estimators obtained by fitting standard models retain their large-sample normal distributions, but with an inflated variance matrix. This is very convenient, because standard software can then be used for fitting, with minor modification to the output.

Unrecognised overdispersion is a form of model misspecification, so one starting-point is to apply the ideas of Section 7.2, treating the generalized linear model score statistic (10.18) as an estimating function $g(Y; \beta)$ for β . An estimator $\tilde{\beta}$ is obtained

X is the $n \times p$ matrix
whose j th row is x_j^T .

by solving the quasi-likelihood equation

$$g(Y; \beta) = X^T u(\beta) = \sum_{j=1}^n x_j u_j(\beta) = \sum_{j=1}^n x_j \frac{Y_j - \mu_j}{g'(\mu_j) \phi_j V(\mu_j)} = 0, \quad (10.35)$$

where the link function gives $g(\mu_j) = \eta_j = x_j^T \beta$. Now if the mean structure has been chosen correctly, then $E(Y_j) = \mu_j$ and the estimating function is unbiased, that is $E\{g(Y; \beta)\} = 0$ for all β . Then the quasi-likelihood estimator $\tilde{\beta}$ is consistent under mild regularity conditions.

In large samples $\tilde{\beta}$ is normal with variance matrix (Section 7.2.1)

$$E \left\{ -\frac{\partial g(Y; \beta)}{\partial \beta^T} \right\}^{-1} \text{var} \{g(Y; \beta)\} E \left\{ -\frac{\partial g(Y; \beta)^T}{\partial \beta} \right\}^{-1}. \quad (10.36)$$

In order to compute this we require $E\{-\partial g(Y; \beta)/\partial \beta^T\}$ and $\text{var}\{g(Y; \beta)\}$. Now

$$\begin{aligned} \frac{\partial u_j(\beta)}{\partial \beta^T} &= \frac{\partial \eta_j}{\partial \beta^T} \frac{\partial \mu_j}{\partial \eta_j} \frac{\partial u_j(\beta)}{\partial \mu_j} \\ &= x_j^T \frac{1}{g'(\mu_j)} \left\{ -\frac{g''(\mu_j)}{g'(\mu_j)^2} u_j(\beta) - \frac{V'(\mu_j)}{V(\mu_j)} u_j(\beta) - \frac{1}{g'(\mu_j) \phi_j V(\mu_j)} \right\}, \end{aligned}$$

and as $E\{u_j(\beta)\} = 0$, it follows that

$$\begin{aligned} E \left\{ -\frac{\partial g(Y; \beta)}{\partial \beta^T} \right\} &= -\sum_{j=1}^n x_j E \left\{ \frac{\partial u_j(\beta)}{\partial \beta^T} \right\} \\ &= \sum_{j=1}^n x_j x_j^T \frac{1}{g'(\mu_j)^2 \phi_j V(\mu_j)} = X^T W X, \end{aligned}$$

where W is the $n \times n$ diagonal matrix with j th element $\{g'(\mu_j)^2 \phi_j V(\mu_j)\}^{-1}$. Moreover if in addition the variance function has been correctly specified, then $\text{var}(Y_j) = \phi_j V(\mu_j)$, and hence

$$\text{var}\{g(Y; \beta)\} = X^T \text{var}\{u(\beta)\} X = \sum_{j=1}^n x_j x_j^T \frac{\text{var}(Y_j)}{g'(\mu_j)^2 \phi_j^2 V(\mu_j)^2} = X^T W X.$$

Thus (10.36) equals $(X^T W X)^{-1}$. Had the variance function been wrongly specified, the variance matrix of $\tilde{\beta}$ would have been of sandwich form $(X^T W' X)^{-1} (X^T W' X)(X^T W X)^{-1}$, where W' is a diagonal matrix involving the true and assumed variance functions. Only if the variance function has been chosen very badly will this sandwich matrix differ greatly from $(X^T W X)^{-1}$, which therefore provides useful standard errors unless a plot of absolute residuals against fitted means is markedly non-random. In that case the choice of variance function should be reconsidered.

Quasi-likelihood estimates and standard errors are easily obtained using software that fits generalized linear models. Usually $\phi_j = a_j \phi$, where the a_j are known constants and $\phi = 1$ corresponds to a model such as the Poisson or binomial, for which the software finds estimates and standard errors by solving (10.35) with $\phi = 1$. As ϕ cancels from (10.35), the quasi-likelihood estimate $\tilde{\beta}$ equals the maximum likelihood estimate. Software that sets $\phi = 1$ will yield a variance matrix that is too small by a

factor ϕ , however, so the usual standard errors must be multiplied by $\hat{\phi}^{1/2}$, where $\hat{\phi}$ is defined at (10.20).

Under an exponential family model, the quantity $g(Y; \beta)$ in (10.35) is the score statistic, so estimators based upon it are asymptotically optimal. Even if that model is false, inference based on $g(Y; \beta)$ is valid provided the mean and its relation with the variance $V(\mu)$ have been correctly specified. Moreover the argument on page 322 shows that $\tilde{\beta}$ is optimal among estimators based on linear combinations of the $Y_j - \mu_j$, in analogy with the Gauss–Markov theorem. The essential requirement for this is that the $u_j(\beta)$ satisfy the two key properties

$$E(\partial \ell / \partial \mu) = 0, \quad \text{var}(\partial \ell / \partial \mu) = E(-\partial^2 \ell / \partial \mu^2)$$

of a log likelihood derivative. In fact, $g(Y; \beta)$ is the derivative with respect to β of the *quasi-likelihood* function

Strictly $Q(\beta; Y)$ is a quasi-log likelihood.

$$Q(\beta; Y) = \sum_{j=1}^n \int_{Y_j}^{\mu_j} \frac{Y_j - u}{\phi a_j V(u)} du,$$

and we can define a deviance as $-2\phi Q(\beta; Y)$. This is positive by construction and can be used to compare nested models under overdispersion.

Example 10.27 (Weighted least squares) The simplest example of quasi-likelihood estimation arises when $V(\mu) = 1$, $\phi_j = \phi a_j$, and the mean of Y_j is $\mu_j = x_j^\top \beta$. Then (10.35) becomes

$$\sum_{j=1}^n x_j \frac{Y_j - x_j^\top \beta}{\phi a_j} = X^\top W(Y - X\beta) = 0,$$

where W is the diagonal matrix $\phi^{-1} \text{diag}(1/a_1, \dots, 1/a_n)$, and

$$\tilde{\beta} = (X^\top W X)^{-1} X^\top W Y$$

is the weighted least squares estimator of β , found using weights a_j^{-1} . This estimator is the maximum likelihood estimator only if the Y_j are independent and normal, but even if not, $\tilde{\beta}$ is the minimum variance unbiased estimator linear in the Y_j (Section 8.4).

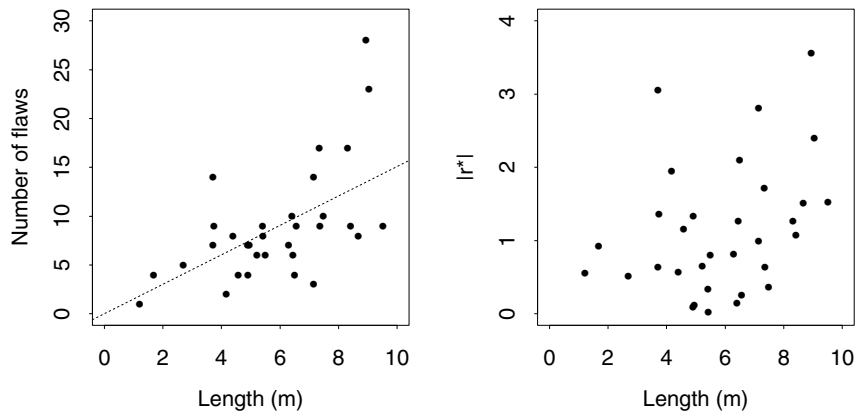
Integration shows that the deviance $Q(\beta; Y)$ equals the weighted sum of squares $(Y - X\beta)^\top W(Y - X\beta)$, while

$$\hat{\phi} = \frac{1}{n-p} \sum_{j=1}^n \frac{(Y_j - x_j^\top \tilde{\beta})^2}{a_j} = \frac{1}{n-p} (Y - X\tilde{\beta})^\top W(Y - X\tilde{\beta});$$

see Section 8.2.4. ■

Example 10.28 (Cloth fault data) The left panel of Figure 10.11 shows the numbers of flaws in $n = 32$ cloth samples of various lengths. A plausible model is that the number of faults y in a sample of length x has a Poisson distribution with mean βx . A maximum likelihood fit of this model gives $\hat{\beta} = 1.51$ with standard error 0.09. However the deviance of 64.5 on 31 degrees of freedom and the right panel of the figure suggest that the data are more variable than the Poisson model might indicate.

Figure 10.11 Cloth data analysis (Bissell, 1972). The left panel shows the numbers of flaws in 32 cloth samples of various lengths (m). The dotted line shows the fitted mean number of faults under the model. The right panel shows that absolute residuals for the fit are overdispersed relative to the standard normal distribution appropriate under the Poisson assumption.



On reflection this is not surprising, as the rate β is likely to vary from one sample to another.

For quasi-likelihood estimation with $\text{var}(Y) = \phi\mu$, (10.35) is given by

$$\sum_{j=1}^n x_j \frac{y_j - \mu_j}{\phi\mu_j} = 0, \quad \mu_j = x_j\beta,$$

and as $\sum(y_j - \hat{\mu}_j)^2/\hat{\mu}_j = 68.03$ on 31 degrees of freedom, $\hat{\phi} = 68.03/31 = 2.19$. The standard error for $\hat{\beta}$ is then $0.09\hat{\phi}^{1/2} = 0.13$, appreciably larger than under the Poisson model.

When the negative binomial model with variance function $\mu(1 + \xi\mu)$ is fitted, the maximum likelihood estimates of β and ξ are 1.51 and 0.115 with standard errors 0.13 and 0.056. The maximized log likelihood is -87.73 , and $\sum(y_j - \hat{\mu}_j)^2/\{\hat{\mu}_j(1 + \hat{\xi}\hat{\mu}_j)\} = 32.57$ on 31 degrees of freedom, giving no evidence of poor fit. For the alternative negative binomial model with variance function $(1 + \xi)\mu$, the maximized log likelihood is -88.63 , so the fit is slightly worse. This is borne out by the right panel of Figure 10.11, which suggests that the variance of the residuals increases with x , as would be the case if the linear variance function was fitted when the quadratic was more appropriate. ■

The discussion above shows how standard errors should be modified in the presence of overdispersion. A similar adjustment applies when using deviance differences for model selection. Let model A be nested within a more complicated model B , with deviances $D_A > D_B$ and parameters $p_A < p_B$. For binomial and Poisson data, the usual approach is to compare $D_A - D_B$ with the $\chi^2_{p_B - p_A}$ distribution. In the presence of overdispersion this is modified by analogy with F tests in linear regression: if $\hat{\phi}_B$ is the estimate of ϕ under the more complex model, then the adequacy of model A relative to model B is assessed by referring $\{(D_A - D_B)/(p_B - p_A)\}/\hat{\phi}_B$ to the $F_{p_B - p_A, n - p_B}$ distribution.

Example 10.29 (Toxoplasmosis data) Table 10.19 gives data on the relation between rainfall and the proportions of people with toxoplasmosis for 34 cities in

City	Rain	r/m	City	Rain	r/m	City	Rain	r/m	City	Rain	r/m
1	1735	2/4	11	2050	7/24	21	1756	2/12	31	1780	8/13
2	1936	3/10	12	1830	0/1	22	1650	0/1	32	1900	3/10
3	2000	1/5	13	1650	15/30	23	2250	8/11	33	1976	1/6
4	1973	3/10	14	2200	4/22	24	1796	41/77	34	2292	23/37
5	1750	2/2	15	2000	0/1	25	1890	24/51			
6	1800	3/5	16	1770	6/11	26	1871	7/16			
7	1750	2/8	17	1920	0/1	27	2063	46/82			
8	2077	7/19	18	1770	33/54	28	2100	9/13			
9	1920	3/6	19	2240	4/9	29	1918	23/43			
10	1800	8/10	20	1620	5/18	30	1834	53/75			

Table 10.19
Toxoplasmosis data: rainfall (mm) and the numbers of people testing positive for toxoplasmosis, r , out of m people tested, for 34 cities in El Salvador (Efron, 1986).

Terms	df	Deviance
Constant	33	74.21
Linear	32	74.09
Quadratic	31	74.09
Cubic	30	62.63

Table 10.20 Analysis of deviance for polynomial logistic models fitted to the toxoplasmosis data.

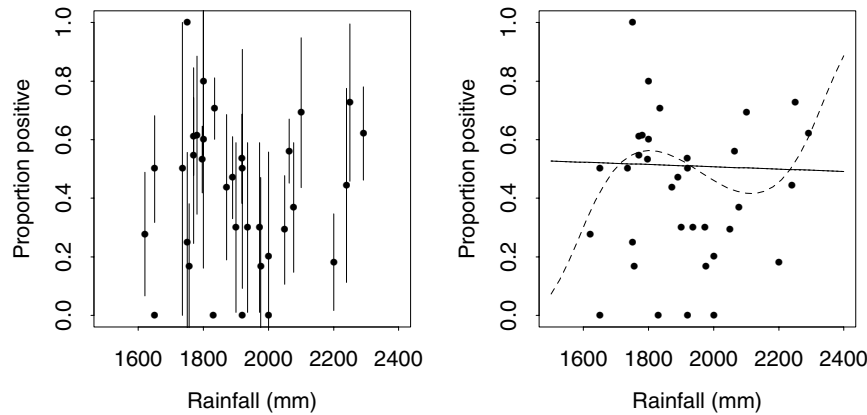


Figure 10.12
Toxoplasmosis data. The left panel shows the proportion of people testing positive, r/m , plus/minus $2\{r(m-r)/m^3\}^{1/2}$, as a function of rainfall in 34 cities in El Salvador. The right panel shows the data and linear (solid), quadratic (dots, almost identical to the linear fit), and cubic (dashes) polynomial models fitted on the logistic scale.

El Salvador. There is wide variation in the numbers tested, as well as in the proportions testing positive, and the left panel of Figure 10.12 indicates a possible nonlinear relation between rainfall and toxoplasmosis incidence.

The right panel shows fitted proportions for logistic regression models in which the linear predictor contains terms linear, quadratic, and cubic in rainfall. Table 10.20 contains the analysis of deviance when the polynomial terms are included successively. The residual deviance of 62.63 on 30 degrees of freedom indicates overdispersion by a factor of roughly two.

Under the binomial assumption, the cubic model is tested against the constant model by comparing the deviance difference $74.21 - 62.63 = 11.58$ with the χ^2_3

distribution, giving significance level 0.009. This overstates the significance of the test because it makes no allowance for overdispersion. Under quasi-likelihood with $\text{var}(R) = \phi m \pi(1 - \pi)$ we obtain $\tilde{\phi} = 1.94$, and our general discussion suggests that we should compare the F statistic $(11.58/3)/\tilde{\phi} = 1.99$ with the $F_{3,30}$ distribution. This gives significance level 0.14, only weak evidence of a relationship between rainfall and incidence. We return to these data in Example 10.32. ■

If the responses are dependent, the above discussion can be extended by taking as estimating function $X^T V(\mu)^{-1}(Y - \mu)$, where $V(\mu)$ is an $n \times n$ covariance matrix for Y ; see page 507. This is a common technique for modelling longitudinal data, in which short, often irregular time series are available on independent individuals. In such cases there may be no function whose derivatives with respect to β give the estimating function, and then no quasi-likelihood exists.

In some cases the response variance may be expressed as $\phi(\gamma)V(\mu; \xi)$, with γ and ξ unknown. An example is the quadratic variance function $\mu + \xi \mu^2$ in Example 10.26. The definition of the deviance depends on ξ , so models with different values of ξ cannot be compared using differences of deviances. An extended quasi-likelihood can be defined as the sum of the contributions

$$-\frac{1}{2} \log\{\phi_j(\gamma)V(\mu_j; \xi)\} - \frac{1}{2} \int_{Y_j}^{\mu_j} \frac{Y_j - u}{\phi_j(\gamma)a_j V(u; \xi)} du,$$

however, and used for inference about the unknown parameters. Unfortunately this definition is ambiguous: for example $\mu + \xi \mu^2$ can be written as $\phi(\gamma) = 1$, $V(\mu; \xi) = \mu + \xi \mu^2$ or as $\phi(\gamma) = \mu$, $V(\mu; \xi) = 1 + \xi \mu$, and these give different extended quasi-likelihoods. Uniqueness can be imposed by insisting that $\phi(\gamma)$ not involve μ or that $V(\mu) = 1$, leading to two different systems of estimating equations. The first system gives inconsistent estimators and the second gives consistent estimators. However simulation shows that for sample sizes of most interest the second estimators are worse than the first. Thus in practice the solutions to the first system are preferable, though neither is really satisfactory.

Exercises 10.6

- Use (2.8) to establish (10.33). Give formulae for the corresponding deviance residuals when $v = 1/\xi$ and when $v = \mu/\xi$.
Suppose that independent counts $y_1, \dots, y_n$ arise with means $\mu_j = \exp(x_j^T \beta)$. Under the model with constant $v = 1/\xi$, write down the negative binomial log likelihood for β and ξ . Explain why the likelihood equations become more complicated if the shape parameter changes for each observation, so $v_j = \mu_j/\xi$.
If we estimate ξ by equating the Pearson statistic $\sum (y_j - \hat{\mu}_j)^2 / V(\hat{\mu}_j)$ to $n - p$, where $V(\mu_j) = \text{var}(Y_j)$, discuss how to obtain the estimate under the above two variance functions.
- Let I be a binary variable with success probability π , and suppose that π is given a density h . Show that I remains a binary variable whatever the choice of h , and hence explain the form of the variance in (10.34).
Against what variable should the squared Pearson residual be plotted if it is desired to assess if (10.34) gives a suitable fit to data?

- 3 Find $Q(\beta; Y)$ when $u_j(\beta) = (Y_j - \mu_j)/\{\phi g'(\mu_j)V(\mu_j)\}$ and $V(\mu)$ equals μ , $\mu(1 - \mu)$, and μ^2 .
- 4 One standard model for over-dispersed binomial data assumes that R is binomial with denominator m and probability π , where π has the beta density

$\Gamma(a)$ is the gamma function.

$$f(\pi; a, b) = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} \pi^{a-1} (1-\pi)^{b-1}, \quad 0 < \pi < 1, a, b > 0.$$

(a) Show that this yields the *beta-binomial* density

$$\Pr(R = r; a, b) = \frac{\Gamma(m+1)\Gamma(r+a)\Gamma(m-r+b)\Gamma(a+b)}{\Gamma(r+1)\Gamma(m-r+1)\Gamma(a)\Gamma(b)\Gamma(m+a+b)}, \quad r = 0, \dots, m.$$

(b) Let μ and σ^2 denote the mean and variance of π . Show that in general,

$$E(R) = m\mu, \quad \text{var}(R) = m\mu(1-\mu) + m(m-1)\sigma^2,$$

and that the beta density has $\mu = a/(a+b)$ and $s^2 = ab/\{(a+b)(a+b+1)\}$. Deduce that the beta-binomial density has mean and variance

$$E(R) = ma/(a+b), \quad \text{var}(R) = m\mu(1-\mu)\{1 + (m-1)\delta\}, \quad \delta = (a+b+1)^{-1}.$$

Hence re-express $\Pr(R = r; a, b)$ as a function of μ and δ . What is the condition for uniform overdispersion?

- 5 Conditional on ε , the observation Y has a generalized linear model density with canonical parameter $\eta + \tau\varepsilon$, where $\tau > 0$. If ε is standard normal, show that the marginal density of Y can be written

$$f(y; \eta, \tau) = \frac{1}{(2\pi)^{1/2}} \int_{-\infty}^{\infty} \exp \left\{ \frac{y\eta + y\tau\varepsilon - b(\eta + \tau\varepsilon)}{a(\phi)} + c(y; \phi) - \varepsilon^2/2 \right\} d\varepsilon.$$

By second-order Taylor series expansion of $b(\eta + \tau\varepsilon)$ for small τ , or otherwise, show that $f(y; \eta, \tau)$ equals

$$f(y; \eta, 0) \exp \left[\frac{\tau^2}{2} \frac{\{y - b'(\eta)\}^2}{a(\phi)^2 \{1 + \tau^2 b''(\eta)/a(\phi)\}} \right] \{1 + \tau^2 b''(\eta)/a(\phi)\}^{-1/2} + o_p(\tau^2).$$

Prove that this approximation is exact when the conditional density of Y given ε is normal, and then find the unconditional mean and variance of Y .

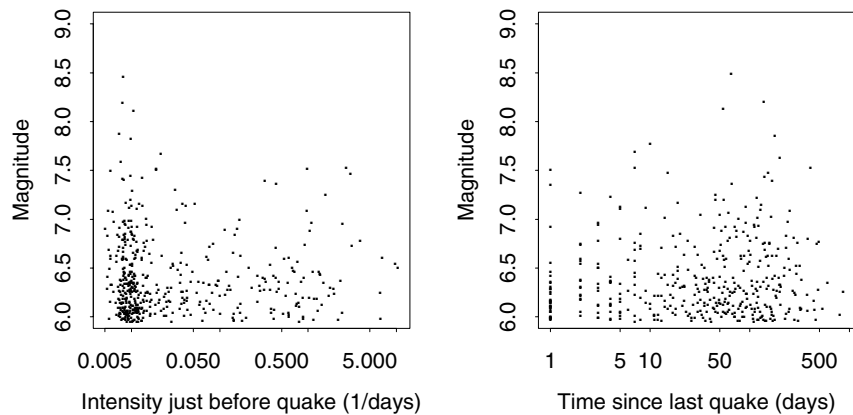
10.7 Semiparametric Regression

Our earlier regression models have involved responses that depend on explanatory variables x through simple parametric functions such as $\beta_0 + \beta_1 x + \beta_2 x^2$. Their conciseness and direct interpretation gives such formulations great appeal, but they are not flexible enough to cater for all the situations met in practice and more general approaches are desirable, especially for exploratory analysis, model-checking, and other situations where the data should be allowed to ‘speak for themselves’. Many ways to do this have been proposed in recent years, under the heading of *nonparametric* or *semiparametric* models, the aim typically being to extract a smooth curve from the data. An algorithm that does this is often termed a *smoother*. In fact smoothing operations typically do involve parameters, but in less prescriptive ways than before, and the results are best understood graphically. There are many approaches to semiparametric modelling, and below we merely sketch the possibilities by extending our previous discussion in two directions.

The adjective ‘smoother’ has become a noun in this context.

Figure 10.13

Earthquake magnitudes plotted against fitted intensity just before the earthquake shock and time since the preceding shock. Note the log scales. The magnitudes have been jittered to reduce overplotting.



Example 10.30 (Japanese earthquake data) Figure 6.19 shows data on 483 earthquake shocks, of magnitude at least 6 on the Richter scale, offshore from Japan from 1885–1980. In Example 6.38 a self-exciting point process model was fitted to the data, in which the intensity at time t was given by

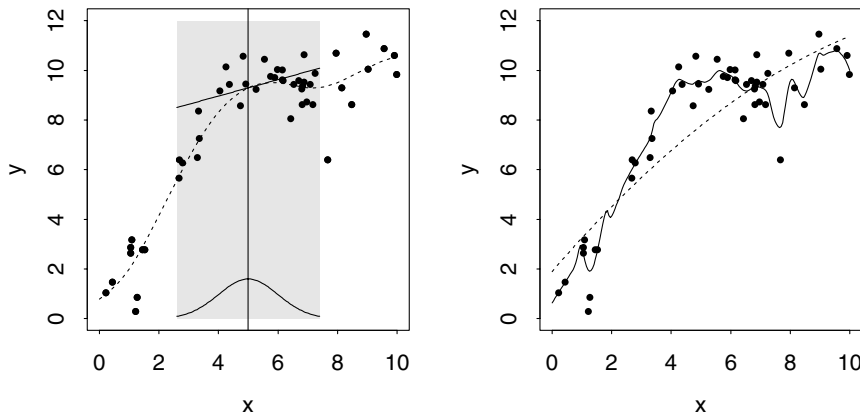
$$\lambda_{\mathcal{H}}(t) = \mu + \kappa \sum_{j:t_j < t} \frac{e^{\beta(m_j - 6)}}{t - t_j + \gamma},$$

where m_j is the magnitude and t_j the time of the j th earthquake. This fits the times adequately, but $\lambda_{\mathcal{H}}(t)$ models only the time of the next shock and not its magnitude. It is natural to ask if the magnitude of a shock depends on the past, for example on the value of $\lambda_{\mathcal{H}}(t)$ just before the shock occurs, or on the time elapsed since the previous shock.

Figure 10.13 contains scatterplots of m_i against $\lim_{\delta \rightarrow 0} \hat{\lambda}_{\mathcal{H}}(t_i - \delta)$ and against $t_i - t_{i-1}$. The lack of pattern in both panels suggests that magnitude is unrelated to these other quantities, but the clustering of points at the left of the left panel makes it hard to be sure, and a smooth curve would sharpen our judgement. Fitting a particular parametric model seems difficult to justify, so the curve should be defined flexibly. ■

10.7.1 Local polynomial models

Suppose that the response y equals $g(x) + \varepsilon$, where g is a smooth function of the scalar x , and ε has mean zero and variance σ^2 . Although we assume that g has as many derivatives as we need, it will not usually have a simple form. The data consist of pairs $(x_1, y_1), \dots, (x_n, y_n)$, and initially we shall suppose that we wish to make inferences about $g(x)$ at a single point $x = x_0$ in the interval spanned by the design points x_j . One approach is to fit a polynomial to all n pairs and then read off its value at x_0 . However, such fits are often unconvincing — see the right panel of Figure 10.12, for example — and furthermore they can be sensitive to observations distant from x_0 . Rather than treat all data pairs equally, it seems natural to attach more importance to observations close to x_0 . This is closely related to kernel density estimation (Section 7.1.2).

**Figure 10.14**

Construction of a local linear smoother. Left panel: observations in the shaded part of the panel are weighted using the kernel shown at the foot, with $h = 0.8$, and the solid straight line is fitted by weighted least squares. The local estimate is the fitted value when $x = x_0$, shown by the vertical line. Two hundred local estimates formed using equi-spaced x_0 were interpolated to give the dotted line, which is the estimate of $g(x)$. Right panel: local linear smoothers with $h = 0.2$ (solid) and $h = 5$ (dots).

Recall that a kernel function $w(u)$ is a unimodal density function symmetric about $u = 0$ and with unit variance. One choice of w is the standard normal density. Another is a rescaled form of the *tricube* function

$$w(u) = \begin{cases} (1 - |u|^3)^3, & |u| \leq 1, \\ 0, & \text{otherwise,} \end{cases} \quad (10.37)$$

and there are many others.

When estimating $g(x)$ at $x = x_0$, we attach weight $w_j = h^{-1}w\{h^{-1}(x_j - x_0)\}$ to (x_j, y_j) , where h is a bandwidth, and fit the polynomial

$$\beta_0 + \beta_1(x - x_0) + \cdots + \beta_k(x - x_0)^k$$

to the responses by weighted least squares (Section 8.2.4). The weights decrease the influence of points for which $|x_j - x_0|/h$ is big: for large h points far from x_0 will affect the fit, whereas as $h \rightarrow 0$ the regression becomes ever more local. The number of data included will vary as x_0 changes, with fewer points when x_0 is near the limits of its range or where the x_j are sparse.

The estimate of $g(x_0)$ is obtained by fitting the linear model

$$\begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix} = \begin{pmatrix} 1 & (x_1 - x_0) & \cdots & (x_1 - x_0)^k \\ \vdots & \vdots & & \vdots \\ 1 & (x_n - x_0) & \cdots & (x_n - x_0)^k \end{pmatrix} \begin{pmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_k \end{pmatrix} + \begin{pmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{pmatrix},$$

that is $y = X\beta + \varepsilon$, with weight matrix $W = \text{diag}(w_1, \dots, w_n)$. This results in the weighted least squares estimate $\hat{\beta} = (X^T W X)^{-1} X^T W y$, of which the component of interest is $\hat{\beta}_0$, the value of the fitted polynomial when $x = x_0$.

In practice a smooth estimate $\hat{g}(x)$ of the entire function $g(x)$ is usually required. It is obtained by interpolating the estimates $\hat{\beta}_0$ for different values of x_0 , as in the left panel of Figure 10.14. The right panel shows how the curve estimate depends on h . When h is large, $\hat{g}$ is too smooth to capture the pattern of the data, and hence it will be biased. When h is small, $\hat{g}$ follows the data better but wiggles implausibly and so has a high variance. The intermediate estimate shown in the left panel balances

bias and variance more satisfactorily. The choice of h is important and we discuss it below.

The dips in $\hat{g}$ towards the right of both panels suggest that local polynomial fits are sensitive to outliers. This is no surprise because they involve least squares estimation, which is non-robust. Some implementations robustify the fit, for example using the Huber estimator or discounting outliers by making the weights depend on the residuals $y_j - \hat{g}(x_j)$ from an initial fit. The popular iterative algorithm *lowess* takes this approach, though it uses nearest neighbours, weighting the proportion p of the data nearest to x_0 instead of using a fixed bandwidth. Typically $p = 2/3$ in standard implementations. Nearest neighbour fitting automatically allows for changes in the density of the x_j and hence reduces fluctuations like those at $x = 2$ in the right panel of Figure 10.14; it amounts to taking a locally-varying bandwidth.

Lowess stands for locally weighted scatterplot smoother.

Local polynomial estimators can be studied using ideas from least squares. The estimator of $g(x_0)$ is a weighted average of the y_j , that is

$$\hat{g}(x_0) = \hat{\beta}_0 = \sum_{j=1}^n S(x_0; x_j, h) y_j,$$

where the elements of the *effective kernel* $S(x_0; x_1, h), \dots, S(x_0; x_n, h)$ form the first row of the $(k+1) \times n$ matrix $(X^T W X)^{-1} X^T W$. This depends on the design matrix, the bandwidth, and x_0 , but not on y , so

$$E\{\hat{g}(x_0)\} = \sum_{j=1}^n S(x_0; x_j, h) g(x_j), \quad \text{var}\{\hat{g}(x_0)\} = \sigma^2 \sum_{j=1}^n S(x_0; x_j, h)^2. \quad (10.38)$$

The second expression here gives a finite-sample variance for $\hat{g}(x_0)$, provided that σ^2 is replaced by an estimate. One natural choice is

$$s^2(h) = \frac{1}{n - 2\nu_1 + \nu_2} \sum_{j=1}^n \{y_j - \hat{g}(x_j)\}^2, \quad (10.39)$$

where ν_1 and ν_2 are defined below. The corresponding estimator is unbiased when g is a polynomial of degree k , but otherwise is biased upwards. A simple way to reduce the bias is to construct $s^2(h)$ using a smaller h than that used for the curve estimate.

For theoretical and practical purposes it is fruitful to represent the smoothing operation in matrix form. The fitted values $\hat{g}(x_i)$ may be obtained by setting $x_0 = x_1, \dots, x_n$, and to each of these corresponds an effective kernel $S_{i1}(h), \dots, S_{in}(h)$, where $S_{ij}(h)$ is an abbreviation of $S(x_i; x_j, h)$. Let us stack these as an $n \times n$ *smoothing matrix* S_h . Then the vector of fitted values $\hat{g} = (\hat{g}(x_1), \dots, \hat{g}(x_n))^T$ may be written as $\hat{g} = S_h y$. There is an analogy with the hat matrix $H = X(X^T X)^{-1} X^T$ for a linear regression with fitted values $\hat{y} = Hy$. Unlike a hat matrix, however, S_h is not idempotent and it may be asymmetric. Any sensible smoothing operation will leave a constant unchanged, so $S_h 1_n = 1_n$. Equivalently any effective kernel sums to one.

Although kernel smoothers are most conveniently defined in terms of their bandwidth, it is useful to know their degrees of freedom when comparing their output.

In a standard linear model the trace of the hat matrix equals the number of parameters, suggesting by analogy that we define the degrees of freedom of the smoother to be $\nu_1 = \text{tr}(S_h)$. The analogy is not perfect, however, and alternatives such as $\nu_2 = \text{tr}(S_h^T S_h)$ have been proposed. Both ν_1 and ν_2 decrease as h increases, and $k \leq \nu_2 \leq \nu_1 \leq n$.

Three things must be chosen when implementing a local polynomial fit: the kernel function w , the degree k of the polynomial, and the bandwidth h . Experience shows that the choice of w is rarely important, though the best-looking curves are obtained when w descends smoothly to zero. Sharp-edged kernels such as the uniform density give rough and visually unappealing fits.

Choice of polynomial

It turns out to be sensible to take k odd, at least in theory. To see this in qualitative terms, consider the simplest form of local smoothing, with $k = 0$. Then the local polynomial is a constant and the least squares estimator of $g(x_0)$ is a weighted average, the *Nadaraya–Watson estimator*

$$\hat{\beta}_{0,NW} = \frac{\sum_{j=1}^n w_j y_j}{\sum_{j=1}^n w_j}. \quad (10.40)$$

This is a simple estimator, but not a very good one, because of its bias at the ends of the data. Figure 10.15 shows what happens when local constant and local linear estimators are fitted to data, shown noiseless for clarity. The left panels show that local linear and local constant fits give similar estimates when x_0 is central; the effective kernel is almost the same. In the right panels x_0 lies at the right-hand edge of the data, and as $\hat{\beta}_{0,NW}$ does not allow for $g'(x_0)$ it is badly biased upwards. The effective kernel for the local linear fit is smaller away from the boundary values of x_j , however, and so its bias is smaller. The fact that local polynomial fits adapt automatically to the presence of a boundary gives them a large practical advantage over many other approaches.

An asymptotic argument given at the end of this section, under which $n \rightarrow \infty$ and $h \rightarrow 0$ with $nh \rightarrow \infty$, shows that when x_0 is away from a boundary, both local constant and local linear fits have approximate bias and variance

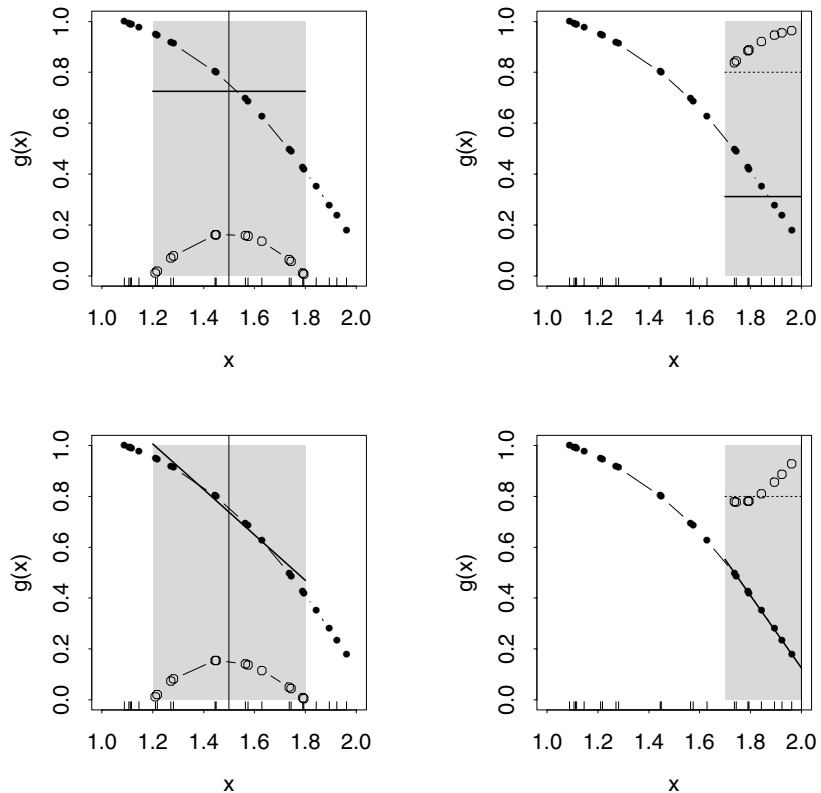
$$\frac{1}{2}h^2 g''(x_0), \quad \frac{\sigma^2}{nhf(x_0)} \int w(u)^2 du, \quad (10.41)$$

where $f(x)$ represents the limiting density of design points. The bias increases if $g(x)$ has high curvature or if h is large; see Figure 10.14. The bias has the desirable property of not depending on the pattern of design points, at least asymptotically. One interpretation of the variance formula is that the effective number of observations that contribute to the weighted average $\hat{\beta}_0$ is $nhf(x_0)/\int w(u)^2 du$, and this explains why we must have $nh \rightarrow \infty$ for a sensible asymptotic framework, while a small value of $f(x_0)$ — that is, few points near x_0 — decreases the precision with which $g(x_0)$ may be estimated.

When x_0 is near a boundary it may be shown that the bias of the local constant estimator increases to $O(h)$. The local linear estimator retains its $O(h^2)$ bias, however,

Geoffrey Stuart Watson (1921–1998) was educated in Melbourne, North Carolina and Cambridge, and held posts in Australia and North America, the last being at Princeton. He made important contributions to time series, to directional data analysis, and to mathematical biology. See Beran and Fisher (1998). E. A. Nadaraya (1935–) worked at Tbilisi State University. Both men published papers describing this estimator in 1964.

Figure 10.15 Local polynomial fitting by least squares. In each panel the function $g(x)$ is shown by a line joining the solid blobs (x_j, y_j) , shown without error for clarity, and the target value x_0 at which g is to be estimated is given by the vertical line; $x_0 = 1.5$ for the left panels and $x_0 = 2$ for the right panels. Only observations falling inside the shaded region contribute to the fit, and the effective kernel is shown by the circles; in the right panels the effective kernel has been shifted upwards by 0.8. The heavy solid lines show the local polynomial fits, which are constant in the upper panels and linear in the lower panels. The local constant fit is more biased than the local linear fit, especially at the edge $x_0 = 2$.



confirming what is suggested by Figure 10.15. Near a boundary both variances remain of order $(nh)^{-1}$. Thus in asymptotic terms the variance of the local linear estimator is no worse than that of the local constant estimator, while its bias is of smaller order. The same argument applies in general: whenever k is even, the variance of $\hat{\beta}_0$ is not increased asymptotically by fitting a polynomial of order $k + 1$, and the bias of $\hat{\beta}_0$ is reduced thereby.

Asymptotic arguments are useful backstops, but the theoretical benefits of higher-order fitting are outweighed by a finite-sample increase in variance. In practice local linear and quadratic polynomials are commonest, but local cubic curves are also sometimes fitted, particularly to data showing high-frequency variation.

Choice of smoothing parameter

One possibility is to choose the bandwidth h to minimize the asymptotic mean squared error

$$\frac{h^4}{4} g''(x_0)^2 + \frac{\sigma^2}{nhf(x_0)} \int w(u)^2 du$$

of $\hat{g}(x_0)$ with respect to h , giving a local bandwidth at x_0 . An overall bandwidth could be found using the integrated asymptotic mean squared error; see (7.7) and (7.8). Unfortunately both local and overall choices of h involve the unknowns g'' and f ,

and further bandwidths are needed to estimate them. A quagmire stands before us, and we must skirt it if we can.

A better, finite-sample, approach is to trade off the bias and variance when each observation is predicted from the rest, choosing h to minimize the cross-validation sum of squares

$$CV(h) = \sum_{j=1}^n \{y_j - \widehat{g}_{-j}(x_j)\}^2,$$

where $\widehat{g}_{-j}(x)$ is obtained by applying the smoother with bandwidth h but dropping the j th case from the data. At first sight it appears that n fits are needed to compute $CV(h)$, but as (Exercise 10.7.5)

$$y_j - \widehat{g}_{-j}(x_j) = \frac{y_j - \widehat{g}(x_j)}{1 - S_{jj}(h)}, \quad (10.42)$$

it turns out that

$$CV(h) = \sum_{j=1}^n \left\{ \frac{y_j - \widehat{g}(x_j)}{1 - S_{jj}(h)} \right\}^2$$

may be computed from components of the overall fit.

In practice $CV(h)$ must be minimized over a grid of values of h , so fast computation of the $S_{jj}(h)$ is essential. An alternative is to minimize the generalized cross-validation criterion

$$GCV(h) = \sum_{j=1}^n \left\{ \frac{y_j - \widehat{g}(x_j)}{1 - \text{tr}(S_h)/n} \right\}^2$$

instead. This replaces the individual $S_{jj}(h)$ by their average, which is found more speedily.

Cross-validation and information criteria are closely related, so it is not surprising that the values of h found by minimizing $CV(h)$ and $GCV(h)$ tend to be too small, analogous to the overfitting seen when AIC is used. In Section 8.7.3 we saw that for linear models this may be remedied by using the corrected information criterion AIC_c . For semiparametric models this suggests that we choose h to minimize $AIC_c(h)$, given by

$$AIC_c(h) = n \log \widehat{\sigma}^2(h) + n \frac{1 + \text{tr}(S_h)/n}{1 - \{\text{tr}(S_h) + 2\}/n}, \quad \widehat{\sigma}^2(h) = n^{-1} \sum \{y_j - \widehat{g}(x_j)\}^2,$$

where the number of regressors p in the linear model is replaced by $\text{tr}(S_h)$. Simulation shows that this procedure reduces the overfitting.

Semiparametric versions of other criteria also exist, but in practice an automatic choice of h should typically be used as a starting-point for investigation, rather than as a black box procedure.

Inference

At first sight it seems that approximate pointwise confidence intervals for $g(x_0)$ can be constructed using the mean and variance (10.38), simply by supposing that

$$\frac{\widehat{g}(x_0) - E\{\widehat{g}(x_0)\}}{\widehat{\text{var}}\{\widehat{g}(x_0)\}^{1/2}} \sim N(0, 1).$$

Unfortunately, however, this leads to confidence statements for $E\{\widehat{g}(x_0)\}$ rather than for the usual quantity of interest $g(x_0)$. If the bias $E\{\widehat{g}(x_0)\} - g(x_0)$ is substantial, the confidence interval may be centred in quite the wrong place. One way to deal with this is to correct the interval using a bias estimate, for example taken from (10.41) or found by comparing $\widehat{g}(x_0)$ and a less biased estimate obtained using a smaller bandwidth or higher k . These solutions tend to be complicated and can work poorly in small samples. A simpler approach ignores the bias issue and constructs a $(1 - 2\alpha)$ *variability band*, in which $\pm z_\alpha \widehat{\text{var}}\{\widehat{g}(x_0)\}^{1/2}$ is added to $\widehat{g}(x_0)$ to give an idea of its variability. In effect this is a confidence band for $E\{\widehat{g}(x_0)\}$.

Sometimes it is useful to construct an overall $(1 - 2\alpha)$ confidence band for g over the set $\mathcal{A}$ of x -values. This requires two curves $L(x)$ and $U(x)$ such that

$$\Pr\{L(x) \leq g(x) \leq U(x), x \in \mathcal{A}\} = 1 - 2\alpha,$$

which may be found by probability approximation to the distribution of $\sup_{x \in \mathcal{A}} |\widehat{g}(x) - g(x)| / \widehat{\text{var}}\{\widehat{g}(x)\}^{1/2}$. References to this are given in the bibliographic notes.

It is useful to have a means of testing the overall significance of an apparent departure from a given parametric form. Suppose that a non-local linear model has yielded the fitted values $\widehat{y} = Hy$, where H is the corresponding hat matrix, and that a more comprehensive, smooth, model has smoothing matrix S_h . Then it is natural to compute a P-value for departures from the non-local model using the ratio of the corresponding residual sums of squares

$$R(h) = \frac{\sum_{j=1}^n (y_j - \widehat{y}_j)^2}{\sum_{j=1}^n \{y_j - \widehat{g}(x_j)\}^2} = \frac{y^\top (I_n - H)y}{y^\top (I_n - S_h)^\top (I_n - S_h)y}.$$

If the observations are normal it turns out to be possible to compute good approximations to the corresponding P-value

$$p_{\text{obs}}(h) = \Pr_0\{R(h) \geq r_{\text{obs}}\}, \quad (10.43)$$

where r_{obs} is the observed value of $R(h)$ (Problem 10.13). It is useful to plot a *significance trace* of $p_{\text{obs}}(h)$ as a function of h , to assess the strength of evidence against the parametric model at different bandwidths.

Example 10.31 (Japanese earthquake data) The left panel of Figure 10.13 suggests that the magnitude of an earthquake might depend on the conditional intensity $\widehat{\lambda}_{\mathcal{H}}(t)$ just before it, with a larger release of energy and hence a bigger shock when the $\widehat{\lambda}_{\mathcal{H}}(t)$ is small. To assess this we fit a local linear smoother with tricube kernel

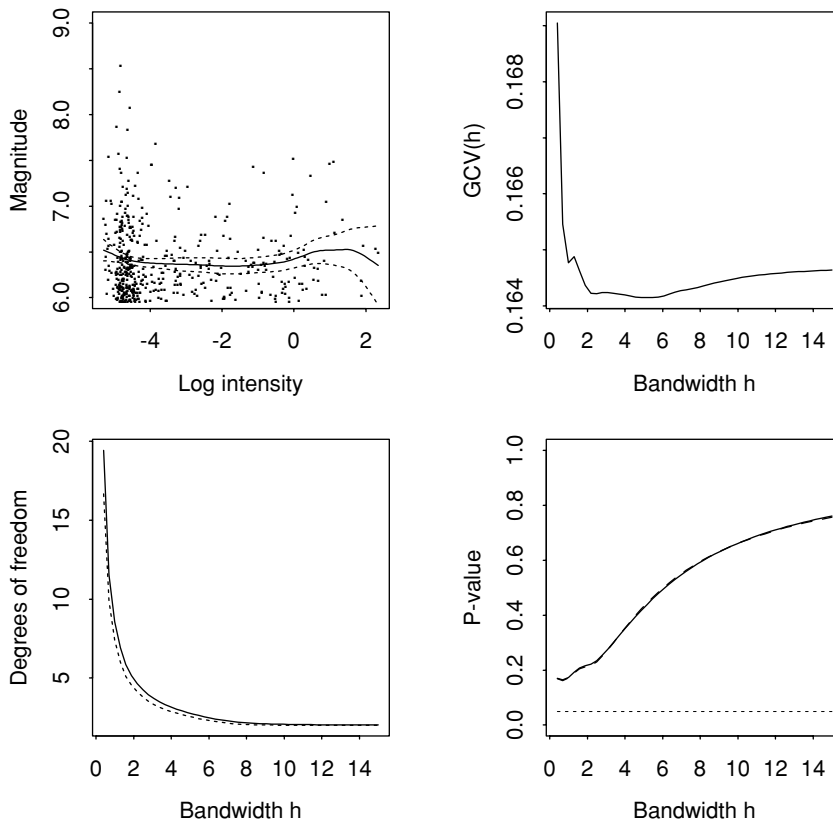


Figure 10.16 Smooth analysis of earthquake data. Upper left: local linear regression of magnitude on log intensity just before quake (solid), with 0.95 pointwise confidence bands (dots). Upper right: generalized cross-validation criterion $GCV(h)$ as a function of bandwidth h . Lower left: relation between degrees of freedom v_1 (solid), v_2 (dots), and h . Lower right: significance traces for test of no relation between magnitude and log intensity, based on chi-squared approximation (dots) and saddlepoint approximation (solid). The horizontal line shows the conventional 0.05 significance level.

(10.37) and bandwidth $h = 2$. The upper left panel of Figure 10.16 shows this fit, with a 0.95 pointwise confidence interval. The width of the interval increases at the boundaries and at the right of the panel, owing to the lower density of design points x_j . There is a suggestion of an increased magnitude at very low and high intensities, but the evidence is not compelling.

The upper right panel shows that although the cross-validation criterion $GCV(h)$ is minimized when $h \doteq 5.5$, the minimum changes little for $h > 2$. In the lower left panel we see that $h > 2$ corresponds to fitting curves with at most 4 or so equivalent degrees of freedom, while for $h > 8$ there are essentially two degrees of freedom, corresponding to straight-line regression. The corresponding plots for $AIC(h)$ and $AIC_c(h)$ decrease sharply and then tail off slowly, and also suggest that large bandwidths are appropriate.

The lower right panel of the figure shows two approximations to the significance trace for an overall test of no relation between log intensity and magnitude. The values of $p_{\text{obs}}(h)$ suggest that the evidence for such a relation varies from weak to non-existent. The approximations rest on the assumption that the data are normal. The large number of observations should mitigate the fact that this assumption is plainly incorrect, and it is unlikely to be critical, at least in this case.

These data show no relation between the fitted intensity just prior to an earthquake and its magnitude. This conclusion is of course very tentative, because seismological knowledge has not been incorporated. ■

Extensions

The locally weighted polynomial fit arises naturally from a modified form of likelihood. For if the ε_j were independent and normal, the contribution from (x_j, y_j) to the overall log likelihood for a polynomial fit of degree k centred at x_0 would be

$$\ell_j(\beta, \sigma; x_0) \equiv -\frac{1}{2\sigma^2} \{y_j - \beta_0 - \beta_1(x_j - x_0) - \cdots - \beta_k(x_j - x_0)^k\}^2 - \frac{1}{2} \log \sigma^2,$$

and $\hat{\beta}$ maximizes the *local log likelihood*

$$\ell(\beta, \sigma; x_0) = \sum_{j=1}^n \frac{1}{h} w \left(\frac{x_j - x_0}{h} \right) \ell_j(\beta, \sigma; x_0).$$

This idea extends fairly directly, for example to generalized linear and stochastic process models, using the appropriate log likelihood contribution and estimating $\hat{\beta}$ by iterative weighted least squares. The ideas described above then go through largely unchanged, though for a generalized linear model $\text{AIC}_c(h)$ must be changed to

$$\text{AIC}_c(h) = \sum_{j=1}^n d_j\{y_j; \hat{\mu}_j(h)\} + n \frac{1 + \text{tr}(S_h)/n}{1 - \{\text{tr}(S_h) + 2\}/n}, \quad (10.44)$$

where $d_j\{y_j; \hat{\mu}_j(h)\}$ is the deviance contribution from y_j when the fitted value is $\hat{\mu}_j(h)$. This is a large topic, to which the bibliographic notes give some entry points. The key ideas are summarized in the following example.

Example 10.32 (Toxoplasmosis data) Example 10.29 described how allowance might be made for the overdispersion of the data in Table 10.19, to which a logistic regression model with cubic dependence on rainfall was fitted. In view of the implausibility of the cubic model shown in the right panel of Figure 10.12, we consider local fitting with binomial probability $\pi(x) = \exp\{\theta(x)\}/[1 + \exp\{\theta(x)\}]$ depending on a local log odds $\theta(x)$. We fit a Taylor series expansion, $\theta(x) \doteq \beta_0 + \beta_1(x - x_0) + \cdots + \beta_k(x - x_0)^k/k!$, and take $\hat{\beta}_0$ as the estimate of $\theta(x_0)$.

The local log likelihood is

$$\ell(\beta; x_0, h) \equiv \sum_{j=1}^n \frac{w_j m_j}{\phi} \{y_j x_j^T \beta - \log(1 + e^{x_j^T \beta})\},$$

where m_j is the binomial denominator, $y_j = r_j/m_j$ is the observed proportion positive, $x_j^T = (1, x_j - x_0, \dots, (x_j - x_0)^k)$, and taking $\phi > 0$ will allow for overdispersion relative to the binomial model. The kernel function reduces the effective value of m_j to $m_j w_j$, so towns whose rainfall x_j is far from x_0 count for less in the estimation of β .

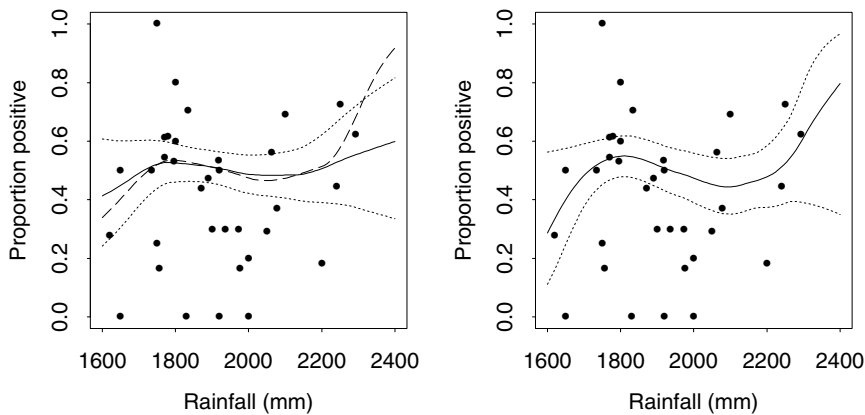


Figure 10.17 Local fits to the toxoplasmosis data. The left panel shows fitted probabilities $\hat{\pi}(x)$, with the fit of local linear logistic model with $h = 400$ (solid) and 0.95 pointwise confidence bands (dots). Also shown is the local linear fit with $h = 300$ (dashes). The right panel shows the local quadratic fit with $h = 400$ and its 0.95 confidence band. Note the increased variability due to the quadratic fit, and its stronger curvature at the boundaries.

The local score function may be written $X^T W u(\beta)$, to be compared with (10.18), and $\hat{\beta}$ is obtained by applying iterative weighted least squares to the binomial model with artificial denominators $w_j m_j$. A sandwich variance matrix $(X^T W_1 X)^{-1} X^T W_2 X (X^T W_1 X)^{-1}$ is required, where the j th elements of the diagonal matrices W_1 and W_2 are $w_j m_j \hat{\pi}_j (1 - \hat{\pi}_j)$ and $w_j^2 m_j \hat{\pi}_j (1 - \hat{\pi}_j)$, with $\hat{\pi}_j$ the fitted probabilities. The dispersion parameter ϕ does not appear in the local score equation, and plays no role in the estimation of β , in the effective kernel, in the smoothing matrix or the degrees of freedom. An estimator $\hat{\phi}$ is obtained by replacing the divisor $n - p$ in (10.20) by its counterpart $n - 2v_1 + v_2$, hence generalizing (10.39) to accommodate the binomial variance function.

Figure 10.17 shows linear and quadratic local fits and their 0.95 pointwise confidence bands, obtained with $h = 400$; the left panel also shows the fit with $h = 300$. The confidence bands for the quadratic fit are appreciably wider, and the fit itself is more curved, particularly at the boundaries. As might be expected, taking $h = 300$ gives a more locally adapted fit, whose effect is similar to increasing the order of the polynomial. All the fits are more plausible than the polynomial shown in Figure 10.12.

The confidence bands are appreciably narrower when no allowance is made for the overdispersion, and they suggest that the probability depends on rainfall. Overdispersion makes this much less plausible, and indeed a horizontal line would lie inside the bands in both panels of Figure 10.17. Any evidence for a relation between the probability and rainfall seems weak, though an analogue of (10.43) would be required for a more definite conclusion. ■

In some applications trigonometric or other expansions may be more appropriate than polynomial expansions; they too may be fitted locally using kernel or nearest neighbour weighting.

Similar ideas may be applied for smoothing in several dimensions, though the curse of dimensionality can then become heavy. It is useful to scale the covariates so that a common bandwidth can be used for them all, for example by using bandwidth $h s_r$ on the r th axis, where s_r is the standard deviation of the r th covariate.

Computation of bias and variance

This can be omitted on a first reading.

To express the lessons of Figure 10.15 in algebraic terms, we compute the mean and variance of $\hat{\beta}$. Taylor series expansion gives

$$\begin{aligned} g(x) &= g(x_0) + (x - x_0)g'(x_0) + \cdots + \frac{1}{k!}(x - x_0)^k g^{(k)}(x_0) \\ &\quad + \frac{1}{(k+1)!}(x - x_0)^{k+1} g^{(k+1)}(x) + \cdots \\ &= \beta_0 + (x - x_0)\beta_1 + \cdots + (x - x_0)^k \beta_k + b(x), \end{aligned}$$

say, where the final term is the remainder. Consequently

$$\begin{pmatrix} g(x_1) \\ g(x_2) \\ \vdots \\ g(x_n) \end{pmatrix} = \begin{pmatrix} 1 & (x_1 - x_0) & \cdots & (x_1 - x_0)^k \\ \vdots & \vdots & & \vdots \\ 1 & (x_n - x_0) & \cdots & (x_n - x_0)^k \end{pmatrix} \begin{pmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_k \end{pmatrix} + \begin{pmatrix} b(x_1) \\ b(x_2) \\ \vdots \\ b(x_n) \end{pmatrix},$$

or equivalently $g = X\beta + b$, where b is the $n \times 1$ vector whose j th element is $b(x_j)$. Let y , g , and ε represent the $n \times 1$ vectors whose j th elements are y_j , $g(x_j)$, and ε_j , and recall that the ε_j are independent with mean zero and variance σ^2 . Then $y = g + \varepsilon = X\beta + b + \varepsilon$, giving

$$\begin{aligned} E(\hat{\beta}) &= E\{(X^T W X)^{-1} X^T W (X\beta + b + \varepsilon)\} = \beta + (X^T W X)^{-1} X^T W b, \\ \text{var}(\hat{\beta}) &= \sigma^2 (X^T W X)^{-1} X^T W^2 X (X^T W X)^{-1}, \end{aligned}$$

Hence $\hat{\beta}$ has a bias that depends on the polynomial terms of degree $k + 1$ and higher. If $g(x)$ is indeed a polynomial of degree k or lower then $b = 0$ and $\hat{\beta}$ is unbiased.

For the local linear fit, $k = 1$, and the bias of $\hat{\beta}$ is

$$(X^T W X)^{-1} X^T W b = \left(\begin{array}{cc} \sum w_j & \sum w_j(x_j - x_0) \\ \sum w_j(x_j - x_0) & \sum w_j(x_j - x_0)^2 \end{array} \right)^{-1} \left(\begin{array}{c} \sum w_j b_j \\ \sum w_j(x_j - x_0) b_j \end{array} \right).$$

Hence the bias of $\hat{\beta}_0$ is

$$\frac{\sum w_j(x_j - x_0)^2 \sum w_j b_j - \sum w_j(x_j - x_0) \sum w_j(x_j - x_0) b_j}{\sum w_j \sum w_j(x_j - x_0)^2 - \{\sum w_j(x_j - x_0)\}^2}.$$

To approximate this, we suppose that the x_j are sufficiently dense to have a well-behaved smooth density, $f(x)$, let $n \rightarrow \infty$ and $h \rightarrow 0$ in such a way that $nh \rightarrow \infty$, and replace the sums by integrals. We then see, for example, that

$$\begin{aligned} \sum w_j(x_j - x_0)^2 &\doteq n \int \frac{1}{h} w\left(\frac{x - x_0}{h}\right) (x - x_0)^2 f(x) dx \\ &= nh^2 \int w(u) u^2 f(x_0 + hu) du \\ &\doteq nh^2 \int w(u) u^2 \{f(x_0) + hu f'(x_0) + \cdots\} du \\ &\doteq nh^2 f(x_0) + O(nh^4), \end{aligned}$$

on changing the variable of integration to $u = (x - x_0)/h$ and recalling that w has unit variance and is symmetric. This calculation presupposes that x_0 is sufficiently far from the boundary relative to h that the range of integration for integrals such as $\int w(u)u \, du$ is effectively infinite; otherwise odd powers of h do not vanish and the result is $anh^2 f(x_0) + O(nh^3)$, with $0 < a < 1$. Provided the odd terms do cancel, similar calculations give

$$\begin{aligned} \sum w_j b_j &\doteq \frac{1}{2} nh^2 f(x_0) g''(x_0), & \sum w_j (x_j - x_0) &\doteq nh^2 f'(x_0), \\ \sum w_j (x_j - x_0) b_j &\doteq O(nh^4), & \sum w_j &\doteq nf(x_0) \end{aligned} \quad (10.45)$$

and on putting the pieces together we find that $\hat{\beta}_0$ has bias whose leading term is $\frac{1}{2} h^2 g''(x_0)$. It turns out that the bias has order h^2 even when x_0 is near the boundary, but a similar calculation for the Nadaraya–Watson estimator (10.40) shows that its bias near the boundary is $O(h)$ (Exercise 10.7.6).

To get a handle on the variance of $\hat{\beta}_0$, it is simplest to orthogonalize X by replacing the j th element of its second column with $x_j - \bar{x}_w$, where $\bar{x}_w = \sum w_j x_j / \sum w_j$. In this parametrization the weighted least squares estimators are

$$\hat{\gamma} = \begin{pmatrix} \sum w_j & 0 \\ 0 & \sum w_j (x_j - \bar{x}_w)^2 \end{pmatrix}^{-1} \begin{pmatrix} \sum w_j y_j \\ \sum w_j (x_j - \bar{x}_w) y_j \end{pmatrix},$$

and $\hat{\beta}_0 = \hat{\gamma}_0 + (x_0 - \bar{x}_w) \hat{\gamma}_1$. This gives a simple explicit formula for $\hat{\beta}_0$, useful for numerical work. Its variance is

$$\text{var}(\hat{\beta}_0) = \sigma^2 \left[\frac{\sum w_j^2}{(\sum w_j)^2} + (x_0 - \bar{x}_w)^2 \frac{\sum w_j^2 (x_j - \bar{x}_w)^2}{\{\sum w_j (x_j - \bar{x}_w)^2\}^2} \right],$$

the first term of which equals the variance of the local constant estimator (10.40). It turns out that $x_0 - \bar{x}_w \doteq h^2 f'(x_0)/f(x_0)$ away from the boundary, and is $O(h)$ otherwise, and calculations like those above show that away from the boundary, both local linear and constant estimators have the approximate variance given in (10.41).

10.7.2 Roughness penalty methods

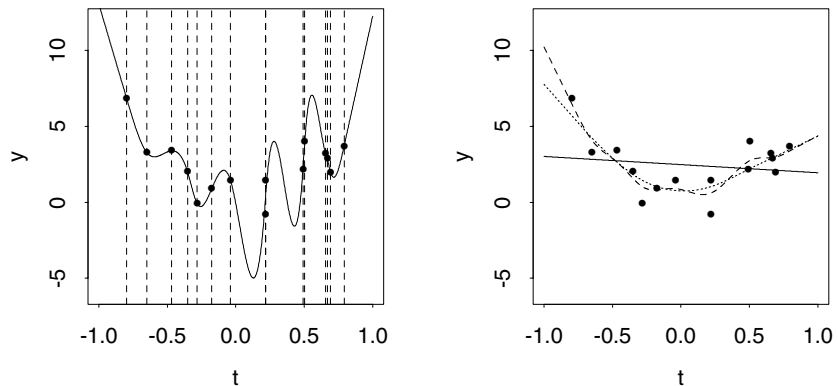
Local polynomial fitting is brought under the likelihood umbrella by local weighting of the log likelihood contributions. A different approach to curve estimation is based on fitting a family of flexible functions to the data, with the most appropriate of these specified indirectly by penalizing the roughness of the result. The idea is to fit a model with potentially as many parameters as there are observations, but to constrain these parameters to the extent desired.

To see how this might be done, we first consider suitably parametrized families of smooth functions. Let the data consist of pairs $(t_1, y_1), \dots, (t_n, y_n)$, where $a = t_0 < t_1 < \dots < t_n < t_{n+1} = b$. We seek a smooth summary $g(t)$ of how the response y depends on t over the interval $[a, b]$.

One approach is to use a *natural cubic spline* $g(t)$ with knots $t_1, \dots, t_n$. Such a function consists of separate cubic polynomials on each of the intervals $[t_1, t_2], \dots,$

We denote the covariate by t rather than x for ease of generalization below.

Figure 10.18 Natural cubic spline fits to $n = 15$ data pairs simulated from the model $y = 8x^2 + \varepsilon$. Left panel: fit with 15 degrees of freedom (solid) that interpolates the data, with values of t_j shown by the vertical dashed lines. Right panel: fits with degrees of freedom 2 (solid), 7 (dashes), and 3.7 (dots); the latter is chosen by cross-validation.



$[t_{n-1}, t_n]$, constrained to be continuous and to have continuous first and second derivatives at each knot. The spline is linear on the extreme intervals $[a, t_1]$ and $[t_n, b]$.

As there are $2 + 4(n - 1) + 2$ coefficients for these polynomial pieces and $3n$ constraints, just n numbers specify the spline. It turns out to be convenient to express it both in terms of its values $g^T = (g_1, \dots, g_n)$ at the knots and the second derivatives $\gamma^T = (\gamma_2, \dots, \gamma_{n-1})$ at $t_2, \dots, t_{n-1}$, where $g_j = g(t_j)$ and $\gamma_j = g''(t_j)$. The second derivatives at t_1 and t_n are zero, so $\gamma_1 = \gamma_n = 0$. In fact there exist $n \times (n - 2)$ and $(n - 2) \times (n - 2)$ matrices Q and R , depending only on $t_1, \dots, t_n$, such that $Q^T g = R\gamma$; R is positive definite and hence invertible, and both Q and R have simple structure that makes numerical work with them very efficient (Problem 10.14). Note that $\gamma = R^{-1} Q^T g$, so $g(t)$ is completely determined by its values at the knots.

An example is shown in the left panel of Figure 10.18. As outlined above, the spline $g(t)$ is linear outside (t_1, t_n) and cubic between the vertical lines that show the t_j , with smooth joins between cubic portions. One way to imagine this is that the spline adjusts to pass smoothly through beads — the y_j — that move on vertical wires fixed at the t_j .

Penalized log likelihood

Although perhaps a useful numerical summary of the data in Figure 10.18, the spline in the left panel is a poor statistical summary: we need a smoother fit. Suppose that $y_j = g(t_j) + \varepsilon_j$, where the ε_j are independent normal errors with common variance σ^2 , and $g(t)$ is a natural cubic spline with its n parameters $g^T = (g_1, \dots, g_n)$, where $g_j = g(t_j)$; let y denote the $n \times 1$ vector of observed responses. Maximization of the likelihood over g boils down to minimization of the sum of squares $(y - g)^T(y - g)$. This is achieved when $g_j = y_j$, clearly overfitted.

When a similar difficulty arose in our discussion of model selection (Section 4.7), we dealt with it by penalizing the log likelihood to account for model complexity, and we apply this idea here as well. If we judge that a straight line is the acme of simplicity, then one measure of the complexity of $g(t)$ over the interval $[a, b]$ is $\int_a^b \{g''(t)\}^2 dt$, which would be zero for a linear fit. Rather than maximize the usual log likelihood,

Here and below, $g''(t)$ is the second derivative of $g(t)$.

therefore, we maximize the *penalized log likelihood*

$$\ell_\lambda(g, \sigma^2) = \sum_{j=1}^n \log f\{y_j; g(t_j), \sigma^2\} - \frac{\lambda}{2\sigma^2} \int_a^b \{g''(t)\}^2 dt, \quad \lambda \geq 0. \quad (10.46)$$

This trades off the increase in the log likelihood term for more complex g against the second term, which penalizes nonlinearity. The extent of the trade-off is controlled by λ , a dimensionless quantity related to the degrees of freedom of the maximizing curve $\hat{g}_\lambda(t)$. When $\lambda = 0$, no penalty is applied and there are n degrees of freedom, corresponding to unconstrained variation of each element of the vector g . As $\lambda \rightarrow \infty$, the penalty becomes so large that $g(t)$ becomes a straight line, which has two degrees of freedom. Intermediate values give curves lying between these extremes. For now we suppose that λ is fixed, deferring discussion of how to choose it.

It turns out that when $g(t)$ is a natural cubic spline, the integral in (10.46) may be expressed as $\gamma^\top Q^\top g = g^\top Q R^{-1} Q^\top g = g^\top K g$, say, where K is a $n \times n$ symmetric matrix of rank $n - 2$ (Problem 10.14). For normal errors the j th log likelihood contribution is

$$\log f\{y_j; g(t_j), \sigma^2\} \equiv -\frac{1}{2}\sigma^{-2}\{y_j - g(t_j)\}^2 - \frac{1}{2}\log \sigma^2,$$

so $\hat{g}_\lambda(t)$ is determined by the vector $\hat{g}$ that minimizes the *penalized sum of squares*

$$(y - g)^\top (y - g) + \lambda g^\top K g \quad (10.47)$$

with respect to g . On completing the square we find that $\hat{g}$ minimizes

$$\{(I + \lambda K)^{-1} y - g\}^\top (I + \lambda K) \{(I + \lambda K)^{-1} y - g\}, \quad (10.48)$$

which differs from (10.47) only by a constant independent of g . It is straightforward to see that $\hat{g} = (I + \lambda K)^{-1} y$ is the unique natural cubic spline that minimizes (10.48); furthermore, as it turns out that it does so among all functions that are differentiable on $[a, b]$ and have absolutely continuous first derivative, $\hat{g}$ is optimal in a large class of smooth functions.

The structure of the matrix K can be exploited to give a fast algorithm for fitting the spline. Recall that $\gamma = R^{-1} Q^\top g$ and $K = Q R^{-1} Q^\top$. Hence $\hat{g}$ is the solution to $(I + \lambda Q R^{-1} Q^\top)g = y$. Equivalently the corresponding $\hat{\gamma}$ solves the system $(R + \lambda Q^\top Q)\gamma = Q^\top y$, which can be solved in $O(n)$ operations because both R and Q are band matrices; their only non-zero elements lie on the diagonal or just above and below it.

Like a local polynomial fit, the spline is a linear smoother. Its smoothing matrix is $S_\lambda = (I + \lambda K)^{-1}$. Note that $Kg = 0$ for vectors of form $g = \beta_0 1_n + \beta_1(t_1, \dots, t_n)^\top$, because the roughness penalty is zero when $g(t)$ is linear, and it follows that $S_\lambda g = g$ for such vectors. Once again there are several definitions of the degrees of freedom for the smooth fit, the most obvious being $\text{tr}(S_\lambda)$.

The right panel of Figure 10.18 shows three fits, of which the linear fit is evidently too smooth, and the one with seven degrees of freedom too rough. The fit with

3.7 degrees of freedom, chosen by cross-validation as described below, seems more plausible.

For later development we must deal with two complications. The first arises when weights w_j are attached to the cases (t_j, y_j) . Then $(y - g)^T(y - g)$ must be replaced by $(y - g)^T W(y - g)$, where $W = \text{diag}\{w_1, \dots, w_n\}$; see Section 8.2.4. The second occurs when some of the t_j are tied. If so, we let $s_1 < \dots < s_q$ denote the ordered distinct values among $t_1, \dots, t_n$ and denote by N the $n \times q$ incidence matrix whose (j, k) element indicates whether $t_j = s_k$; obviously $q \geq 2$, and $N = I$ if the t_j are distinct and ordered. With these changes (10.47) alters to

$$(y - Ng)^T W(y - Ng) + \lambda g^T K g,$$

which is minimized at $\hat{g} = N(N^T W N + \lambda K)^{-1} N^T W y$. The smoothing matrix $S_\lambda = N(N^T W N + \lambda K)^{-1} N^T W$ reduces to the previous expression when $W = N = I$.

How much smoothing?

The smoothing parameter λ plays the same role as the bandwidth in a local polynomial model, and it too is typically chosen by minimizing information or cross-validation criteria such as

$$\text{CV}(\lambda) = \sum_{j=1}^n \left\{ \frac{y_j - \hat{g}(t_j)}{1 - S_{jj}(\lambda)} \right\}^2, \quad \text{GCV}(\lambda) = \sum_{j=1}^n \left\{ \frac{y_j - \hat{g}(t_j)}{1 - \text{tr}(S_\lambda)/n} \right\}^2,$$

where both the diagonal elements $S_{jj}(\lambda)$ of the smoothing matrix S_λ and the fitted values $\hat{g}(t_j)$ depend on λ . As with other applications of smoothing, the goal is to trade off fidelity to the data against smoothness of the fit. Once again a caveat is needed: the results from an automatic procedure cannot always be trusted, and it is often valuable to apply different levels of smoothing. As mentioned above, it is useful to know the degrees of freedom of a smooth fit.

Example 10.33 (Spring barley data) Table 10.21 gives standardized yields from an agricultural field trial in which three blocks of long narrow plots were sown with 75 varieties of spring barley in a random order within each block. The yield from variety 27 in the third block is missing, but otherwise there are three replicates for each variety. The plot of the yields in the left panel of Figure 10.19 shows strong spatial patterns owing to fertility trends within each block, in addition to the variety effects. For the moment we ignore differences among the varieties, and illustrate how fitting a natural cubic spline can account for the fertility gradient in the first block.

The left panel of Figure 10.19 shows some of the disadvantages of polynomial fitting. The lower curve, for example, wiggles implausibly compared to a spline fit with the same degrees of freedom, shown in the upper right panel. The lower right panel shows how $\text{CV}(\lambda)$ and $\text{GCV}(\lambda)$ vary with the equivalent degrees of freedom, for the three blocks. The fit to block 2 seems fairly reasonable, but block 3 is evidently overfitted with 40 degrees of freedom, and block 1 is probably also overfitted. We reconsider these data in Example 10.35. ■

Location <i>t</i>	Block 1		Block 2		Block 3	
	Variety	Yield <i>y</i>	Variety	Yield <i>y</i>	Variety	Yield <i>y</i>
1	57	9.29	49	7.99	63	11.77
2	39	8.16	18	9.56	38	12.05
3	3	8.97	8	9.02	14	12.25
4	48	8.33	69	8.91	71	10.96
5	75	8.66	29	9.17	22	9.94
6	21	9.05	59	9.49	46	9.27
7	66	9.01	19	9.73	6	11.05
8	12	9.40	39	9.38	30	11.40
9	30	10.16	67	8.80	16	10.78
10	32	10.30	57	9.72	24	10.30
11	59	10.73	37	10.24	40	11.27
12	50	9.69	26	10.85	64	11.13
13	5	11.49	16	9.67	8	10.55
14	23	10.73	6	10.17	56	12.82
15	14	10.71	47	11.46	32	10.95
16	68	10.21	36	10.05	48	10.92
17	41	10.52	64	11.47	54	10.77
18	1	11.09	63	10.63	37	11.08
19	64	11.39	33	11.03	21	10.22
20	28	11.24	74	10.85	29	10.59
21	46	10.65	13	11.35	62	11.35
22	73	10.77	43	10.25	5	11.39
23	37	10.92	3	10.08	70	10.59
24	55	12.07	53	10.25	13	11.26
25	19	11.03	23	9.57	11	11.79
26	10	11.64	62	11.34	44	12.25
27	35	11.37	52	10.19	36	12.23
28	26	10.34	12	10.80	52	10.84
29	17	9.52	2	10.04	60	10.92
30	71	8.99	32	9.69	68	10.41
31	8	8.34	22	9.36	3	10.96
32	62	9.25	42	9.43	19	9.94
33	44	9.86	72	11.46	67	11.27
34	53	9.90	73	9.29	59	11.79
35	74	11.04	25	10.10	2	11.51
36	20	10.30	45	9.53	75	11.64
37	56	11.56	15	10.55	27	—
38	29	9.69	35	11.34	43	9.78
39	2	10.68	66	11.36	51	8.86
40	47	10.91	5	10.88	10	10.28
41	11	10.05	56	11.61	35	12.15
42	38	10.80	46	10.33	74	10.36
43	65	10.06	71	10.53	66	9.59
44	13	10.04	51	8.67	34	10.53
45	31	10.50	21	9.56	18	11.26
46	40	9.51	1	9.95	50	10.37
47	4	9.20	31	11.10	42	10.10
48	67	9.74	11	10.11	1	9.95
49	22	8.84	41	9.36	58	9.80
50	49	9.33	61	10.23	26	10.58
51	58	9.51	55	11.38	41	9.31
52	43	9.35	14	11.30	25	9.29

Table 10.21 Spring barley data (Besag *et al.*, 1995). Spatial layout and plot yield at harvest *y* (standardized to have unit crude variance) in a final assessment trial of 75 varieties of spring barley. The varieties are sown in three blocks, with each variety replicated thrice in the design. The yield for variety 27 is missing in the third block.

Table 10.21 (cont.)

Location t	Block 1		Block 2		Block 3	
	Variety	Yield y	Variety	Yield y	Variety	Yield y
53	7	9.01	44	10.90	33	10.03
54	25	10.58	34	10.97	9	9.49
55	61	11.03	54	12.22	17	11.52
56	16	9.89	24	10.10	57	12.24
57	52	11.39	4	11.22	65	11.64
58	70	11.24	65	10.01	49	10.74
59	34	12.18	75	10.29	73	10.29
60	42	10.21	38	10.95	7	10.25
61	24	11.08	17	9.66	23	11.39
62	33	11.05	68	9.31	72	13.34
63	51	10.29	7	8.84	55	12.73
64	60	10.57	27	10.64	31	12.62
65	69	10.42	58	9.45	39	10.19
66	15	10.49	48	9.66	47	11.61
67	6	10.00	28	9.85	15	10.52
68	63	9.23	60	9.24	20	9.07
69	54	10.57	30	10.11	61	10.76
70	18	10.27	70	9.63	28	9.91
71	45	8.86	20	9.04	53	10.17
72	72	9.45	9	8.43	69	8.68
73	9	8.03	40	10.97	45	8.74
74	36	9.22	50	8.98	12	9.15
75	27	8.70	10	9.88	4	9.39

10.7.3 More general models

We now consider how the discussion above should be modified when there are explanatory variables as well as a smooth variable, treating certain covariates nonparametrically and others not, and allowing the response to have a density other than the normal.

Let the data consist of independent triples $(x_1, t_1, y_1), \dots, (x_n, t_n, y_n)$, with j th log likelihood contribution $\ell_j(\eta_j, \kappa)$, where $\eta_j = x_j^T \beta + g(t_j)$; for now we suppress dependence on κ . Then the analogue of (10.47) is the *penalized log likelihood*

$$\ell_\lambda(\beta, g) = \sum_{j=1}^n \ell_j(\eta_j) - \frac{1}{2} \lambda \int_a^b \{g''(t)\}^2 dt, \quad \lambda > 0, \quad (10.49)$$

where a and b are chosen so that $a < t_1, \dots, t_n < b$. If all the t_j are distinct and $\lambda = 0$, the maximum is obtained by choosing $g_j = g(t_j)$ to maximise the j th log likelihood contribution, but this is not useful because the resulting model has n parameters and is too rough. The integral in (10.49) penalizes roughness of $g(t)$, so λ has the same interpretation as before.

If the ordered distinct values of $t_1, \dots, t_n$ are $s_1 < \dots < s_q$ and if $g(t)$ is a natural cubic spline with knots at the s_i , then the integral in (10.49) may be written $g^T K g$, where the $q \times 1$ vector g has i th element $g_i = g(s_i)$. Given a value of λ , our aim

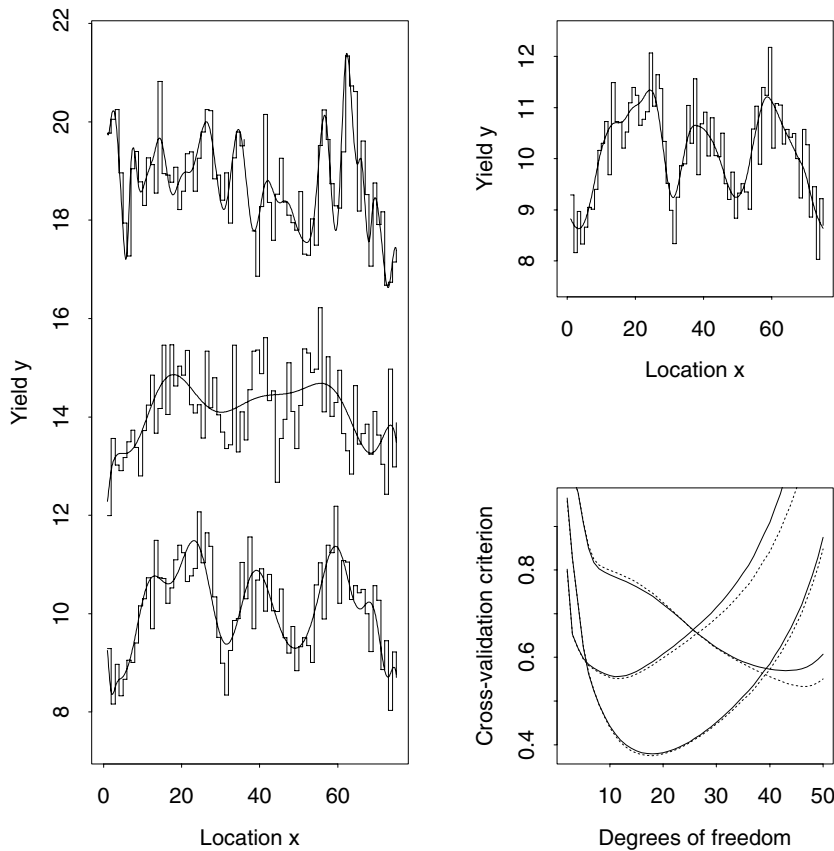


Figure 10.19 Spring barley data analysis. Left panel: yield y as a function of location x for the three blocks. Yields for blocks 2, 3 have been offset by adding 4, 8 respectively. The smooth solid lines are the fits of polynomials of degree 20, 10 and 40 to the data from blocks 1, 2 and 3. Upper right: yields for block 1, with smoothing spline fit with 18 degrees of freedom. Lower right: cross-validation (solid) and generalized cross-validation (dots) criteria for smoothing spline fits to blocks 1, 2 and 3, with minima at roughly 20, 10 and 40 equivalent degrees of freedom.

is to find values of β and g that maximize $\ell_\lambda(\beta, g)$. As the $n \times 1$ vector of linear predictors may be written $\eta = X\beta + Ng$, where N is the $n \times q$ incidence matrix for the elements of g and the $n \times p$ matrix X has j th row $x_j^\top$, the score equations are

$$\begin{pmatrix} \partial \ell_\lambda(\hat{\beta}, \hat{g}) / \partial \beta \\ \partial \ell_\lambda(\hat{\beta}, \hat{g}) / \partial g \end{pmatrix} = \begin{pmatrix} X^\top u(\hat{\eta}) \\ N^\top u(\hat{\eta}) - \lambda K \hat{g} \end{pmatrix} = 0, \quad (10.50)$$

where the $n \times 1$ vector $u(\eta)$ has j th element $\partial \ell_j(\eta_j) / \partial \eta_j$. The usual Taylor series expansion (Sections 4.4.1, 10.2.2) then gives

$$\begin{pmatrix} X^\top W X & X^\top W N \\ N^\top W X & N^\top W N + \lambda K \end{pmatrix} \begin{pmatrix} \hat{\beta} \\ \hat{g} \end{pmatrix} \doteq \begin{pmatrix} X^\top W z \\ N^\top W z \end{pmatrix}, \quad (10.51)$$

where $W = \text{diag}\{w_1, \dots, w_n\}$, with $w_j = E\{-\partial^2 \ell_j(\eta_j) / \partial \eta_j^2\}$ and where $z = W^{-1}u(\eta) + \eta$ is the $n \times 1$ adjusted dependent variable. Fisher scoring would solve (10.51) iteratively starting from suitable initial values of β and g , but here there are $p + q$ regressors, where q is typically comparable to n , and an approach known as *backfitting* is generally used instead. The idea is to alternate between the two matrix

equations in (10.51), rewritten as

$$(X^T W X) \hat{\beta} \doteq X^T W (z - N \hat{g}), \quad (10.52)$$

$$(N^T W N + \lambda K) \hat{g} \doteq N^T W (z - X \hat{\beta}). \quad (10.53)$$

Given initial values β_0 and g_0 of β and g , we calculate η , W , and z , replace $\hat{g}$ in (10.52) by g_0 , and then obtain an approximate value of $\hat{\beta}$ by regressing $z - N g_0 = W^{-1}u + X \beta_0$ on the columns of X with weights W . We then recalculate $\eta = X \hat{\beta} + N g_0$, W , and z , and solve (10.53) by applying the matrix $(N^T W N + \lambda K)^{-1} N^T W$ to $z - X \hat{\beta} = W^{-1}u + N g_0$, thus obtaining an approximate value of $\hat{g}$. We then set β_0 and g_0 equal to $\hat{\beta}$ and $\hat{g}$ and iterate the cycle to convergence.

Example 10.34 (Partial spline model) The case where y_j is normal with mean η_j and variance σ^2 is known as a *partial spline model*. Here $u(\eta) = \sigma^{-2}(y - \eta)$ and $W = \sigma^{-2}I_n$, so $z = y$ (Example 10.4).

The first backfitting step is least squares regression of $y - N g_0$ on the columns of X . The second applies the linear smoother $(N^T N + \sigma^2 \lambda K)^{-1} N^T$ to the residual $y - X \hat{\beta}$ from the first step; the effective penalty is thus $\lambda' = \sigma^2 \lambda$. At each step of the iteration either least squares or linear smoothing is applied to the residual from the previous operation, continuing until any systematic structure has been removed from both $y - X \hat{\beta}$ and $y - N \hat{g}$. ■

Example 10.36 gives a further illustration of such fitting.

Backfitting yields parameter estimates, but unless the fit is purely exploratory other quantities are required for inference. The deviance is defined in the usual way, and the error degrees of freedom of a fit are

$$n - \nu_\lambda = n - \text{tr}(S_\lambda) - \text{tr}[\{X^T W (I - S_\lambda) X\}^{-1} X^T W (I - S_\lambda)^2 X],$$

where this and the smoothing matrix

$$S_\lambda = N(N^T W N + \lambda K)^{-1} N^T W$$

are computed at convergence. The usual chi-squared theory is often used for approximate comparison of nested models, even though it has no firm theoretical basis. Standard errors for elements of $\hat{\beta}$ and the fitted $\hat{g}(t)$ are generally obtained using the approximate linearization entailed by (10.51).

As usual in semiparametric modelling, the degree of smoothing is critical. Various forms of the cross-validation and generalized cross-validation criteria for choice of λ have been suggested. One approach takes

$$\text{CV}(\lambda) = \sum_{j=1}^n \frac{(y_j - \hat{y}_j)^2}{v_j \{1 - h_{jj}(\lambda)\}^2}, \quad \text{GCV}(\lambda) = \sum_{j=1}^n \frac{(y_j - \hat{y}_j)^2}{v_j \{1 - \text{tr}(H_\lambda)/n\}^2},$$

where v_j is the estimated variance of y_j , $\hat{y}_j$ is the j th fitted value, and $h_{jj}(\lambda)$ is the partial derivative of $\hat{y}_j$ with respect to y_j , analogous to the earlier role of $S_{jj}(h)$. Another possibility is to cross-validate the approximating linear problem obtained at each step of the fitting algorithm.

When several covariates might be treated nonparametrically, say t , u , and v , we can take the linear predictor to be $x^\top \beta + g_1(t) + g_2(u) + g_3(v)$ and fit smoothing splines or other nonparametric curves with a version of backfitting in which β and the g s are iteratively estimated in succession. Such a setup is known as a *generalized additive model*, to which the same ideas apply as outlined above. The degrees of smoothing are controlled by separate penalties for each of the g s, and the corresponding λ s may be estimated by minimizing a cross-validation or similar criterion. Surfaces $g(t, u)$ can also be fitted using similar ideas.

In some cases it is necessary to diminish the computational burden by reducing the number of knots and hence the number of parameters q that specify the fitted curve. Although the resulting fit no longer has the optimality properties of the natural cubic spline, this is typically unimportant in practice.

Example 10.35 (Spring barley data) In addition to their strong spatial dependence, the spring barley yields in Table 10.21 depend on variety effects. The simplest model that would accomodate these is a two-way layout with variety and block effects, in which the response is

$$y_{vb} = \tau_b + \beta_v + \varepsilon_{vb}, \quad v = 1, \dots, 75, b = 1, 2, 3, \quad \text{where } \varepsilon_{vb} \stackrel{\text{iid}}{\sim} N(0, \sigma^2).$$

This has residual sum of squares 94.87 on 147 degrees of freedom, giving $\hat{\sigma}^2 = 0.645$, while the standard error for a difference of variety effects $\hat{\beta}_{v_1} - \hat{\beta}_{v_2}$ is 0.655.

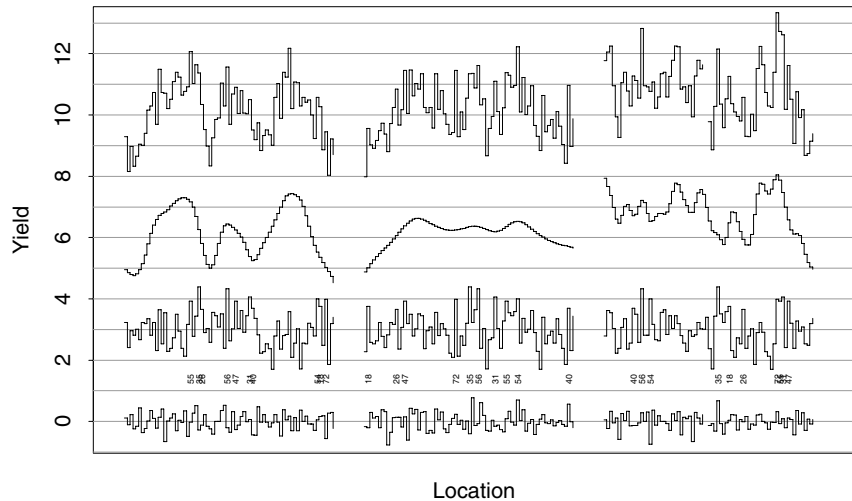
As the two-way layout ignores the spatial variation, it greatly overestimates σ^2 , thereby decreasing the sensitivity of comparisons among the varieties. Moreover the variety effect estimators may be biased if all three replicates of a particular variety happen to fall where the fertilities are higher than average. It seems more sensible to fit a model in which the yield for the v th variety in the b th block depends on its location t_{vb} through

$$y_{vb} = g_b(t_{vb}) + \beta_v + \varepsilon_{vb}, \quad v = 1, \dots, 75, b = 1, 2, 3,$$

where $g_b(t)$ is a smooth function that determines how the fertility pattern in block b depends on location t . When this model is fitted using smoothing splines with 40 knots for each of the g_b , 77 degrees of freedom are needed to account for the variety and block effects, the degrees of freedom that minimize the generalized cross-validation criterion are 16.4, 8.3, and 25.2 for $b = 1, 2$, and 3, the residual sum of squares is 16.85 on $224 - 77 - 16.4 - 8.3 - 25.3 = 97$ degrees of freedom, and $\hat{\sigma}^2 = 0.174$, about one quarter of the value for the two-way layout. The standard errors for differences of variety effects are roughly 0.41, so more precise comparisons are possible than in the simpler fit. Fewer degrees of freedom are needed to model the spatial variation here than the 20, 10, and 40 required for the three blocks in Example 10.33, because allowance for variety effects enables smoother fertility trends to be used.

Figure 10.20 shows how this model decomposes the original data into fertility trends, variety effects, and residuals. As their degrees of freedom would suggest, the estimate $\hat{g}_2(t)$ is appreciably smoother than the fertility trends in blocks 1 and 3. The best-yielding varieties in decreasing order of $\hat{\beta}_v$ are 35, 56, 31, 54, 72, 55, 47, 18,

Figure 10.20 Spring barley data analysis. Block 1 is shown on the left and block 3 on the right. The panel shows, from the top, the original yields y , the fertility trend and variety effect estimates $\hat{g}_b(t)$ and $\hat{\beta}_v$, both offset for display, and the crude residuals. The varieties with the ten largest $\hat{\beta}_v$ are marked.



40, and 26, but as a 0.95 confidence interval for differences of two variety effects has width 1.61, there is no clear-cut best variety.

A probability plot of the residuals shows nothing to undermine normality of the errors, but the correlograms and partial correlograms of residuals from blocks 1 and 3 show slight negative correlations, suggesting that $\hat{g}_1(t)$ and $\hat{g}_3(t)$ may be overfitted. A more complete analysis would try and remedy this by refitting the model with fewer degrees of freedom for the fertility trends in blocks 1 and 3. ■

Exercises 10.7

- 1 Explain how the derivatives $g'(x_0), \dots, g^{(k)}(x_0)$ may be estimated using the least squares estimator $\hat{\beta}$ from a local polynomial fit of degree k .
- 2 What is the bias of the local fit of a polynomial of degree k to a function that is polynomial of degree $l \leq k$? How would you measure the disadvantage of this relative to an unweighted fit?
- 3 By writing $\sum \{y_j - \hat{g}(x_j)\}^2 = (y - \hat{g})^T (y - \hat{g})$ and recalling that $y = g + \varepsilon$ and $\hat{g} = Sy$, where S is a smoothing matrix, show that

$$E \left[\sum_{j=1}^n \{y_j - \hat{g}(x_j)\}^2 \right] = \sigma^2(n - 2v_1 + v_2) + g^T(I - S)^T(I - S)g.$$

Hence explain the use of $s^2(h)$ as an estimator of σ^2 . Under what circumstances is it unbiased?

- 4 (a) If $S_{11}(h), \dots, S_{nn}(h)$ are the leverages of a smoothing matrix S_h , establish

$$v_1 = \sum_{j=1}^n S_{jj}(h), \quad v_2 = \sum_{i,j=1}^n S_{ij}(h)^2 = \sigma^{-2} \sum_{j=1}^n \text{var} \{ \hat{g}(x_j) \}.$$

(b) Show that $S(x_j; x_j, h)$ is proportional to the $(1, 1)$ element of $(X^T W X)^{-1}$, and let the influence function of the smoother be $I(x) = w(0)e_1^T(X^T W X)^{-1}e_1$, where $e_1^T = (1, 0, \dots, 0)$ has length $k + 1$. Show that

$$\sigma^{-2} \text{var} \{ \hat{g}(x) \} \leq I(x),$$

and deduce that $v_2 \leq v_1$.

(c) Let $I_1(x)$ and $I_2(x)$ be the influence functions corresponding to bandwidths $h_1 < h_2$, and let the corresponding weight matrices be W_1 and W_2 . Show that $W_1 \leq W_2$ componentwise and deduce that $I_2(x) \geq I_1(x)$.

- 5 Consider a linear smoother with $n \times n$ smoothing matrix S_h , so $\hat{g} = S_h y$, and show that the function $a_j(u)$ giving the fitted value at x_j as a function of the response u there satisfies

$$a_j(u) = \begin{cases} \hat{g}(x_j), & u = y_j, \\ \hat{g}_{-j}(x_j), & u = \hat{g}_{-j}(x_j). \end{cases}$$

Explain why this implies that $S_{jj}(h)\{y_j - \hat{g}_{-j}(x_j)\} = \hat{g}(x_j) - \hat{g}_{-j}(x_j)$, and hence obtain (10.42).

- 6 (a) Check (10.45), and hence verify that $E(\hat{\beta}_0) - g(x_0) \doteq \frac{1}{2}h^2 g''(x_0)$ far from a boundary. Do the corresponding calculation for x_0 near a boundary.
(b) Show that the bias of the Nadaraya–Watson estimator (10.40) may be expressed as

$$\frac{\sum w_j \left\{ (x_j - x_0)g'(x_0) + \frac{1}{2}(x_j - x_0)^2 g''(x_0) + \cdots \right\}}{\sum w_j},$$

and deduce that this is approximately $hg'(x_0)a$ near a boundary, where $a \neq 0$, and $\frac{1}{2}h^2\{g''(x_0) + f'(x_0)g'(x_0)\}$ elsewhere.

- 7 Develop the details of local likelihood smoothing when a linear polynomial is fitted to Poisson data, using link function $\log \mu = \beta_0 + \beta_1(x - x_0)$.

10.8 Survival Data

10.8.1 Introduction

Survival or event history analysis concerns the times of events. Such data are particularly common in medicine and the social sciences, but also arise in many other domains. As we saw in Section 5.4, the responses may be incompletely observed owing to censoring or truncation. Here we give an introduction to regression analysis of such data in the simplest and most common situation, with just one event per individual. Throughout we use the term ‘failure’ to describe the event of interest, and refer to the time to failure as a survival time.

Let the data available on the j th of n independent individuals be (x_j, y_j, d_j) , where x_j is a $p \times 1$ vector of explanatory variables, $d_j = 1$ indicates that y_j is an observed survival time, and $d_j = 0$ indicates that the survival time is right-censored at y_j . Consider a parametric model under which the survival time has density $f(y; x, \beta)$, survivor function $\mathcal{F}(y; x, \beta) = 1 - F(y; x, \beta)$, and hazard and cumulative hazard functions $h(y; x, \beta)$ and $H(y; x, \beta) = -\log \mathcal{F}(y; x, \beta)$. Assume that the censoring mechanism is uninformative, that is, independent of the failure time and uninformative about β . Then the discussion in Section 5.4 implies that the log likelihood may be written as

$$\ell(\beta) = \sum_{j=1}^n \{d_j \log h(y_j; x_j, \beta) - H(y_j; x_j, \beta)\}, \quad (10.54)$$

from which maximum likelihood estimates $\hat{\beta}$ may be obtained by iterative weighted

least squares. Any additional parameters ϕ may be estimated by interleaving updates to $\hat{\beta}$ and $\hat{\phi}$. As usual with incomplete data, confidence intervals should be based on observed information, as in Example 4.47.

Residuals are important for model checking. The relation $F(y; x, \beta) = 1 - \exp\{-H(y; x, \beta)\}$ implies that if Y is continuous and uncensored with distribution function $F(y; x, \beta)$, then $H(Y; x, \beta)$ is exponentially distributed with unit mean. This suggests that for diagnostic purposes the *Cox–Snell residuals* $H(y_j; x, \hat{\beta})$ can be regarded as an exponential random sample. For observations censored at c , we argue that as $E\{H(Y; x, \beta) \mid Y > c\} = H(c; x, \beta) + 1$, an appropriate residual is $H(c; x, \hat{\beta}) + 1$. This yields modified Cox–Snell residuals

$$r_j = H(y_j; x, \hat{\beta}) + 1 - d_j. \quad (10.55)$$

Other residuals may be defined for the proportional hazards model, described below. Case diagnostics discussed in Section 10.2.3 may also be useful.

Accelerated life models

One notion used particularly in reliability studies is that time to failure may be accelerated or retarded relative to some baseline. Let Y and Y_0 denote failure times for individuals with covariates x and $x = 0$. Then the accelerated life model posits the existence of a positive function $\tau(\beta; x)$ such that Y and $\tau(\beta; x)Y_0$ have the same distribution; equivalently $Y/\tau(\beta; x) \stackrel{D}{=} Y_0$, a baseline random variable. An individual with $\tau(\beta; x) < 1$ will ‘wear out’ at a faster rate than the baseline, and conversely. If Y_0 has survivor, density, and hazard functions $\mathcal{F}_0(y)$, $f_0(y)$, and $h_0(y)$, the corresponding functions for Y are

$$\mathcal{F}_0\{y/\tau(\beta; x)\}, \tau(\beta; x)^{-1}f_0\{y/\tau(\beta; x)\}, \tau(\beta; x)^{-1}h_0\{y/\tau(\beta; x)\}. \quad (10.56)$$

This is a scale model, so obvious possibilities are to let Y_0 be an exponential, gamma, Weibull, log-normal, or log-logistic variable. If $\tau(\beta; x) = \exp(x^T \beta)$, then $\log Y \stackrel{D}{=} x^T \beta + \varepsilon$, where $\varepsilon = \log Y_0$. The regression-scale model $x^T \beta + \sigma \varepsilon$ is equivalent to taking $(Y/\tau)^{1/\sigma} \stackrel{D}{=} Y_0$. Any of these gives a linear model for the log responses, and if there is no censoring this can be fitted using least squares, though typically information will be lost by doing so. However there is no special difficulty with maximum likelihood estimation by iterative weighted least squares, if the density of ε or equivalently of Y_0 is known.

Example 10.36 (Leukaemia data) Table 10.22 contains data on the survival of acute leukaemia victims. The covariate x is $\log_{10}$ white blood cell count at time of diagnosis, and the patients are grouped according to the presence or not of a morphologic characteristic of their white blood cells. Within each group suppose that survival time Y is exponential with mean $\tau = \exp(\eta)$, where $\eta = \beta_0 + \beta_1 I(\text{Group} = 1) + \beta_2 x$. This is a generalized linear model with gamma errors, log link function, and dispersion parameter $\phi = 1$.

After Cox and Snell
(1968).

Group 1						Group 2					
	<i>x</i>	<i>y</i>		<i>x</i>	<i>y</i>		<i>x</i>	<i>y</i>		<i>x</i>	<i>y</i>
1	3.36	65	10	3.85	143	18	3.64	56	27	4.45	3
2	2.88	156	11	3.97	56	19	3.48	65	28	4.49	8
3	3.63	100	12	4.51	26	20	3.60	17	29	4.41	4
4	3.41	134	13	4.54	22	21	3.18	7	30	4.32	3
5	3.78	16	14	5.00	1	22	3.95	16	31	4.90	30
6	4.02	108	15	5.00	1	23	3.72	22	32	5.00	4
7	4.00	121	16	4.72	5	24	4.00	3	33	5.00	43
8	4.23	4	17	5.00	65	25	4.28	4			
9	3.73	39				26	4.43	2			

Table 10.22 Survival times *y* (weeks) for two groups of acute leukaemia patients, together with $x = \log_{10}$ white blood cell count at time of diagnosis (Feigl and Zelen, 1965). Patients in group 1 had Auer rods and/or significant granulation of the leukaemic cells in the bone marrow at the time of diagnosis; those in group 2 did not.

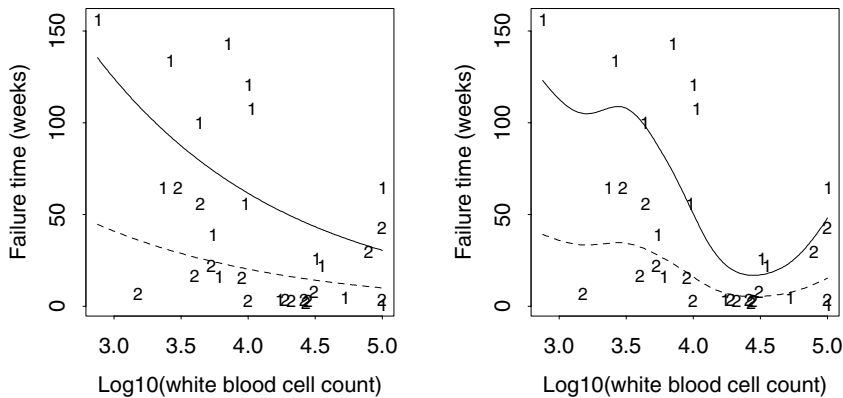


Figure 10.21 Plots of data and fitted means for generalized linear (left) and generalized additive (right) models fitted to two groups of survival times for leukaemia patients: group 1 (solid); group 2 (dashed).

When this model is fitted the deviance drops by 17.82 to 40.32, and the degrees of freedom drop from 32 to 30. The parameter estimates and standard errors are $\hat{\beta}_0 = 5.81$ (1.29), $\hat{\beta}_1 = 1.02$ (0.35), and $\hat{\beta}_2 = -0.70$ (0.30). The mean survival time drops rapidly with x , but is increased in group 1; both effects are significant. The fitted means $\hat{\tau}$ and data are shown in the left panel of Figure 10.21. An exponential probability plot of the residuals $y/\hat{\tau}$ casts no doubt on the model. This can be verified more formally by fitting gamma, Weibull, and log-logistic distributions, none of which improves on the exponential.

Inspection of the left panel of Figure 10.21 suggests some lack of fit of the systematic part of the model and we use this to illustrate the fitting of a generalized additive model with linear predictor $\eta = \beta_0 + \beta_1 I(\text{Group} = 1) + s(x)$, where the smooth function $s(x)$ is a natural cubic spline. The right panel shows smooth dependence on x with 3.36 degrees of freedom, found by generalized cross-validation starting from a cubic spline with 6 knots equi-spaced along the range of x . With this model $\hat{\beta}_0 = 2.80$ (0.24) and $\hat{\beta}_1 = 1.15$ (0.31); the value of $\hat{\beta}_0$ changes because $s(x)$ is parametrized to be orthogonal to a constant. The deviance is 31.68 with 27.63 equivalent degrees of freedom, so the F statistic for comparison of this with the fully parametric model is $(40.32 - 31.68363)/2.37/(31.68/27.63) = 3.18$ on 2.37 and

27.63 degrees of freedom, giving significance level 0.05. Here chi-squared asymptotics are of dubious relevance, and as simulation from the parametric model gives a rather larger significance level, there is no reason to choose the more complex generalized additive model, particularly as increased mean survival time at the highest white blood cell counts seems implausible. ■

Often called the Cox model, because introduced by Cox (1972).

10.8.2 Proportional hazards model

In medical applications the focus of interest is typically on how treatments or classifications of the units affect survival, the form of the survival distribution being of secondary importance. This suggests that we seek inferences that will be valid for any such distribution. This is difficult for accelerated life models, and instead we let the covariates act directly on the hazard. Suppose that an individual with baseline covariate x_0 has hazard function $h_0(y)$ after a time y on trial, while an individual with covariate x has hazard function $h(y) = \xi(\beta; x)h_0(y)$, where $\xi(\beta; x)$ is a positive function sometimes called a risk score; usually $\xi(\beta; x) = \exp(x^T\beta)$. The ratio $h(y)/h_0(y)$ does not involve h_0 . This *proportional hazards* assumption turns out to be crucial, but it is strong and must be checked in practice.

We can take $x_0 = 0$ without loss of generality.

The basic relationship between the survivor and hazard functions, $\mathcal{F}(y) = \exp\{-H(y)\}$, where $H(y) = \int_0^y h(u) du$, implies that the survival time for an individual with covariate x has survivor and density functions

$$\mathcal{F}_0(y)^{\xi(\beta; x)}, \quad \xi(\beta; x)h_0(y)\mathcal{F}_0(y)^{\xi(\beta; x)}.$$

Thus whereas accelerated life models scale the axis of a baseline survivor function, $\mathcal{F}_0$, proportional hazards raise the baseline survivor function to a power.

The action of the covariates being of primary interest, we seek a likelihood on which to base inference for β , regardless of $h_0(y)$. To motivate the argument below, note that if the hazard function was entirely arbitrary, then inference could only be based on events where failures actually occurred, because the hazard might in principle be zero at every other time. Thus it suffices to estimate the baseline cumulative hazard function by a step function $H_0(y) = \sum_{j: y_j \leq y} h_j$, where $h_j = h_0(y_j) > 0$ only at observed failure times.

Suppose there are no ties, take $0 < y_1 < \dots < y_n$ without loss of generality, and let $\mathcal{R}_j$ denote the *risk set* of individuals still available to fail at the instant before y_j , that is, all except those who have previously failed or been censored; see Figure 5.8. For brevity set $\xi_j = \xi(\beta; x_j)$. Then the log likelihood is

$$\begin{aligned} \sum_{j=1}^n \{d_j \log(\xi_j h_j) - \xi_j H_0(y_j)\} &= \sum_{j=1}^n \left\{ d_j \log(\xi_j h_j) - \xi_j \sum_{i=1}^j h_i \right\} \\ &= \sum_{j=1}^n \left(d_j \log \xi_j + d_j \log h_j - h_j \sum_{i \in \mathcal{R}_j} \xi_i \right). \end{aligned}$$

With β fixed the h_j have maximum likelihood estimators $\hat{h}_j = d_j / \sum_{i \in \mathcal{R}_j} \xi_i$, positive

only when $d_j = 1$, so the profile log likelihood for β is

$$\ell_p(\beta) = \max_{h_1, \dots, h_n} \ell(\beta, h_1, \dots, h_n) \equiv \sum_{j=1}^n d_j \log \left(\frac{\xi_j}{\sum_{i \in \mathcal{R}_j} \xi_i} \right). \quad (10.57)$$

The corresponding profile likelihood is

$$\prod_{j=1}^n \left\{ \frac{\xi(\beta; x_j)}{\sum_{i \in \mathcal{R}_j} \xi(\beta; x_i)} \right\}^{d_j} = \prod_{\text{failures}} \frac{\xi(\beta; x_j)}{\sum_{i \in \mathcal{R}_j} \xi(\beta; x_i)}. \quad (10.58)$$

Alternatively we may reason that the probability of the particular failure observed to occur at y_j , conditional on a failure occurring then, is

$$\frac{\xi_j h_0(y_j)}{\sum_{i \in \mathcal{R}_j} \xi_i h_0(y_j)} = \frac{\xi_j}{\sum_{i \in \mathcal{R}_j} \xi_i}, \quad (10.59)$$

and hence (10.58) is the probability that failures occur in the observed order, conditional on their occurrence times and margining over times of censoring. Thus (10.58) is the product of a nested sequence of multinomial variables. There is a close connection to the discussion of Poisson variables and log-linear models on page 501.

Expression (10.58) is known as a *partial likelihood*. In Section 12.2 it is derived as a marginal likelihood based on the observed ranking of failure times. Although a mathematically complete derivation is beyond our scope, it turns out that despite the maximization over n nuisance parameters, (10.58) can be treated as an ordinary likelihood: the maximum partial likelihood estimator $\hat{\beta}$ is consistent for β under mild conditions, and standard errors can be based on the inverse observed information matrix.

Information contained in the failure times is lost, because they are treated as fixed in constructing the partial likelihood. The loss of information compared to using the correct parametric model turns out to be small in most cases, however, so standard errors from partial likelihood are close to those obtained under the true model. Partial likelihood inferences make essentially no assumptions about h_0 , and are in this sense semiparametric.

Tied failure times have probability zero for continuous distributions, but nevertheless they arise in data due to rounding. Three possible modifications of the partial likelihood to adjust for the simultaneous failure of a elements of the risk set $\mathcal{R}_j$ at time y_j are to include a term corresponding to each failure occurring first, to compute the exact probability of a failures, and to use an approximation to this. Thus (10.59) is replaced by one of

$$\prod_{i=1}^a \frac{\xi_i}{\sum_{k \in \mathcal{R}_j} \xi_k}, \quad \frac{\prod_{i=1}^a \xi_i}{\sum \prod_{k=1}^a \xi_{l_k}}, \quad \prod_{i=1}^a \frac{\xi_i}{\sum_{k \in \mathcal{R}_j} \xi_k - \frac{i-1}{a} \sum_{k=1}^a \xi_k}, \quad (10.60)$$

where the sum in the exact central formula is over all subsets of $\mathcal{R}_j$ of size a . The first of these arises from applying the profile likelihood argument above to tied data. In practice these corrections often give similar results.

Example 10.37 (Leukaemia data) Consider the data in Table 10.22 with $\xi = \exp\{\beta_0 + \beta_1 I(\text{Group} = 1) + \beta_2 x\}$. Now β_0 cancels from the partial likelihood, maximization of which gives $\hat{\beta}_1 = -1.07$ (0.43) and $\hat{\beta}_2 = 0.85$ (0.31). These are similar to the values for the exponential model, apart from the sign change because the hazard and mean survival time are inversely related. Note in particular that the standard errors barely differ, confirming our comments about the efficiency of partial likelihood estimation.

These data have 17 ties. The estimates above result from using the third, approximate, term in (10.60), while the second, exact, formula gives $\hat{\beta}_1 = -1.08$ (0.45) and $\hat{\beta}_2 = 0.90$ (0.34), and the simple first approximation gives $\hat{\beta}_1 = -1.02$ (0.42) and $\hat{\beta}_2 = 0.83$ (0.31). There is little to choose among these, but rather more to choose among the likelihood ratio statistics for inclusion of the two covariates, which are 15.6 using the second and third terms, and 14.6 using the first. The third term in (10.60) thus seems preferable to the first, as both require the same computational effort.

It may be useful to contrast two types of semiparametric procedure. Partial likelihood inference requires no assumptions about the baseline hazard and distribution function, and in a sense relaxes the vertical axis of Figure 10.21. Use of splines or other smoothing procedures can also be described as semiparametric, but it relaxes the horizontal axis of the figure, which relates the covariate and response. Spline terms can be introduced into the linear predictor of the proportional hazards model, replacing $\beta_2 x$ by $s(x)$, but this does not improve significantly on the exponential model. ■

The baseline cumulative hazard and survivor functions may be estimated by

$$\hat{H}_0(y) = \sum_{j: y_j \leq y} \frac{d_j}{\sum_{i \in \mathcal{R}_j} \hat{\xi}_i}, \quad \hat{\mathcal{F}}_0(y) = \prod_{j: y_j \leq y} \left(1 - \frac{d_j}{\sum_{i \in \mathcal{R}_j} \hat{\xi}_i}\right), \quad (10.61)$$

where $\hat{\xi}_j = \xi(\hat{\beta}; x_j)$. These are needed to assess fit and to predict survival probabilities for individuals from a fitted model. The estimated survivor function for an individual with covariates x_+ is $\mathcal{F}_+(y) = \exp\{-\xi(\hat{\beta}; x_+) \hat{H}_0(y)\}$, from which the probability of survival beyond a given point can be read off, with standard errors found using the delta method.

The construction above extends to stratified data, with the baseline hazard varying between strata but the parameter being common to all strata. This is useful in checking proportionality of hazards.

Log rank test

We now briefly discuss use of the proportional hazards model to construct tests for equality of survival distributions. When $\xi(\beta; x) = \exp(x^\top \beta)$, the log partial likelihood (10.57) equals

$$\ell_p(\beta) = \sum_{\text{failures}} \left[x_j^\top \beta - \log \left\{ \sum_{i \in \mathcal{R}_j} \exp(x_i^\top \beta) \right\} \right] = \sum_{\text{failures}} \{x_j^\top \beta - \log A_j(\beta)\},$$

say, with first derivative

$$U(\beta) = \sum_{\text{failures}} \left\{ x_j - \frac{\sum_{i \in \mathcal{R}_j} x_i \exp(x_i^T \beta)}{A_j(\beta)} \right\} = \sum_{\text{failures}} \left\{ x_j - \frac{B_j(\beta)}{A_j(\beta)} \right\}, \quad (10.62)$$

say, and negative second derivative

$$J(\beta) = \sum_{\text{failures}} \left\{ \frac{\sum_{i \in \mathcal{R}_j} x_i x_i^T \exp(x_i^T \beta)}{A_j(\beta)} - \frac{B_j(\beta) B_j(\beta)^T}{A_j(\beta)^2} \right\}. \quad (10.63)$$

Suppose the data fall into two groups with respective hazard functions $h_0(y)$ and $h(y) = e^\beta h_0(y)$. Then a score test for $\beta = 0$, that is, equality of survival distributions, is obtained by letting x_j be scalar, with $x_j = 1$ or 0 indicating that failure j belongs to groups 1 or 0, and taking $U(0) \sim N\{0, J(0)\}$ or equivalently $U(0)^2/J(0) \sim \chi_1^2$. This is known as the *log rank test*.

Now $A_j(0)$ and $B_j(0)$ respectively equal

$$\sum_{i \in \mathcal{R}_j} \exp(x_i^T \beta) \Big|_{\beta=0} = m_{0j} + m_{1j}, \quad \sum_{i \in \mathcal{R}_j} x_i \exp(x_i^T \beta) \Big|_{\beta=0} = m_{1j},$$

the total number of individuals and the number of group 1 individuals available to fail at time y_j . Thus

$$U(0) = \sum_{\text{failures}} \left(R_j - \frac{m_{1j}}{m_{0j} + m_{1j}} \right), \quad J(0) = \sum_{\text{failures}} \frac{m_{0j} m_{1j}}{(m_{0j} + m_{1j})^2},$$

where $R_j = 1$ if the individual failing at time y_j belongs to group 1 and $R_j = 0$ otherwise. Hence the score statistic is a sum of centred binary variables. These are not independent but under mild conditions the normal limiting distribution above will nonetheless hold.

An alternative argument proceeds by cross-classifying the risk set at each failure time by group membership and failure/survival, and using the hypergeometric distribution for the number of group 1 failures conditional on the row and column totals in the resulting 2×2 table. This applies also when there are ties, and yields

$$U(0) = \sum_{\text{failures}} \left(R_j - \frac{m_{1j} a_j}{m_{0j} + m_{1j}} \right),$$

$$J(0) = \sum_{\text{failures}} \frac{m_{0j} m_{1j} a_j (m_{0j} + m_{1j} - a_j)}{(m_{0j} + m_{1j})^2 (m_{0j} + m_{1j} - 1)},$$

with R_j now the number of group 1 failures at y_j among the total number of failures a_j at the j th failure time; see the discussion after (10.28). This reduces to the previous version when there are no ties, that is, $a_j \equiv 1$.

Example 10.38 (Mouse data) Figure 5.11 compares cumulative hazard functions for subsets of the data of Table 5.7. The values of $U(0)^2/J(0)$ for the left and right panels are 3.3 and 40.1, each to be treated as χ_1^2 . The first has significance level 0.07,

weak evidence that the distributions differ. The second strongly supports the visual impression of quite different distributions. ■

The log rank test generalizes to quantitative covariates x , to multiple survival distributions, and to weighted sums

$$\sum_{\text{failures}} w_j \left(R_j - \frac{m_{1j} a_j}{m_{0j} + m_{1j}} \right),$$

where the w_j can depend on the failure times, on m_{0j} , m_{1j} , and on a_j . Such statistics can give better power against alternatives other than proportional hazards. Their variances may be found using ideas from Section 7.2.3.

Time-dependent covariates

Thus far we have supposed that the covariate vector x_j takes the same value throughout the period over which the j th individual is observed. This is appropriate for variables such as age on entry, sex, and summaries of medical history prior to entry to the study, but it is also necessary to be able to accommodate explanatory variables that vary during the study. Quantities such as a patient's blood pressure may be available at various points over the observation period, for example, or a treatment may be not allocated until well after the study has begun, or changed during the trial. In reliability trials the key explanatory variable may be cumulative stress, or perhaps instantaneous stress, both of which may change during the experiment.

The interpretation of effects of covariates that may be influenced by the treatments demands careful thought. Consider for example a study in which treatments for hypertension are compared, blood pressure being an explanatory variable. Use of initial blood pressure as a covariate should increase the precision with which the treatment effects can be estimated, but interest would focus on the treatments, the estimated effect of blood pressure being of little direct interest. Use of blood pressure monitored after treatment allocation, by contrast, would allow the analyst to assess the extent to which treatments affect survival by influencing blood pressure; the estimate might then be of prime concern.

Time-varying covariates may also be constructed for technical reasons, for instance to check adequacy of the proportional hazards assumption by including y or $\log y$ in the linear predictor.

Whatever the interpretation, use of time-dependent covariates leads to replacement of the p elements $x_{j1}, \dots, x_{jp}$ of x_j by functions $x_{j1}(y), \dots, x_{jp}(y)$, $0 \leq y \leq y_j$. These may be indicator variables, for example showing the treatment being applied at time y . The covariates are typically measured only at certain times, so the function $x_{jr}(y)$ is usually obtained by interpolation. Let $x_j(y)$ denote the $p \times 1$ vector $(x_{j1}(y), \dots, x_{jp}(y))^T$. The hazard function $\xi\{\beta; x_j\}h_0(y)$ becomes $\xi\{\beta; x_j(y)\}h_0(y)$, and our previous argument shows that the log partial likelihood is

$$\ell_p(\beta) = \sum_{j=1}^n d_j \left(\xi\{\beta; x_j(y_j)\} - \log \left[\sum_{i \in \mathcal{R}_j} \xi\{\beta; x_i(y_j)\} \right] \right),$$

the outer sum being over failure times y_j . Thus rather than x_j , the covariates needed for the j th individual are $\{x_j(y_i) : y_i \leq y_j\}$, where the failure times y_i are those at which case j lies in the risk set.

Standard large-sample likelihood results may be used for inference on β .

Model checking

When the data contain two groups, a graphical check on proportional hazards may be based on their estimated cumulative hazard functions. If the cumulative hazard functions for the two groups are $H_0(y)$ and $H(y)$, then proportional hazards asserts that $H(y) = \xi(\beta; x)H_0(y)$. Thus $\log \hat{H}(y) - \log \hat{H}_0(y)$ should appear independent of y .

Various residuals can be defined. The modified Cox–Snell residuals (10.55) equal $r_j = \hat{H}_j(y_j) + 1 - d_j$, where the estimated cumulative hazard function for the j th individual under a proportional hazards model with constant covariates is $\hat{H}_j(y) = \xi(\hat{\beta}; x_j)\hat{H}_0(y_j)$ and $\hat{H}_0$ is given at (10.61). Plots of the r_j for subsets of the observations may cast light on interactions, but are not useful for assessing distributional assumptions in the proportional hazards model because $H_0(y)$ is not specified parametrically.

If Y_j is a continuous random variable with censoring indicator D_j and cumulative hazard function $H_j(y)$, then $I(Y_j \leq y, D_j = 1) - H_j\{\min(y, Y_j)\}$ is a zero-mean continuous-time martingale; see page 552. With $y = \infty$ this gives $D_j - H_j(Y_j)$, and implies that a *martingale residual* may be constructed as $d_j - \hat{H}_j(y_j) = 1 - r_j$, just the residual above apart from a location and sign change. The functional form of a covariate in a proportional hazards model with $\xi(\beta; x) = \exp(x^T \beta)$ can be checked by plotting $1 - r_j$ computed with the covariate omitted against the covariate itself.

The strong negative skewness of martingale residuals can be reduced by transformation, giving *deviance residuals*

$$\text{sign}\{d_j - \hat{H}_j(y_j)\}[2\{\hat{H}_j(y_j) - d_j - d_j \log \hat{H}_j(y_j)\}]^{1/2},$$

which are useful for checking for outliers; they are formally equivalent to treating the D_j as Poisson variables with means $H_j(Y_j)$.

An approach based on (10.62) uses components of the contributions

$$x_j - \frac{\sum_{i \in \mathcal{R}_j} x_i \exp(x_i^T \hat{\beta})}{\sum_{i \in \mathcal{R}_j} \exp(x_i^T \hat{\beta})},$$

to the score vector, thus giving a residual for each covariate and for each individual seen to fail. These $p \times 1$ vectors can be scaled by pre-multiplication by $J_j(\hat{\beta})$, where the $p \times p$ matrix $J_j(\beta)$ is the contribution to (10.63) from the j th failure. They are closely related to the influence measures (8.29) and (10.13), and plots of their components help to determine which of the observations are influential for elements of $\hat{\beta}$. They are also useful for assessing adequacy of proportional hazards. A natural way in which hazards might not be proportional is $h(y) = h_0(y) \exp\{x^T \beta(y)\}$, that is, the coefficient of x depends on time. If this is the case, and there are no tied failures, then $E(S_j) + \hat{\beta} \doteq \beta(y_j)$, where S_j is a standardized version of the score contributions

computed using only the risk set at time y_j . A non-constant plot of observed S_j against y_j suggests this type of model failure.

These and other diagnostics for the proportional hazards model can be extended to time-dependent covariates.

Example 10.39 (PBC data) Primary biliary cirrhosis (PBC) is a chronic fatal disease of the liver, with an incidence of about 50 cases per million. Controlled clinical trials are hard to perform with very rare diseases, so the double-blinded randomized trial conducted at the Mayo Clinic from 1974–1984 is a valuable resource for liver specialists. A total of 424 patients were eligible for the trial, and the 312 who consented to take part were randomized to be treated either with the drug D-penicillamine or with a placebo. Although basic data are available on all 424 patients, we consider only these 312 individuals. Covariates available on each of them at recruitment include the demographic variables sex and age; clinical variables, namely presence or absence of ascites, hepatomegaly, spiders, and a ternary variable *edtrt* whose values 0, 1/2, 1 indicate no, mild, and severe edema; and biochemical variables, namely levels of serum bilirubin (mg/dl), serum cholesterol (mg/dl), albumin (gm/dl), urine copper ($\mu\text{g/day}$), alkaline phosphatase (U/ml), SGOT (U/ml), and triglycerides (mg/dl), platelet count (coded), prothombin time (seconds), and the histologic stage of the disease (1–4). There are 28 missing values of serum cholesterol and 30 of triglycerides, and we ignore these covariates. Four missing values of platelets and two of urine copper were replaced by the medians of the remaining values; this should have little effect on the analysis. At the time at which the data considered here became available, 125 patients had died, with just 11 deaths not due to PBC, eight patients had been lost to follow-up, and 19 had undergone a liver transplant. As the response is time to death, these patients are regarded as censored.

Edema is the accumulation of fluids in body tissues.

The upper left panel of Figure 10.22 shows that estimated survivor functions for the patients with the drug and the placebo are very close, and it is no surprise that the log-rank statistic has value 0.1, insignificant when treated as χ_1^2 . This is borne out by the estimated treatment effect of -0.057 (0.179) for a fit of the proportional hazards model with treatment effect only. Analysis stratified by sex gives an estimate of -0.045 (0.179). Neither differs significantly from zero. The corresponding baseline survival function estimates in the upper right panel of Figure 10.22 suggest no need to stratify.

Similar analyses for subgroups of the data and the corresponding log-rank statistics also show no significant treatment effects.

Having established that treatment has no effect on survival, we try constructing a model for prediction of survivor functions for new patients. This should be useful in assessing for whom liver transplant is a priority. The first step is to see which readily accessible covariates are highly predictive of survival. We exclude histologic stage, which requires a liver biopsy, and urine copper and SGOT, which are frequently unmeasured. The product-limit estimates and log rank statistics show strong dependence of failure on the other variables individually, so we fit a proportional hazards model

Variable	Estimate (SE)			
	Full	Reduced	Transformed	Final
age	0.028 (0.009)	0.030 (0.009)	0.033 (0.009)	0.041 (0.009)
alb	−0.97 (0.027)	−1.09 (0.24)	−3.06 (0.72)	−3.07 (0.72)
alkphos	0.015 (0.035)			
ascites	0.29 (0.31)			
bili	0.11 (0.02)	0.11 (0.02)	0.88 (0.10)	0.88 (0.10)
edtrt	0.69 (0.32)	0.77 (0.31)	0.79 (0.30)	0.69 (0.30)
hepmeg	0.49 (0.22)	0.50 (0.22)	0.25 (0.22)	
platelet	−0.61 (1.02)			
protime	0.24 (0.08)	0.25 (0.08)	3.01 (1.02)	3.57 (1.13)
sex	−0.48 (0.26)	−0.55 (0.25)		
spiders	0.29 (0.21)	0.30 (0.21)		

Table 10.23 Parameter estimates and standard errors for proportional hazards models fitted to the PBC data. The full fit is reduced by backwards elimination. In the last two columns log transformation is applied to alb, bili, and protime.

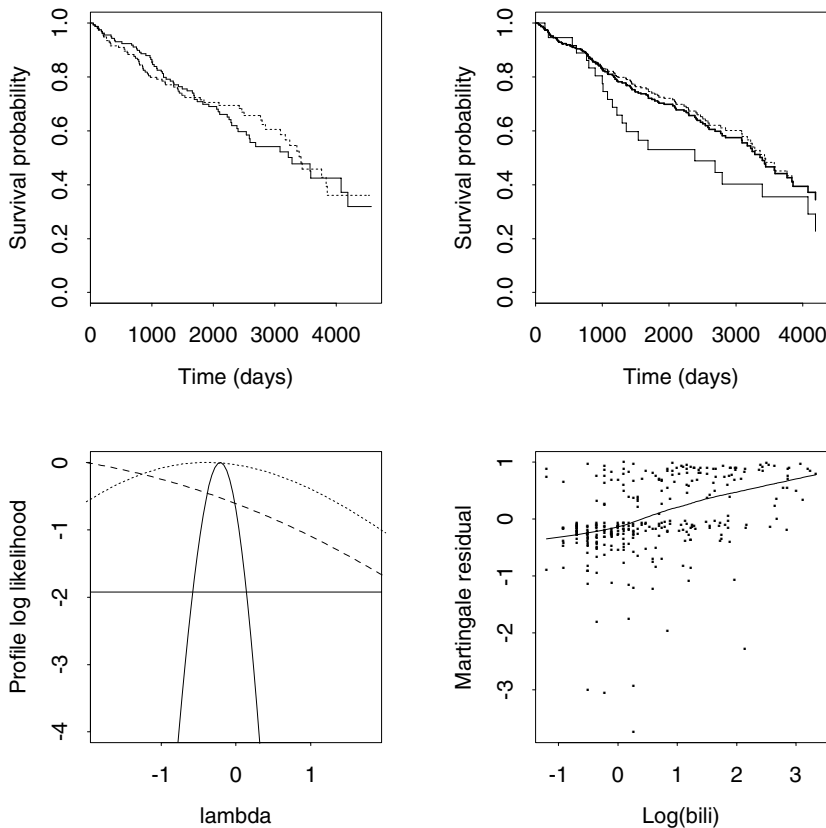


Figure 10.22 PBC data analysis (Fleming and Harrington, 1991). Top left: product-limit estimates for control (solid) and treatment (dots) groups. Top right: estimates of baseline survivor function for data stratified by sex, men (dots), women (solid). The heavy line shows the unstratified estimate. Lower left: profile likelihood for Box–Cox transformations of bilirubin (solid), albumin (dots), and prothrombin time (dashes); the horizontal line indicates 95% confidence limits for the transformation parameter. Lower right: martingale residuals from the model with terms age, log(alb), edtrt, log(protime) against log bilirubin, and lowess smooth with $p = 2/3$.

with all but the excluded covariates. Table 10.23 suggests that serum bilirubin is most significant and that several other covariates can be dropped. Backward selection based on AIC leads to the reduced model in the table. The likelihood ratio statistic for comparison of the two models is 1.22, plainly insignificant. Dropping sex and

spiders also leads to a likelihood ratio statistic of 7.29, with significance level 0.20 when treated as χ^2_5 . Bearing in mind the tendency of AIC to overfit, we now ignore these covariates.

To investigate whether transformation is worthwhile we apply the Box–Cox approach (Example 8.23) to *alb*, *bili*, and *protime*. The lower left panel of Figure 10.22 clearly indicates log transformation of *bili*, but not of the other variables. The need for transformation of *bili* can also be assessed through the plot of martingale residuals obtained when it is dropped, given in the lower right panel of the figure. Note the strong negative skewness of the residuals. The near-linearity of the lowess smooth shows the appropriateness of the transformation. The corresponding plot against *bili* itself is harder to read because the points are bunched towards zero. The plots for *alb* and *protime* are more ambiguous. If we take logs of all three variables, then the maximized log partial likelihood increases by 13.8 and *hepmeg* can be dropped; see Table 10.23.

A model with terms $\text{age} + \log(\text{alb}) + \log(\text{bili}) + \text{edtrt} + \log(\text{protime})$ is medically plausible. As the disease progresses, the liver's ability to produce albumin decreases, leading to the negative coefficient for *alb*, while damage to the bile ducts reduces excretion of bilirubin and so increases its level in the body. Edema is often associated with the later stages of the disease, while prothrombin is decreased, leading to slower clotting of the blood. Finally and unsurprisingly, risk increases with age.

The upper panels of Figure 10.23 show deviance residuals plotted against age and prothrombin time. Inspection of those in the left panel lying outside the 0.01 and 0.99 normal quantiles reveals an error in the data coding; case 253 has residual -2.55 but his age should be 54.4 rather than 78.4. The right panel shows an unusually high prothrombin time of 17.1, which should have been 10.7. The estimates after these corrections are shown in the final column of Table 10.23.

The lower left panel of Figure 10.23 shows the scaled scores plotted against prothrombin time. There is some suggestion of non-proportionality, but it is too limited to suggest model failure. Such plots for the other variables cast no doubt on proportionality of hazards, and we accept the model.

To illustrate prediction, consider an individual with $\text{age}=60$, $\text{alb}=4$, $\text{bili}=1$, $\text{edtrt}=0$, and $\text{protime}=8$, for whom $x^T \hat{\beta} = -1.618$ and whose hazard is reduced by a factor $\exp(x^T \hat{\beta}) = 0.20$ compared to baseline. Setting $\text{edtrt}=1$ and $\text{bili}=20$ gives estimated risk scores of 0.4 and 2.8. The lower right panel of Figure 10.23 shows how the survivor functions then vary. The median estimated lifetime in each case can be found by solving for y the equation $\hat{\mathcal{F}}_0(y)^{\exp(x^T \hat{\beta})} = 0.5$. ■

The proportional hazards model has been broadened in many directions. Suppose, for instance, that individuals move between states 1 and 2 and back again, baseline time-dependent transition rates $\gamma_{12}(y)$ and $\gamma_{21}(y)$ being modified to $\gamma_{12}(y)\xi_{12}(\beta; x)$ and $\gamma_{21}(y)\xi_{21}(\beta; x)$ for an individual with explanatory variables x . The partial likelihood for β is a product of terms corresponding to each of the observed transitions between states. For instance, the contribution from transition $1 \rightarrow 2$ at time y by an

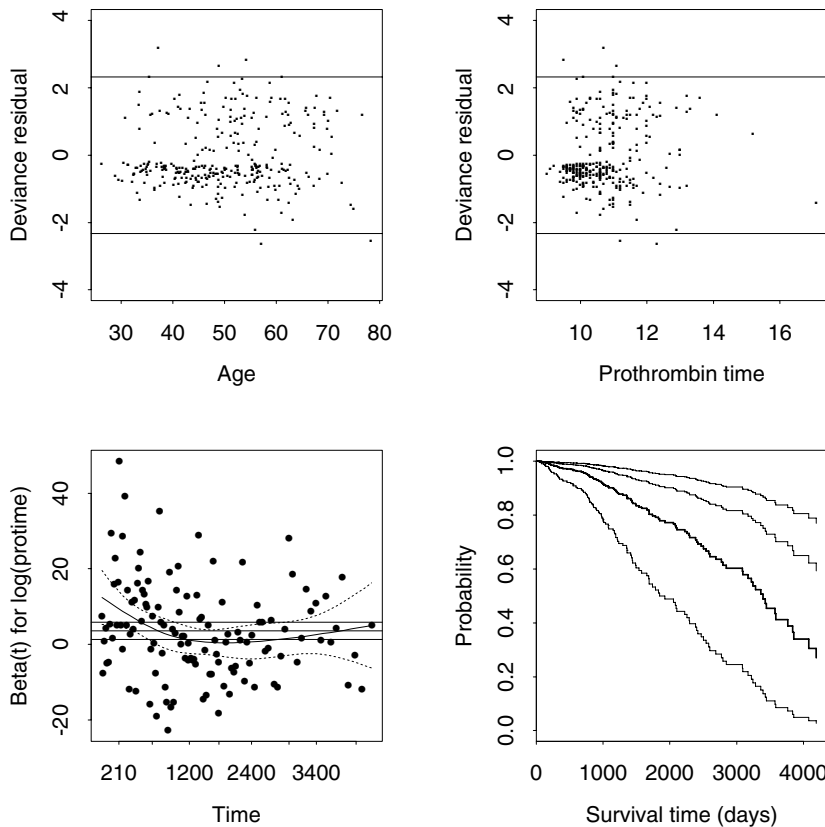


Figure 10.23 PBC data analysis. Upper panels: deviance residuals plotted against age and prothrombin time, with horizontal lines showing 0.01 and 0.99 standard normal quantiles. Lower left: scaled scores S_j^* plotted against prothrombin time, with lowest smooth and approximate 0.95 pointwise confidence bands (curved lines). Also shown are overall estimate and 0.95 confidence interval (horizontal lines). Lower right: baseline survivor function estimate (heavy), with predicted survivor functions for individuals with risk factors 0.2, 0.4, and 2.8 (top to bottom).

individual with covariates x_j is

$$\frac{\gamma_{12}(y)\xi_{12}(\beta; x_j)}{\sum \gamma_{12}(y)\xi_{12}(\beta; x_k)} = \frac{\xi_{12}(\beta; x_j)}{\sum \xi_{12}(\beta; x_k)},$$

the sum being over individuals in state 1 at time y . Individuals unobserved at y , or not in state 1, do not appear in the sum. Such extensions of partial likelihood enable inference for many types of partially observed and censored multi-state data, but details cannot be given here.

Counting processes and martingale residuals

Consider a random variable Y with censoring indicator D and hazard function $h(y)$, and let

$$V(y) = I(Y \geq y), \quad N(y) = I(Y \leq y, D = 1),$$

be random variables that indicate whether Y is in view at time y , and whether failure has been observed by y . As $V(y)$ is left-continuous, its value at time y can be predicted the moment before, y^- , whereas the counting process $N(y)$ is right-continuous and so is not predictable. Let $\{\mathcal{H}_y : y \geq 0\}$ denote the history of the process up to time y . This is known as a filtration or increasing collection of sigma-algebras: $\mathcal{H}_x \subset \mathcal{H}_y$ for

This can be skipped at a first reading.

$x \leq y$; knowledge accumulates. Define also $dN(y) = N\{(y + dy)^-\} - N(y)$, which equals 1 if failure is observed to occur at y , and otherwise equals 0. Then

$$E\{dN(y) \mid \mathcal{H}_{y^-}\} = \Pr\{dN(y) = 1 \mid \mathcal{H}_{y^-}\} = h(y)V(y), \quad y \geq 0; \quad (10.64)$$

the mean failure rate at y can be predicted from the history to y^- . However potential dependence on $\mathcal{H}_{y^-}$ makes $h(y)V(y)$ a random variable.

Now (10.64) implies that $dM(y) = dN(y) - h(y)V(y)$ is a zero-mean continuous-time martingale with respect to $\mathcal{H}_{y^-}$, for all $y > 0$. This implies that

$$M(y) = \int_0^y dM(u) = N(y) - \int_0^y h(u)V(u) du = N(y) - H\{\min(y, Y)\},$$

has the property that for any $y \geq x$,

$$\begin{aligned} E\{M(y) \mid \mathcal{H}_x\} - M(x) &= E\left\{\int_x^y dM(u) \mid \mathcal{H}_x\right\} \\ &= \int_x^y E[E\{dM(u) \mid \mathcal{H}_{u^-}\} \mid \mathcal{H}_x] = 0, \end{aligned}$$

and is therefore a martingale. Thus $E\{M(y)\} = 0$ for all y , and in particular $E\{M(\infty)\} = E\{D - H(Y)\} = 0$.

Let independent variables $Y_1, \dots, Y_n$ with cumulative hazard functions $H_j(y)$ and censoring indicators $D_1, \dots, D_n$ be observed, and set $V_j(y) = I(Y_j \geq y)$ and $N_j(y) = I(Y_j \leq y, D_j = 1)$. The corresponding continuous-time martingale is $M_j(y) = N_j(y) - H_j\{\min(y, Y_j)\}$, from which martingale residuals are obtained by setting $y = \infty$ and replacing unknowns with estimates.

Developments of the above formulation are central to mathematical treatments of time-to-event data, references to which are given in Section 10.9.

Exercises 10.8

- 1 Show that if Y is continuous with cumulative hazard function $H(y)$, then $H(Y)$ has the unit exponential distribution. Hence establish that $E\{H(Y) \mid Y > c\} = 1 + H(c)$, and explain the reasoning behind (10.55).
- 2 Let Y be a positive continuous random variable with survivor and hazard functions $\mathcal{F}(y)$ and $h(y)$. Let $\psi(x)$ and $\chi(x)$ be arbitrary continuous positive functions of the covariate x , with $\psi(0) = \chi(0) = 1$. In a proportional hazards model, the effect of a non-zero covariate is that the hazard function becomes $h(y)\psi(x)$, whereas in an accelerated life model, the survivor function becomes $\mathcal{F}\{y\chi(x)\}$. Show that the survivor function for the proportional hazards model is $\mathcal{F}(y)^{\psi(x)}$, and deduce that this model is also an accelerated life model if and only if

$$\log \psi(x) + G(\tau) = G\{\tau + \log \chi(x)\},$$

where $G(\tau) = \log\{-\log \mathcal{F}(e^\tau)\}$. Show that if this holds for all τ and some non-unit $\chi(x)$, we must have $G(\tau) = \kappa\tau + \alpha$, for constants κ and α , and find an expression for $\chi(x)$ in terms of $\psi(x)$. Hence or otherwise show that the classes of proportional hazards and accelerated life models coincide if and only if Y has a Weibull distribution.

- 3 In the usual notation for a linear regression model, $X^T(y - X\hat{\beta}) = 0$. By writing the partial likelihood corresponding to (10.62) as $\sum_{j=1}^n d_j \{x_j^T \beta - \log A_j(\beta)\}$, show that

$$\sum_{j=1}^n x_j \left\{ d_j - \exp(x_j^T \hat{\beta}) \hat{H}_0(y_j) \right\} = 0.$$

Which type of residual for a proportional hazards model is analogous to the raw residual in a linear model?

- 4 Suppose that survival data consist of independent observations (Y_j, C_j) , $j = 1, \dots, n$, where Y_j is an exponential random variable with mean $\exp(x_j^T \beta)$, censored at random, and the censoring indicator is C_j , which equals 0 if Y_j is censored and equals 1 otherwise. Show that the likelihood for these data is the same as if the counts C_j had Poisson distributions with means $y_j \exp(-x_j^T \beta)$. Hence show that maximum likelihood estimates for the censored data model, and their standard errors based on observed information, can be obtained by regarding the censoring variable as having the Poisson distribution with log link function and offset $\log y_j$.
- 5 Write down the partial likelihood contributions from failure times $y = 1, 2$, for the data in Table 10.22, using the model $\xi = \exp\{\beta_0 + \beta_1 I(\text{Group} = 1) + \beta_2 x\}$.
- 6 Suppose that the continuous-time proportional hazards model holds, but that the failure times are grouped into intervals $0 = u_0 < u_1 < \dots < u_m = \infty$. Show that the corresponding grouped hazards

$$h_i(x) = \Pr(Y < u_i \mid Y \geq u_{i-1}; x), \quad i = 1, \dots, m,$$

satisfy

$$\log\{1 - h_i(x)\} = \xi(\beta; x) \log\{1 - h_i(0)\},$$

and write down the corresponding log likelihood when $\xi(\beta; x_j) = \exp(x_j^T \beta)$. Hence find the maximum likelihood estimator of β when the $h_i(0)$ are treated as nuisance parameters. Does this have the usual properties if $n \rightarrow \infty$ and m is fixed? (Prentice and Gloeckler, 1978)

10.9 Bibliographic Notes

Driven by the needs of applications, the literature on regression models has expanded hugely over the last 30 years, and most of the development has been in nonlinear modelling. Generalized linear models were first explicitly formulated by Nelder and Wedderburn (1972), though others had previously suggested special cases. The resulting conceptual unification of apparently disparate models has had a major influence on subsequent developments, not least because of the part played by the iterative weighted least squares algorithm, for which Green (1984) is a standard reference. McCullagh and Nelder (1989) give an excellent account of generalized linear models and their ramifications, while Dobson (1990) is more elementary. Shorter accounts of generalized linear models and corresponding diagnostics are Firth (1991), [?], and Davison and Tsai (1992). Jørgensen (1997b) describes more general classes of exponential family-like distributions.

Data with binary responses are discussed by Collett (1991) and by Cox and Snell (1989).

Bishop *et al.* (1975) and Fienberg (1980) are standard references to log-linear models, though their approach is rather different to that adopted here. Log-linear and

marginal models are discussed in Chapter 6 of McCullagh and Nelder (1989), with more recent work by Liang *et al.* (1992), Glonek and McCullagh (1995), and others. Generalized estimating equations and marginal modelling are of great importance in longitudinal data, a good discussion of which is given in Diggle *et al.* (1994). Agresti (1984) discusses models for ordinal data.

Quasi-likelihood was introduced by Wedderburn (1974) in a seminal article. For subsequent developments see McCullagh (1991), the useful survey by Firth (1993), and Davison (2001). Heyde (1997) gives a longer more theoretical account. See also the bibliographic notes for Chapter 7.

There are now many books on semiparametric regression. Bowman and Azzalini (1997) give an elementary account of kernel methods, with an applied emphasis, while Wand and Jones (1995) contains a more theoretical treatment, and Simonoff (1996) gives an excellent general discussion. Fan and Gijbels (1996) describe the theory of local polynomial modelling in detail, while the more practical account by Loader (1999) includes references to purpose-written software for local fitting. Hastie and Loader (1993) give a shorter more intuitive account of the properties of these methods. The account of spline methods in Section 10.7.2 is based on Green and Silverman (1994). Hastie and Tibshirani (1990) give a book-length account of generalized additive models. Wood (2000) gives a recent account of smoothing parameter selection for penalized likelihood procedures.

Survival data analysis has developed very rapidly over the last three decades. A major impetus was given by the introduction of the proportional hazards model by Cox (1972), which led to greatly increased interest in the area, the use of point process methods by Aalen (1978), and a flood of subsequent work. Fleming and Harrington (1991) and Andersen *et al.* (1993) are standard references to this topic, the latter also treating event history analysis in other areas. Therneau and Grambsch (2000) is an excellent recent book highlighting computation for proportional hazards models and their extensions, while Hougaard (2000) is an account of more advanced topics such as frailty and multistate models. See also Klein and Moeschberger (1997). Although most developments have centred on the proportional hazards model, it is not always suitable in practice, and many other possibilities have been suggested. See also the bibliographic notes to Chapter 5.

10.10 Problems

- 1 Suppose that Y has a density with generalized linear model form

$$f(y; \theta, \phi) = \exp \left\{ \frac{y\theta - b(\theta)}{a(\phi)} + c(y; \phi) \right\},$$

where $\theta = \theta(\eta)$ and $\eta = \beta^T x$.

- (a) Show that the weight for iterative weighted least squares based on expected information is

$$w = b''(\theta)(d\theta/d\eta)^2/a(\phi),$$

and deduce that $w^{-1} = V(\mu)a(\phi)\{dg(\mu)/d\mu\}^2$, where $V(\mu)$ is the variance function, and that the adjusted dependent variable is $\eta + (y - \mu)dg(\mu)/d\mu$.

Note that initial values are not required for β , since w and z can be determined in terms of η and μ ; initial values can be found from y as $\mu^1 = y$ and $\eta^1 = g(y)$.

(b) Give explicit formulae for the weight and adjusted dependent variable when $R = mY$ is binomial with denominator m and probability $\pi = e^\eta / (1 + e^\eta)$.

- 2 The independent observations Y_j , $j = 1, \dots, n$, have Poisson distributions with means μ_j , where $g(\mu_j) = \eta_j$, $g(\cdot)$ is the link function, and η_j is the linear predictor $x_j^T \beta$. The x_j are $p \times 1$ vectors of known covariates such that the matrix X whose j th row is x_j^T has rank p . Show that the likelihood equation for the maximum likelihood estimator $\hat{\beta}$ of β can be written

$$X^T s(\hat{\beta}) = 0,$$

and hence derive the iterative weighted least squares algorithm for estimation of β , giving explicit formulae for the weight matrix and the adjusted dependent variable.

In a set of data on faults in lengths of textile, there were y faults in independent samples of length x . Five pairs (y, x) were (6, 5.5), (4, 6.5), (17, 8.3), (9, 3.8), and (14, 7.2). Suppose that the Y_j are independent Poisson variables with means η_j , and $\eta_j = \beta_0 + \beta_1 x_j$. Give the link function for this model, verify that the maximum likelihood estimates are $\hat{\beta}_0 = 1.006$ and $\hat{\beta}_1 = 1.437$, and calculate their asymptotic covariance matrix. Is there evidence that $\beta_0 \neq 0$?

- 3 For a generalized linear model with known dispersion parameter ϕ and canonical link function, write the deviance as $\sum_{j=1}^n d_j^2$, where d_j^2 is the contribution from the j th observation. Also let

$$u_j(\beta) = \partial \log f(y_j; \eta_j, \phi) / \partial \eta_j, \quad w_j = -\partial^2 \log f(y_j; \eta_j, \phi) / \partial \eta_j^2,$$

denote the elements of the score vector and observed information, let X denote the $n \times p$ matrix whose j th row is x_j^T , where $\eta_j = \beta^T x_j$, and let H denote the matrix $W^{1/2} X (X^T W X)^{-1} X^T W^{1/2}$, where $W = \text{diag}\{w_1, \dots, w_n\}$.

Recall Exercise 8.5.2.

(a) Let $\hat{\beta}_{(k)}$ be the solution of the likelihood equation when case k is deleted,

$$\sum_{j \neq k} x_j u_j(\hat{\beta}_{(k)}) = 0, \quad (10.65)$$

and let $\hat{\beta}$ be the maximum likelihood estimate based on all n observations. Use first-order Taylor series expansion of (10.65) about $\hat{\beta}$ to show that

$$\hat{\beta}_{(k)} \doteq \hat{\beta} - (X^T W X)^{-1} x_k \frac{u_k(\hat{\beta})}{1 - h_{kk}}.$$

Express $\hat{\beta}_{(k)}$ in terms of the standardized Pearson residual $r_{pk} = u_k / \{w_k(1 - h_{kk})\}^{1/2}$.

(b) Use a second order Taylor series expansion of the deviance to show that the change in the deviance when the k th case is deleted is approximately

$r_{Gk} = \text{sign}(y_k - \hat{\mu}_k) \sqrt{r_{Gk}^2}$
is called a jackknifed deviance residual.

$$r_{Gk}^2 = (1 - h_{kk})r_{Dk}^2 + h_{kk}r_{Pk}^2,$$

where r_{Dk} is the standardized deviance residual $d_k / (1 - h_{kk})^{1/2}$.

(c) Suppose models A and B have deviances D_A and D_B . Use (b) to find an expression for the change in the likelihood ratio statistic $D_A - D_B$, when the k th case is deleted.

(d) Show that your results (a)–(c) are exact in models with normal errors.

- 4 In a study on the relation between social class, education, and income, m independently sampled individuals are classified according to the social class of their parents, their income group, and their level of education. m is fixed in advance. The number of individuals with parents in class j , income group k , and with educational level l is y_{jkl} , where $j = 1, \dots, J$, $k = 1, \dots, K$ and $l = 1, \dots, L$. Show that the joint multinomial distribution for the y_{jkl} which is appropriate to this sampling scheme is equivalent to that derived by treating the y_{jkl} as independent Poisson random variables with means μ_{jkl} , conditional on $\sum_{kjl} y_{jkl} = m$, and give the multinomial probabilities in terms of the μ_{jkl} .

One possible model for such data would be that the multinomial probabilities π_{jkl} may be written in the form $\alpha_j(\beta\gamma)_{kl}$, where $\sum_j \alpha_j = \sum_{kl}(\beta\gamma)_{kl} = 1$. Show that the maximum likelihood estimate for α_j is then $y_{j..}/m$, where a dot indicates summation over the corresponding subscript, and find the maximum likelihood estimates of the $(\beta\gamma)_{kl}$. Derive the deviance statistic to test the adequacy of this model, and show that for large m it is equivalent to

$$\frac{1}{m} \sum_{jkl} \frac{(my_{jkl} - y_{j..}y_{..kl})^2}{y_{j..}y_{..kl}},$$

when the model is correct.

- 5 The rate of growth of an epidemic such as AIDS for a large population can be estimated fairly accurately and treated as a known function $g(t)$ of time t . In a smaller area where few cases have been observed the rate is hard to estimate because data are scarce. However predictions of the numbers of future cases in such an area must be made in order to allocate resources such as hospital beds. A simple assumption is that cases in the area arise in a non-homogeneous Poisson process with rate $\lambda g(t)$, for which the mean number of cases in period (t_1, t_2) is $\lambda \int_{t_1}^{t_2} g(t) dt$. Suppose that $N_1 = n_1$ individuals with the disease have been observed in the period $(-\infty, 0)$, and that predictions are required for the number N_2 of cases to be observed in a future period (t_1, t_2) .
- (a) Find the conditional distribution of N_2 given $N_1 + N_2$, and show it to be free of λ . Deduce that a $(1 - 2\alpha)$ prediction interval (n_-, n_+) for N_2 is found by solving approximately the equations

$$\begin{aligned}\alpha &= \Pr(N_2 \leq n_- | N_1 + N_2 = n_1 + n_-), \\ \alpha &= \Pr(N_2 \geq n_+ | N_1 + N_2 = n_1 + n_+).\end{aligned}$$

(b) Use a normal approximation to the conditional distribution in (a) to show that for moderate to large n_1, n_- and n_+ are the solutions to the quadratic equation

$$(1 - p)^2 n^2 + p(p - 1)(2n_1 + z_\alpha^2)n + n_1 p \{n_1 p - (1 - p)z_\alpha^2\} = 0,$$

where $\Phi(z_\alpha) = \alpha$ and

$$p = \int_{t_1}^{t_2} g(t) dt / \left\{ \int_{t_1}^{t_2} g(t) dt + \int_{-\infty}^0 g(t) dt \right\}.$$

(c) Find approximate 0.90 prediction intervals for the special case where $g(t) = 2^{t/2}$, so that the doubling time for the epidemic is two years, $n_1 = 10$ cases have been observed until time 0, and $t_1 = 0, t_2 = 1$ (next year) (Cox and Davison, 1989).

- 6 Let R_0 and R_1 be independent binomial variables with denominators m_0 and m_1 and probabilities π_0 and π_1 , and let $\Delta = \{\pi_1(1 - \pi_0)\}/\{\pi_0(1 - \pi_1)\}$ be the odds ratio for the 2×2 table $(R_0, m_0 - R_0; R_1, m_1 - R_1)$. Let $A = R_0 + R_1$, and let $Y^{(s)} = Y(Y - 1) \cdots (Y - s + 1) = Y!/(Y - s)!$, with $Y^{(s)} = 0$ if $Y + 1 \leq s$.
- (a) Show that $E\{R_1^{(s)}(m_0 - R_0)^{(s)} | A = a\} = \Delta^s E\{R_0^{(s)}(m_1 - R_1)^{(s)} | A = a\}$, and that when $\Delta = 1, E(R_1^{(s)} | A = a) = m_1^{(s)} a^{(s)} / (m_0 + m_1)^{(s)}$.
- (b) When $\Delta = 1$, show that

$$E(R_1 | A = a) = \frac{m_1 a}{m_0 + m_1}, \quad \text{var}(R_1 | A = a) = \frac{m_0 m_1 a(m_0 + m_1 - a)}{(m_0 + m_1)^2 (m_0 + m_1 - 1)}.$$

(c) Show that unconditionally $\{E(R_1)E(m_0 - R_0)\}/\{E(R_0)E(m_1 - R_1)\} = \Delta$, whereas conditionally on A ,

$$\{E(R_1)E(m_0 - R_0) + \text{var}(R_1)\}/\{E(R_0)E(m_1 - R_1) + \text{var}(R_1)\} = \Delta.$$

What does this indicate about the conditional maximum likelihood estimate of Δ relative to the unconditional one?

(d) Show that conditional on A , R_1 has a generalized linear model density with

$$b(\theta) = \log \left\{ \sum_{u=u_-}^{u_+} \binom{m_1}{u} \binom{m_0}{a-u} e^{u\theta} \right\}, \quad u_- = \max\{0, a - m_0\}, \quad u_+ = \min\{m_1, a\}.$$

Deduce that a score test of $\Delta = 1$ based on data from n independent 2×2 tables $(R_{0j}, m_{0j} - R_{0j}; R_{1j}, m_{1j} - R_{1j})$ is obtained by treating $\sum R_{1j}$ as approximately normal with mean and variance

$$\sum_{j=1}^n \frac{m_{1j}a_j}{m_{0j} + m_{1j}}, \quad \sum_{j=1}^n \frac{m_{0j}m_{1j}a_j(m_{0j} + m_{0j} - a_j)}{(m_{0j} + m_{1j})^2(m_{0j} + m_{1j} - 1)},$$

when continuity-corrected this is the *Mantel–Haenszel test*. (Mantel and Haenszel, 1959)

- 7 Suppose that the cumulant-generating function of X can be written in the form $m\{b(\theta + t) - b(\theta)\}$. Let $E(X) = \mu = mb'(\theta)$ and let $\kappa_2(\mu)$ and $\kappa_3(\mu)$ be the variance and third cumulant respectively of X , expressed in terms of μ ; $\kappa_2(\mu)$ is the variance function $V(\mu)$. (a) Show that

$$\kappa_3(\mu) = \kappa_2(\mu)\kappa_2'(\mu) \quad \text{and} \quad \frac{\kappa_3}{\kappa_2^2} = \frac{d}{d\mu} \log \kappa_2(\mu).$$

Verify that the binomial cumulants have this form with $b(\theta) = \log(1 + e^\theta)$.

(b) Show that if the derivatives of $b(\theta)$ are all $O(1)$, then $Y = g(X)$ is approximately symmetrically distributed if g satisfies the second-order differential equation

$$3\kappa_2^2(\mu)g''(\mu) + g'(\mu)\kappa_3(\mu) = 0.$$

Show that if $\kappa_2(\mu)$ and $\kappa_3(\mu)$ are related as in (a), then

$$g(x) = \int^x \kappa_2^{-1/3}(\mu) d\mu.$$

(c) Hence find symmetrizing transformations for Poisson and binomial variables. (McCullagh and Nelder, 1989, Section 4.8)

- 8 Show that the chi-squared density with known degrees of freedom ν ,

$$\frac{y^{\nu/2-1}}{2^{\nu/2}\sigma^\nu\Gamma(\nu/2)} \exp\left(-\frac{y}{2\sigma^2}\right), \quad y > 0, \sigma > 0, \nu = 1, 2, \dots,$$

can be written in generalized linear model form (10.14), where θ and ϕ are functions, to be found, of ν and σ^2 . Hence derive an expression for its r th cumulant, $r \geq 1$.

The yield of an industrial process was measured r_i times independently at m different temperatures t_i . The resulting yields Z_{ij} , $i = 1, \dots, m$, $j = 1, \dots, r_i$ may be assumed to be independent and normally distributed with both means ξ_i and variances τ_i dependent on t_i . Explain how the sums of squares $Y_i = \sum_{j=1}^{r_i} (Z_{ij} - \bar{Z}_i)^2$, where $\bar{Z}_i = r_i^{-1} \sum_{j=1}^{r_i} Z_{ij}$, may be used to assess the dependence of variance on temperature in a suitable generalized linear model. Briefly discuss the advantages and disadvantages of the canonical link function of your model.

- 9 At each of the doses $x_1 < x_2 < \dots < x_n$ of a drug, m animals are tested. At dose x_i , r_i animals respond. Derive the maximum likelihood equation when the linear predictor takes the form $\eta = \beta x$ when a probit link function is used. If only one dosage $x_0 > 0$ is used, show that

$$\hat{\beta} = \frac{1}{x_0} \Phi^{-1}(r/m), \quad \text{var}(\hat{\beta}) \doteq \frac{\Phi(\beta x_0)\{1 - \Phi(\beta x_0)\}}{mx_0^2\{\phi(\beta x_0)\}^2},$$

where ϕ and Φ are the standard normal density and distribution functions. Plot the function $\Phi(\eta)\{1 - \Phi(\eta)\}/\phi(\eta)^2$ for η in the range $-3 \leq \eta \leq 3$, and comment on the implications for the choice of x_0 if there is some prior knowledge of the likely value of β .

Table 10.24 Simulated data with two covariates, binary response, and fitted values.

Case	x_1	x_2	y	$\hat{y}$
1	3.7	0.83	1	0.999
2	3.5	1.09	1	0.999
3	1.25	2.50	1	0.875
4	0.75	1.50	1	0.066
5	0.8	3.2	1	0.886
6	0.7	3.5	1	0.921
7	0.6	0.75	0	0.005
8	1.1	1.70	0	0.320
9	0.9	0.75	0	0.017
10	0.9	0.45	0	0.008

- 10 Let Y be binomial with probability $\pi = e^\lambda / (1 + e^\lambda)$ and denominator m .
 (a) Show that $m - Y$ is binomial with $\lambda' = -\lambda$. Consider

$$\tilde{\lambda} = \log \left(\frac{Y + c_1}{m - Y + c_2} \right)$$

as an estimator of λ . Show that in order to achieve consistency under the transformation $Y \rightarrow m - Y$, we must have $c_1 = c_2$.

- (b) Write $Y = m\pi + \sqrt{m\pi(1-\pi)}Z$, where $Z = O_p(1)$ for large m . Show that

$$E\{\log(Y + c)\} = \log(m\pi) + \frac{c}{m\pi} - \frac{1 - \pi}{2m\pi} + O(m^{-3/2}).$$

Find the corresponding expansion for $E\{\log(m - Y + c)\}$, and with $c_1 = c_2 = c$ find the value of c for which $\tilde{\lambda}$ is unbiased for λ to order m^{-1} .

What is the connection to the empirical logistic transform?
 (Cox, 1970, Section 3.2)

- 11 Arcturian society is surprisingly similar to ours, the main differences being that Arcturians have three eyes (left, centre, and right) and are better at quantum physics. Their statistics is relatively rudimentary. On a recent study visit to our planet an Arcturian statistician encountered marginal models and decided to use one for visual impairment data similar to those in Table 10.16. He set up a $2 \times 2 \times 2$ table of probabilities $(\pi_{000}, \pi_{001}, \pi_{010}, \pi_{011}, \pi_{100}, \pi_{101}, \pi_{110}, \pi_{111})$ and used logistic models with marginal probabilities $\pi_L = \pi_{100} + \pi_{101} + \pi_{110} + \pi_{111}$ and so forth, and odds ratios

$$\gamma_{LC} = \frac{\Pr(L = C = 1)\Pr(L = C = 0)}{\Pr(L = 1, C = 0)\Pr(L = 0, C = 1)},$$

$$\gamma_{LR} = \frac{\Pr(L = R = 1)\Pr(L = R = 0)}{\Pr(L = 1, R = 0)\Pr(L = 0, R = 1)}.$$

Show that the corresponding odds ratio γ_{CR} may be expressed as

$$\frac{\Pr(C = R = 1) \{1 - \pi_C - \pi_R + \Pr(C = R = 1)\}}{\{\pi_C - \Pr(C = R = 1)\} \{\pi_R - \Pr(C = R = 1)\}},$$

and that $\Pr(C = R = 1)$ lies between $\min(\pi_C, \pi_R)$ and $\max(0, \Pr(L = C = 1) + \Pr(L = R = 1) - \pi_L)$. Deduce that if $\pi_L, \pi_C, \pi_R, \gamma_{LC}$ and γ_{LR} are fixed, then the range of values that γ_{CR} can take is limited by the other parameters. What problems of fitting and interpretation might be encountered with such a model? Compare this with the corresponding log-linear model.

- 12 The data in Table 10.24 are from an experiment with a binary response in which two covariates are fitted. The parameter estimates and their standard errors are $\hat{\beta}_0 = -9.530(3.224)$, $\hat{\beta}_1 = 3.882(1.425)$, and $\hat{\beta}_2 = 2.649(0.9121)$. The table gives the data and fitted values for the first ten of the 39 observations. The overall deviance for the model is 29.77.

He usually comes disguised as Elvis, but attends statistical congresses in the guise of an eminent statistician; this may account for the other-worldly discussion.

Give a careful interpretation of the effect of the covariates on the response.

Verify that the fitted value for case 8 is correct.

Is the deviance a useful guide to the fit of the model?

- 13 (a) In (10.43), show that $p_{\text{obs}}(h) = \Pr_0(y^T A y \geq 0)$, where A is a $n \times n$ real symmetric matrix.
 (b) Let $y \sim N_n(\mu, \Omega)$, where $\Omega = LL^T$ and L is a non-singular lower triangular matrix. If $\mu = 0$, then show that $y^T A y \stackrel{D}{=} \sum_{j=1}^n \lambda_j U_j^2$, where $U_1, \dots, U_n \stackrel{\text{iid}}{\sim} N(0, 1)$ and $\lambda_1 \geq \dots \geq \lambda_n$ are the eigenvalues of $L^T A L$. Deduce that the r th cumulant of $y^T A y$ equals $\kappa_r = 2^{r-1}(r-1)!\text{tr}\{(\Omega A)^r\}$. Show that the same is true whenever μ lies in the null space of A .
 (c) For a simple approximation to the distribution of $y^T A y$, we match its first three cumulants with those of the random variable $aW + c$, where $W \sim \chi_b^2$. Show that this gives $a = |\kappa_3|/(4\kappa_2)$, $b = 8\kappa_2^3/\kappa_3^2$ and $c = \kappa_1 - ab$. Outline how this can be used to approximate $p_{\text{obs}}(h)$.
 (d) Compute the cumulant-generating function of $y^T A y$, and develop a saddlepoint approximation to its distribution.
 (Azzalini *et al.*, 1989; Azzalini and Bowman, 1993; Kuonen, 1999)
- 14 In the penalized least squares setup of Section 10.7.2, with $t_0 < t_1 < \dots < t_n < t_{n+1}$, set $h_j = t_{j+1} - t_j$ for each $j = 1, \dots, n-1$, let $g_1, \dots, g_n$ and $\gamma_1, \dots, \gamma_n$ be arbitrary real numbers, and define

$$g(t) = \frac{(t - t_j)g_{j+1} + (t_{j+1} - t)g_j}{h_j} - \frac{1}{6}(t - t_j)(t_{j+1} - t) \left\{ \left(1 + \frac{t - t_j}{h_j}\right) \gamma_{j+1} + \left(1 + \frac{t_{j+1} - t}{h_j}\right) \gamma_j \right\}$$

on each interval $t_j \leq t \leq t_{j+1}$, $j = 1, \dots, n-1$.

(a) Show that on each such interval $g(t)$ is a cubic function with $g(t_j) = g_j$.

(b) Show that

$$\lim_{t \downarrow t_j} g'(t) = \frac{g_{j+1} - g_j}{h_j} - \frac{1}{6}h_j(2\gamma_j + \gamma_{j+1}),$$

$$\lim_{t \uparrow t_{j+1}} g'(t) = \frac{g_{j+1} - g_j}{h_j} + \frac{1}{6}h_j(\gamma_j + 2\gamma_{j+1}),$$

and

$$g''(t) = \frac{(t - t_j)\gamma_{j+1} + (t_{j+1} - t)\gamma_j}{h_j}, \quad g'''(t) = h_j^{-1}(\gamma_{j+1} - \gamma_j), \quad t_j \leq t \leq t_{j+1},$$

and hence deduce that $g''(t_j) = \gamma_j$.

(c) Show that

$$g'(t_1) = \frac{g_2 - g_1}{t_2 - t_1} - \frac{1}{6}(t_2 - t_1)\gamma_2, \quad g'(t_n) = \frac{g_n - g_{n-1}}{t_n - t_{n-1}} - \frac{1}{6}(t_n - t_{n-1})\gamma_{n-1},$$

and deduce that if $g(t)$ is to be a natural cubic spline, then we must define

$$g(t) = \begin{cases} g_1 - (t_1 - t)g'(t_1), & t \leq t_1, \\ g_n + (t - t_n)g'(t_n), & t \geq t_n, \end{cases}$$

independent of the values of t_0 and t_{n+1} . Deduce that $\gamma_1 = \gamma_n = 0$.

(d) We have seen that $g(t)$ is continuous, with continuous first derivative at t_1 and t_n , and that $\lim_{t \uparrow t_j} g''(t) = \lim_{t \downarrow t_j} g''(t) = \gamma_j$ for each j . If $g(t)$ is to be a natural cubic spline, it must also satisfy $\lim_{t \uparrow t_j} g'(t) = \lim_{t \downarrow t_j} g'(t)$ for each $j = 2, \dots, n-1$. Show that this

implies that

$$\frac{g_{j+1} - g_j}{h_j} - \frac{g_j - g_{j-1}}{h_{j-1}} = \frac{1}{6}h_{j-1}\gamma_{j-1} + \frac{1}{3}(h_{j-1} + h_j)\gamma_j + \frac{1}{6}h_j\gamma_{j+1},$$

$$j = 2, \dots, n-1,$$

Recall that $\gamma_1 = \gamma_n = 0$.

and that this system of equations may be rewritten as $Q^T g = R\gamma$, where $g^T = (g_1, \dots, g_n)$, $\gamma^T = (\gamma_2, \dots, \gamma_{n-1})$, and Q and R have dimensions $n \times (n-2)$ and $(n-2) \times (n-2)$; it is necessary to label the columns of Q from 2 to $n-2$ and both rows and columns of R likewise, so their top left elements are respectively q_{12} and r_{22} .

(e) Use integration by parts to show that the integral in (10.46) may be written

$$\int_{t_0}^{t_{n+1}} \{g''(t)\}^2 dt = \sum_{j=1}^{n-1} \frac{\gamma_{j+1} - \gamma_j}{h_j} (g_j - g_{j+1}),$$

and deduce that the integral may be written $\gamma^T Q^T g = g^T K g$.

(f) Write down Q and R when $n = 5$ and $h_1 = \dots = h_{n-1} = 1$. Show that R is then invertible, and give $K = QR^{-1}Q^T$.

(Green and Silverman, 1994, pp. 22–25)

- 15 (a) Let $U_1, \dots, U_n$ be independent exponential variables with parameters $\lambda_1, \dots, \lambda_n$, and let $H_0(u)$ be a differentiable monotone increasing function of $u > 0$, with derivative $h_0(u)$. Show that $Y_1 = H_0(U_1), \dots, Y_n = H_0(U_n)$ have joint density

$$\prod_{j=1}^n \lambda_j h_0(y_j) \exp\{-\lambda_j H_0(y_j)\}.$$

(b) Show that the joint density of $U_1, \dots, U_n$ may be written as

$$\prod_{j=1}^n \frac{\lambda_{(j)}}{\sum_{i=j}^n \lambda_{(i)}} \times \prod_{j=1}^n \left(\sum_{i=j}^n \lambda_{(i)} \right) \exp(-\lambda_{(j)} u_{(j)}), \quad (10.66)$$

where the elements of the rank statistic $R = \{(1), \dots, (n)\}$ are determined by the ordering on the failure times, $U_{(1)} < \dots < U_{(n)}$. Establish that the first term of this product is invariant to transformations $Y = H_0(U)$ but that the second is not.

(c) Suppose that $\lambda_j = \exp(x_j^T \beta)$. Give an argument why inference for β should be based on the first term of (10.66) only.

- 16 In Figure 5.8, let y represent time on trial and t calendar time, and suppose that the hazard function for an individual with covariates x has form $h_0(y)h_0^\dagger(t)\exp(x^T\beta)$, where $h_0(y)$ represents a baseline hazard for time on trial and $h_0^\dagger(t)$ a baseline hazard for calendar time. Discuss how partial likelihood inference might be generalized to account for inclusion of $h_0^\dagger(t)$, which is included to allow for changes in medical practice during the course of the trial.
- 17 Consider independent exponential variables Y_j with densities $\lambda_j \exp(-\lambda_j y_j)$, where $\lambda_j = \exp(\beta_0 + \beta_1 x_j)$, $j = 1, \dots, n$, where x_j is scalar and $\sum x_j = 0$ without loss of generality. (a) Find the expected information for β_0, β_1 and show that the maximum likelihood estimator $\hat{\beta}_1$ has asymptotic variance $(nm_2)^{-1}$, where $m_2 = n^{-1} \sum x_j^2$. (b) Under no censoring, show that the partial log likelihood for β_1 equals

$$-\sum_{j=1}^n \log \left\{ \sum_{i=j}^n \exp(\beta_1 x_{(i)}) \right\},$$

where the elements of the rank statistic $R = \{(1), \dots, (n)\}$ are determined by the ordering on the failure times, $y_{(1)} < \dots < y_{(n)}$. Deduce the information in the partial likelihood is

$$I_R(\beta_1) = \sum_{j=1}^n E_R \{m_{2,j}(\beta_1) - m_{1,j}(\beta_1)^2\},$$

where the expectation is over the distribution of R and

$$m_{k,j}(\beta_1) = \frac{\sum_{i=j}^n x_{(i)}^k \exp(\beta_1 x_{(i)})}{\sum_{i=j}^n \exp(\beta_1 x_{(i)})}.$$

Show that when $\beta_1 = 0$,

$$E_R\{m_{2,j}(\beta_1)\} = m_2, \quad E_R\{m_{1,j}(\beta_1)^2\} = \frac{m_2}{n-1} \sum_{i=1}^n \frac{i-1}{n-i+1},$$

and hence find the efficiency of partial likelihood estimation of β_1 relative to maximum likelihood estimation. Compute this for $n = 2, 5, 10, 20, 50, 100$, and comment.

(c) It can be shown that as $n \rightarrow \infty$ for small β_1 , the relative efficiency equals $\exp(-m_2\beta_1^2)$. Show that in the two-sample problem with equal numbers of observations in each group and $x_j = \pm 1/2$, the relative efficiency exceeds 0.75 when $|\beta_1| < 1.07$, corresponding to a ratio of failure rates between the two groups in the range $(1/3, 3)$. Discuss. (Kalbfleisch, 1974)

- 18 Suppose that n independent Poisson processes of rates $\lambda_j(y)$ are observed simultaneously, and that the m events occur at $0 < y_1 < \dots < y_m < y_0$, in processes $j_1, \dots, j_m$.

(a) Show that the probabilities that the first event occurs at y_1 and that given this it has type j_1 are respectively

$$\left\{ \sum_{j=1}^n \lambda_j(y_1) \right\} \exp \left\{ - \sum_{j=1}^n \int_0^{y_1} \lambda_j(u) du \right\}, \quad \frac{\lambda_{j_1}(y_1)}{\sum_{j=1}^n \lambda_j(y_1)}.$$

Hence interpret the quantities

$$\exp \left\{ - \sum_{j=1}^n \int_0^{y_0} \lambda_j(u) du \right\} \prod_{i=1}^m \left\{ \sum_{j=1}^n \lambda_j(y_i) \right\}, \quad \prod_{i=1}^m \frac{\lambda_{j_i}(y_i)}{\sum_{j=1}^n \lambda_j(y_i)}. \quad (10.67)$$

(b) Now suppose that $\lambda_j(y) = h_0(y) \xi\{\beta; x_j(y)\} V_j(y)$, where $h_0(y)$ is a baseline hazard function, $\xi\{\beta; x_j(y)\}$ depends on parameters β and time-varying covariates $x_j(y)$, and $V_j(y)$ is a predictable process, with $V_j(y) = 1$ if the j th process is in view at time y , and $V_j(y) = 0$ if not. Thus if the j th process is censored at time c_j , $V_j(y) = 0$, $y > c_j$. If $\mathcal{R}_i$ is the set $\{j : V_j(y_i) = 1\}$, show that the second term in (10.67) equals

$$\prod_{i=1}^m \frac{\xi\{\beta; x_{j_i}(y_i)\}}{\sum_{j \in \mathcal{R}_i} \xi\{\beta; x_j(y_i)\}}.$$

How does this specialize for time-varying explanatory variables in the proportional hazards model?

- 19 Two individuals with cumulative hazard functions $uH_1(y_1)$ and $uH_2(y_2)$ are independent conditional on the value u of a frailty U whose density is $f(u)$.

(a) For this shared frailty model, show that

$$\mathcal{F}(y_1, y_2) = \Pr(Y_1 > y_1, Y_2 > y_2) = \int_0^\infty \exp\{-uH_1(y_1) - uH_2(y_2)\} f(u) du.$$

If $f(u) = \lambda^\alpha u^{\alpha-1} \exp(-\lambda u) / \Gamma(\alpha)$, for $u > 0$ is a gamma density, then show that

$$\mathcal{F}(y_1, y_2) = \frac{\lambda^\alpha}{\{\lambda + H_1(y_1) + H_2(y_2)\}^\alpha}, \quad y_1, y_2 > 0,$$

and deduce that in terms of the marginal survivor functions $\mathcal{F}_1(y_1)$ and $\mathcal{F}_2(y_2)$ of Y_1 and Y_2 ,

$$\mathcal{F}(y_1, y_2) = \{\mathcal{F}_1(y_1)^{-1/\alpha} + \mathcal{F}_2(y_2)^{-1/\alpha} - 1\}^{-\alpha}, \quad y_1, y_2 > 0.$$

What happens to this joint survivor function as $\alpha \rightarrow \infty$?

- (b) Find the likelihood contributions when both individuals are observed to fail, when one is censored, and when both are censored.
- (c) Extend this to k individuals with parametric regression models for survival.
- 20 A positive stable random variable U has $E(e^{-sU}) = \exp(-\delta s^\alpha/\alpha)$, $0 < \alpha \leq 1$.
- (a) Show that if Y follows a proportional hazards model with cumulative hazard function $u \exp(x^\top \beta) H_0(y)$, conditional on $U = u$, then Y also follows a proportional hazards model unconditionally. Are β , α , and δ estimable from data with single individuals only?
- (b) Consider a shared frailty model, as in the previous question, with positive stable U . Show that the joint survivor function may be written as

$$\mathcal{F}(y_1, y_2) = \exp\left(-\left[-\log \mathcal{F}_1(y_1)\right]^{1/\alpha} + \left[-\log \mathcal{F}_2(y_2)\right]^{1/\alpha}\right)^\alpha, \quad y_1, y_2 > 0,$$

in terms of the marginal survivor functions $\mathcal{F}_1$ and $\mathcal{F}_2$. Show that if the conditional cumulative hazard functions are Weibull, $u H_r(y) = u \xi_r y^\gamma$, $\gamma > 0$, $r = 1, 2$, then the marginal survivor functions are also Weibull. Show also that the time to the first event has a Weibull distribution.

- 21 Consider individuals arising in k independent clusters of sizes $n_1, \dots, n_k$, and such that conditional on the values $u_1, \dots, u_k$ of unobserved frailties $U_1, \dots, U_k$, the individuals in the i th cluster have survival times independently distributed according to a proportional hazards model with cumulative hazards $u_i \xi_{ij} H_0(y_{ij})$, for $j = 1, \dots, n_i$, where ξ_{ij} is shorthand for $\xi(\beta; x_{ij})$, x_{ij} being a vector of explanatory variables. Let $h_0(y)$ be the derivative of $H_0(y)$, and suppose that the U_i are independent gamma variables with unit means and shape parameter θ .

(a) If the survival times are subject to non-informative censoring, show that the joint density of U_i and the (survival time, censoring indicator) pairs (Y_{ij}, D_{ij}) for the i th cluster is

$$\prod_{j=1}^{n_i} \{u_i \xi_{ij} h_0(y_{ij})\}^{d_{ij}} \times \exp\left\{-\sum_{j=1}^{n_i} u_i \xi_{ij} H_0(y_{ij})\right\} \times \frac{\theta^\theta u_i^{\theta-1}}{\Gamma(\theta)} \exp(-\theta u_i),$$

and deduce that the conditional means of U_i and of $\log U_i$ given the observed data are $w_i(\theta, \beta) = A_i/B_i$ and $\psi(A_i) - \log B_i$, where

$$A_i = \theta + d_i, \quad B_i = \theta + \sum_{j=1}^{n_i} \xi_{ij} H_0(y_{ij}), \quad d_i = \sum_{j=1}^{n_i} d_{ij}, \quad \psi(\alpha) = d \log \Gamma(\alpha)/d\alpha.$$

Discuss the merits and demerits (if any) of inference in terms of $\psi = \theta^{-1}$: what happens as $\psi \rightarrow 0$?

(b) Show that a step of the EM algorithm for estimation of (θ, β) involves updating (θ', β') by maximization of $\ell_1(\beta, H_0) + \ell_2(\theta)$ over β, H_0 , and θ , where

$$\ell_1(\beta, H_0) = \sum_{i=1}^k \sum_{j=1}^{n_i} [d_{ij} \{\log \xi_{ij} + \log h_0(y_{ij})\} - w'_{ij} \xi_{ij} H_0(y_{ij})],$$

$$\ell_2(\theta) = \sum_{i=1}^k \{(\theta + d_i - 1)(\psi(A'_i) - \log B'_i) - A_i \theta / B'_i\} + k\{\theta \log \theta - \log \Gamma(\theta)\},$$

$w'_i = w_i(\theta', \beta')$ and A'_i and B'_i are evaluated at (θ', β') . Extend the argument leading to (10.57) to establish that the step for β involves maximizing the partial likelihood that is a product over individuals of terms

$$\left\{ \frac{w'_i \xi(\beta; x_{ij})}{\sum_{k \in \mathcal{R}_{ij}} w'_i \xi(\beta; x_k)} \right\}^{d_{ij}},$$

with the risk set $\mathcal{R}_{ij}$ containing those individuals from every cluster available to fail at failure time y_{ij} . When $\xi(\beta; x) = \exp(x^\top \beta)$, show that this amounts to using an offset in the proportional hazards model. Find the form of $\hat{H}_0(y)$, and give an algorithm for estimation of β and θ .

(c) Show that the joint survivor function for the individuals in a cluster is

$$\Pr(Y_{i1} > y_{i1}, \dots, Y_{in_i} > y_{in_i}) = \theta^\theta \left\{ \theta + \sum_{j=1}^{n_i} \xi_{ij} H_0(y_j) \right\}^{-\theta},$$

and hence give the log likelihood contribution from (y_{ij}, d_{ij}) , for $j = 1, \dots, n_i$. Explain how to use this to obtain the observed information matrix for θ and β based on the estimates obtained in (b).

(Klein, 1992)

22 Let $Y_1, \dots, Y_n$ be independent exponential variables with hazards $\lambda_j = \exp(\beta^T x_j)$.

(a) Show that the expected information for β is $X^T X$, in the usual notation.

(b) Now suppose that Y_j is subject to uninformative right censoring at time c_j , so that y_j is a censoring time or a failure time as the case may be. Show that the log likelihood is

$$\ell_U(\beta) = \sum_f \beta^T x_j - \sum_{j=1}^n \exp(\beta^T x_j) y_j,$$

where $\sum_f$ denotes a sum over observations seen to fail. If the j th censoring-time is exponentially distributed with rate κ_j , show that the expected information for β is $X^T X - X^T C X$, where $C = \text{diag}\{c_1, \dots, c_n\}$, and $c_j = \kappa_j/(\kappa_j + \lambda_j)$ is the probability that the j th observation is censored. What is the implication for estimation of β if the c_j are constant?

(c) Sometimes a variable W_j has been measured which can act as a surrogate response variable for censored individuals. We formulate this as $W_j = Z_j/U_j$, where Z_j is the unobserved remaining life-time of the j th individual from the moment of censoring, and U_j is a noise component which has a fixed distribution independent of the censoring time and of x_j . Owing to the exponential assumption, the excess life Z_j is independent of Y_j if censoring occurred. If U_j has gamma density

$$\alpha^\kappa u^{\kappa-1} \exp(-\alpha u) / \Gamma(\kappa), \quad \alpha, \kappa > 0, u > 0,$$

show that W_j has density

$$\lambda_j \kappa \alpha^\kappa / (\alpha + \lambda_j w)^{\kappa+1}, \quad w > 0.$$

Show that the log likelihood for the data, including the additional information in the W_j , is

$$\ell(\beta) = L_U(\beta) + \sum_c \left\{ \beta^T x_j + \log \kappa + \kappa \log \alpha - (\kappa + 1) \log (\alpha + e^{\beta^T x_j} w_j) \right\},$$

where $\sum_c$ denotes a sum over censored individuals, and we have assumed that α and κ are known. Show that the expected information for β is

$$X^T X - 2/(\kappa + 2) X^T C X,$$

and compare this with (b). Explain qualitatively in terms of the variability of the distribution of U why the loss of information decreases as κ increases.

(Cox, 1983)